

Derivative of a Composite Function

We begin with examples.

EXAMPLE 1 Relating Derivatives

The function $y = \frac{3}{2}x = \frac{1}{2}(3x)$ is the composite of the functions $y = \frac{1}{2}u$ and $u = 3x$.

How are the derivatives of these functions related?

Solution We have

$$\frac{dy}{dx} = \frac{3}{2}, \quad \frac{dy}{du} = \frac{1}{2}, \quad \text{and} \quad \frac{du}{dx} = 3.$$

Since $\frac{3}{2} = \frac{1}{2} \cdot 3$, we see that

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Is it an accident that

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}?$$

If we think of the derivative as a rate of change, our intuition allows us to see that this relationship is reasonable. If $y = f(u)$ changes half as fast as u and $u = g(x)$ changes three times as fast as x , then we expect y to change $3/2$ times as fast as x . This effect is much like that of a multiple gear train (Figure 3.26). ■

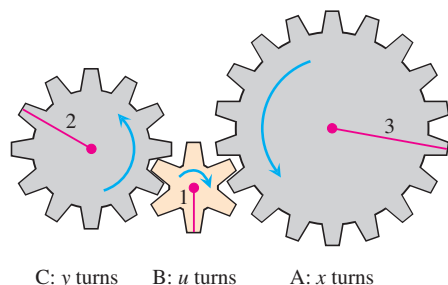


FIGURE 3.26 When gear A makes x turns, gear B makes u turns and gear C makes y turns. By comparing circumferences or counting teeth, we see that $y = u/2$ (C turns one-half turn for each B turn) and $u = 3x$ (B turns three times for A's one), so $y = 3x/2$. Thus, $dy/dx = 3/2 = (1/2)(3) = (dy/du)(du/dx)$.

EXAMPLE 2

The function

$$y = 9x^4 + 6x^2 + 1 = (3x^2 + 1)^2$$

is the composite of $y = u^2$ and $u = 3x^2 + 1$. Calculating derivatives, we see that

$$\begin{aligned} \frac{dy}{du} \cdot \frac{du}{dx} &= 2u \cdot 6x \\ &= 2(3x^2 + 1) \cdot 6x \\ &= 36x^3 + 12x. \end{aligned}$$

Calculating the derivative from the expanded formula, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(9x^4 + 6x^2 + 1) \\ &= 36x^3 + 12x. \end{aligned}$$

Once again,

$$\frac{dy}{du} \cdot \frac{du}{dx} = \frac{dy}{dx}.$$

The derivative of the composite function $f(g(x))$ at x is the derivative of f at $g(x)$ times the derivative of g at x . This is known as the Chain Rule (Figure 3.27). ■

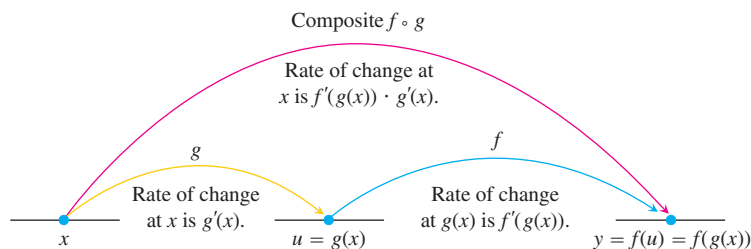


FIGURE 3.27 Rates of change multiply: The derivative of $f \circ g$ at x is the derivative of f at $g(x)$ times the derivative of g at x .

THEOREM 3 The Chain Rule

If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x , and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

where dy/du is evaluated at $u = g(x)$.

Intuitive "Proof" of the Chain Rule:

Let Δu be the change in u corresponding to a change of Δx in x , that is

$$\Delta u = g(x + \Delta x) - g(x)$$

Then the corresponding change in y is

$$\Delta y = f(u + \Delta u) - f(u).$$

It would be tempting to write

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \quad (1)$$

and take the limit as $\Delta x \rightarrow 0$:

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \\ &= \lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \quad (\text{Note that } \Delta u \rightarrow 0 \text{ as } \Delta x \rightarrow 0 \\ &\quad \text{since } g \text{ is continuous.}) \\ &= \frac{dy}{du} \frac{du}{dx}. \end{aligned}$$

The only flaw in this reasoning is that in Equation (1) it might happen that $\Delta u = 0$ (even when $\Delta x \neq 0$) and, of course, we can't divide by 0. The proof requires a different approach to overcome this flaw, and we give a precise proof in Section 3.8. ■

EXAMPLE 3 Applying the Chain Rule

An object moves along the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = \cos(t^2 + 1)$. Find the velocity of the object as a function of t .

Solution We know that the velocity is dx/dt . In this instance, x is a composite function: $x = \cos(u)$ and $u = t^2 + 1$. We have

$$\begin{aligned}\frac{dx}{du} &= -\sin(u) & x &= \cos(u) \\ \frac{du}{dt} &= 2t. & u &= t^2 + 1\end{aligned}$$

By the Chain Rule,

$$\begin{aligned}\frac{dx}{dt} &= \frac{dx}{du} \cdot \frac{du}{dt} \\ &= -\sin(u) \cdot 2t & \frac{dx}{du} \text{ evaluated at } u \\ &= -\sin(t^2 + 1) \cdot 2t \\ &= -2t \sin(t^2 + 1).\end{aligned}$$

As we see from Example 3, a difficulty with the Leibniz notation is that it doesn't state specifically where the derivatives are supposed to be evaluated. ■

"Outside-Inside" Rule

It sometimes helps to think about the Chain Rule this way: If $y = f(g(x))$, then

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x).$$

In words, differentiate the "outside" function f and evaluate it at the "inside" function $g(x)$ left alone; then multiply by the derivative of the "inside function."

EXAMPLE 4 Differentiating from the Outside In

Differentiate $\sin(x^2 + x)$ with respect to x .

Solution

$$\frac{d}{dx} \sin(\underbrace{x^2 + x}_{\text{inside}}) = \cos(\underbrace{x^2 + x}_{\substack{\text{inside} \\ \text{left alone}}}) \cdot \underbrace{(2x + 1)}_{\substack{\text{derivative of} \\ \text{the inside}}}$$

Repeated Use of the Chain Rule

We sometimes have to use the Chain Rule two or more times to find a derivative. Here is an example.

HISTORICAL BIOGRAPHY

Johann Bernoulli
(1667–1748)

EXAMPLE 5 A Three-Link “Chain”

Find the derivative of $g(t) = \tan(5 - \sin 2t)$.

Solution Notice here that the tangent is a function of $5 - \sin 2t$, whereas the sine is a function of $2t$, which is itself a function of t . Therefore, by the Chain Rule,

$$\begin{aligned}
 g'(t) &= \frac{d}{dt}(\tan(5 - \sin 2t)) \\
 &= \sec^2(5 - \sin 2t) \cdot \frac{d}{dt}(5 - \sin 2t) && \text{Derivative of } \tan u \text{ with } u = 5 - \sin 2t \\
 &= \sec^2(5 - \sin 2t) \cdot \left(0 - \cos 2t \cdot \frac{d}{dt}(2t)\right) && \text{Derivative of } 5 - \sin u \text{ with } u = 2t \\
 &= \sec^2(5 - \sin 2t) \cdot (-\cos 2t) \cdot 2 \\
 &= -2(\cos 2t) \sec^2(5 - \sin 2t).
 \end{aligned}$$

The Chain Rule with Powers of a Function

If f is a differentiable function of u and if u is a differentiable function of x , then substituting $y = f(u)$ into the Chain Rule formula

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

leads to the formula

$$\frac{d}{dx} f(u) = f'(u) \frac{du}{dx}.$$

Here's an example of how it works: If n is a positive or negative integer and $f(u) = u^n$, the Power Rules (Rules 2 and 7) tell us that $f'(u) = nu^{n-1}$. If u is a differentiable function of x , then we can use the Chain Rule to extend this to the **Power Chain Rule**:

$$\frac{d}{dx} u^n = nu^{n-1} \frac{du}{dx}. \quad \frac{d}{du} (u^n) = nu^{n-1}$$

EXAMPLE 6 Applying the Power Chain Rule

$$\begin{aligned}
 \text{(a)} \quad \frac{d}{dx} (5x^3 - x^4)^7 &= 7(5x^3 - x^4)^6 \frac{d}{dx} (5x^3 - x^4) && \text{Power Chain Rule with } u = 5x^3 - x^4, n = 7 \\
 &= 7(5x^3 - x^4)^6 (5 \cdot 3x^2 - 4x^3) \\
 &= 7(5x^3 - x^4)^6 (15x^2 - 4x^3) \\
 \text{(b)} \quad \frac{d}{dx} \left(\frac{1}{3x - 2} \right) &= \frac{d}{dx} (3x - 2)^{-1} \\
 &= -1(3x - 2)^{-2} \frac{d}{dx} (3x - 2) && \text{Power Chain Rule with } u = 3x - 2, n = -1 \\
 &= -1(3x - 2)^{-2} (3) \\
 &= -\frac{3}{(3x - 2)^2}
 \end{aligned}$$

In part (b) we could also have found the derivative with the Quotient Rule.

$\sin^n x$ means $(\sin x)^n$, $n \neq -1$.

EXAMPLE 7 Finding Tangent Slopes

- (a) Find the slope of the line tangent to the curve $y = \sin^5 x$ at the point where $x = \pi/3$.
 (b) Show that the slope of every line tangent to the curve $y = 1/(1 - 2x)^3$ is positive.

Solution

$$\begin{aligned} \text{(a)} \quad \frac{dy}{dx} &= 5 \sin^4 x \cdot \frac{d}{dx} \sin x && \text{Power Chain Rule with } u = \sin x, n = 5 \\ &= 5 \sin^4 x \cos x \end{aligned}$$

The tangent line has slope

$$\left. \frac{dy}{dx} \right|_{x=\pi/3} = 5 \left(\frac{\sqrt{3}}{2} \right)^4 \left(\frac{1}{2} \right) = \frac{45}{32}.$$

$$\begin{aligned} \text{(b)} \quad \frac{dy}{dx} &= \frac{d}{dx} (1 - 2x)^{-3} \\ &= -3(1 - 2x)^{-4} \cdot \frac{d}{dx} (1 - 2x) && \text{Power Chain Rule with } u = (1 - 2x), n = -3 \\ &= -3(1 - 2x)^{-4} \cdot (-2) \\ &= \frac{6}{(1 - 2x)^4} \end{aligned}$$

At any point (x, y) on the curve, $x \neq 1/2$ and the slope of the tangent line is

$$\frac{dy}{dx} = \frac{6}{(1 - 2x)^4},$$

the quotient of two positive numbers. ■

EXAMPLE 8 Radians Versus Degrees

It is important to remember that the formulas for the derivatives of both $\sin x$ and $\cos x$ were obtained under the assumption that x is measured in radians, *not* degrees. The Chain Rule gives us new insight into the difference between the two. Since $180^\circ = \pi$ radians, $x^\circ = \pi x/180$ radians where x° means the angle x measured in degrees.

By the Chain Rule,

$$\frac{d}{dx} \sin(x^\circ) = \frac{d}{dx} \sin\left(\frac{\pi x}{180}\right) = \frac{\pi}{180} \cos\left(\frac{\pi x}{180}\right) = \frac{\pi}{180} \cos(x^\circ).$$

See Figure 3.28. Similarly, the derivative of $\cos(x^\circ)$ is $-(\pi/180) \sin(x^\circ)$.

The factor $\pi/180$, annoying in the first derivative, would compound with repeated differentiation. We see at a glance the compelling reason for the use of radian measure. ■

Parametric Equations

Instead of describing a curve by expressing the y -coordinate of a point $P(x, y)$ on the curve as a function of x , it is sometimes more convenient to describe the curve by expressing *both* coordinates as functions of a third variable t . Figure 3.29 shows the path of a moving particle described by a pair of equations, $x = f(t)$ and $y = g(t)$. For studying motion,

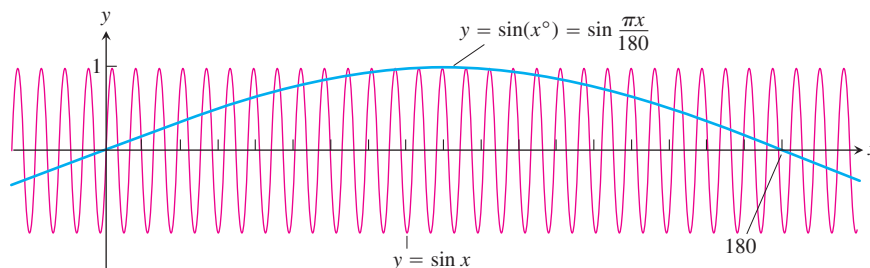


FIGURE 3.28 $\sin(x^\circ)$ oscillates only $\pi/180$ times as often as $\sin x$ oscillates. Its maximum slope is $\pi/180$ at $x = 0$ (Example 8).

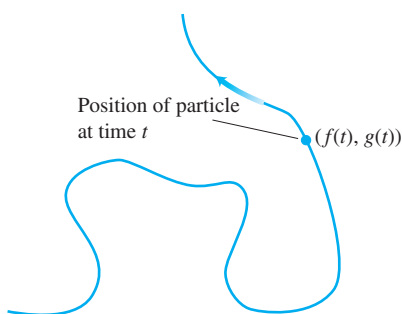


FIGURE 3.29 The path traced by a particle moving in the xy -plane is not always the graph of a function of x or a function of y .

t usually denotes time. Equations like these are better than a Cartesian formula because they tell us the particle's position $(x, y) = (f(t), g(t))$ at any time t .

DEFINITION Parametric Curve

If x and y are given as functions

$$x = f(t), \quad y = g(t)$$

over an interval of t -values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a **parametric curve**. The equations are **parametric equations** for the curve.

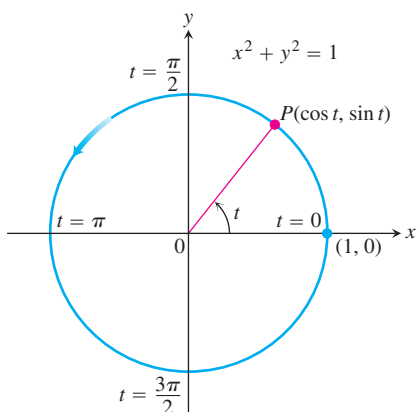


FIGURE 3.30 The equations $x = \cos t$ and $y = \sin t$ describe motion on the circle $x^2 + y^2 = 1$. The arrow shows the direction of increasing t (Example 9).

The variable t is a **parameter** for the curve, and its domain I is the **parameter interval**. If I is a closed interval, $a \leq t \leq b$, the point $(f(a), g(a))$ is the **initial point** of the curve. The point $(f(b), g(b))$ is the **terminal point**. When we give parametric equations and a parameter interval for a curve, we say that we have **parametrized** the curve. The equations and interval together constitute a **parametrization** of the curve.

EXAMPLE 9 Moving Counterclockwise on a Circle

Graph the parametric curves

- (a) $x = \cos t, \quad y = \sin t, \quad 0 \leq t \leq 2\pi.$
 (b) $x = a \cos t, \quad y = a \sin t, \quad 0 \leq t \leq 2\pi.$

Solution

- (a) Since $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$, the parametric curve lies along the unit circle $x^2 + y^2 = 1$. As t increases from 0 to 2π , the point $(x, y) = (\cos t, \sin t)$ starts at $(1, 0)$ and traces the entire circle once counterclockwise (Figure 3.30).
 (b) For $x = a \cos t, y = a \sin t, 0 \leq t \leq 2\pi$, we have $x^2 + y^2 = a^2 \cos^2 t + a^2 \sin^2 t = a^2$. The parametrization describes a motion that begins at the point $(a, 0)$ and traverses the circle $x^2 + y^2 = a^2$ once counterclockwise, returning to $(a, 0)$ at $t = 2\pi$. ■

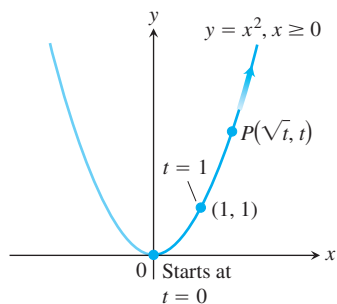


FIGURE 3.31 The equations $x = \sqrt{t}$ and $y = t$ and the interval $t \geq 0$ describe the motion of a particle that traces the right-hand half of the parabola $y = x^2$ (Example 10).

EXAMPLE 10 Moving Along a Parabola

The position $P(x, y)$ of a particle moving in the xy -plane is given by the equations and parameter interval

$$x = \sqrt{t}, \quad y = t, \quad t \geq 0.$$

Identify the path traced by the particle and describe the motion.

Solution We try to identify the path by eliminating t between the equations $x = \sqrt{t}$ and $y = t$. With any luck, this will produce a recognizable algebraic relation between x and y . We find that

$$y = t = (\sqrt{t})^2 = x^2.$$

Thus, the particle's position coordinates satisfy the equation $y = x^2$, so the particle moves along the parabola $y = x^2$.

It would be a mistake, however, to conclude that the particle's path is the entire parabola $y = x^2$; it is only half the parabola. The particle's x -coordinate is never negative. The particle starts at $(0, 0)$ when $t = 0$ and rises into the first quadrant as t increases (Figure 3.31). The parameter interval is $[0, \infty)$ and there is no terminal point. ■

EXAMPLE 11 Parametrizing a Line Segment

Find a parametrization for the line segment with endpoints $(-2, 1)$ and $(3, 5)$.

Solution Using $(-2, 1)$ we create the parametric equations

$$x = -2 + at, \quad y = 1 + bt.$$

These represent a line, as we can see by solving each equation for t and equating to obtain

$$\frac{x + 2}{a} = \frac{y - 1}{b}.$$

This line goes through the point $(-2, 1)$ when $t = 0$. We determine a and b so that the line goes through $(3, 5)$ when $t = 1$.

$$\begin{aligned} 3 &= -2 + a & \Rightarrow & & a = 5 & & x = 3 \text{ when } t = 1. \\ 5 &= 1 + b & \Rightarrow & & b = 4 & & y = 5 \text{ when } t = 1. \end{aligned}$$

Therefore,

$$x = -2 + 5t, \quad y = 1 + 4t, \quad 0 \leq t \leq 1$$

is a parametrization of the line segment with initial point $(-2, 1)$ and terminal point $(3, 5)$. ■

Slopes of Parametrized Curves

A parametrized curve $x = f(t)$ and $y = g(t)$ is **differentiable** at t if f and g are differentiable at t . At a point on a differentiable parametrized curve where y is also a differentiable function of x , the derivatives dy/dt , dx/dt , and dy/dx are related by the Chain Rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

If $dx/dt \neq 0$, we may divide both sides of this equation by dx/dt to solve for dy/dx .

Parametric Formula for dy/dx If all three derivatives exist and $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}. \quad (2)$$

EXAMPLE 12 Differentiating with a ParameterIf $x = 2t + 3$ and $y = t^2 - 1$, find the value of dy/dx at $t = 6$.**Solution** Equation (2) gives dy/dx as a function of t :

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{2} = t = \frac{x-3}{2}.$$

When $t = 6$, $dy/dx = 6$. Notice that we are also able to find the derivative dy/dx as a function of x . ■**EXAMPLE 13** Moving Along the Ellipse $x^2/a^2 + y^2/b^2 = 1$ Describe the motion of a particle whose position $P(x, y)$ at time t is given by

$$x = a \cos t, \quad y = b \sin t, \quad 0 \leq t \leq 2\pi.$$

Find the line tangent to the curve at the point $(a/\sqrt{2}, b/\sqrt{2})$, where $t = \pi/4$. (The constants a and b are both positive.)**Solution** We find a Cartesian equation for the particle's coordinates by eliminating t between the equations

$$\cos t = \frac{x}{a}, \quad \sin t = \frac{y}{b}.$$

The identity $\cos^2 t + \sin^2 t = 1$, yields

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1, \quad \text{or} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

The particle's coordinates (x, y) satisfy the equation $(x^2/a^2) + (y^2/b^2) = 1$, so the particle moves along this ellipse. When $t = 0$, the particle's coordinates are

$$x = a \cos(0) = a, \quad y = b \sin(0) = 0,$$

so the motion starts at $(a, 0)$. As t increases, the particle rises and moves toward the left, moving counterclockwise. It traverses the ellipse once, returning to its starting position $(a, 0)$ at $t = 2\pi$.The slope of the tangent line to the ellipse when $t = \pi/4$ is

$$\begin{aligned} \left. \frac{dy}{dx} \right|_{t=\pi/4} &= \left. \frac{dy/dt}{dx/dt} \right|_{t=\pi/4} \\ &= \left. \frac{b \cos t}{-a \sin t} \right|_{t=\pi/4} \\ &= \frac{b/\sqrt{2}}{-a/\sqrt{2}} = -\frac{b}{a}. \end{aligned}$$

The tangent line is

$$y - \frac{b}{\sqrt{2}} = -\frac{b}{a} \left(x - \frac{a}{\sqrt{2}} \right)$$

$$y = \frac{b}{\sqrt{2}} - \frac{b}{a} \left(x - \frac{a}{\sqrt{2}} \right)$$

or

$$y = -\frac{b}{a}x + \sqrt{2}b. \quad \blacksquare$$

If parametric equations define y as a twice-differentiable function of x , we can apply Equation (2) to the function $dy/dx = y'$ to calculate d^2y/dx^2 as a function of t :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(y') = \frac{dy'/dt}{dx/dt}. \quad \text{Eq. (2) with } y' \text{ in place of } y$$

Parametric Formula for d^2y/dx^2

If the equations $x = f(t)$, $y = g(t)$ define y as a twice-differentiable function of x , then at any point where $dx/dt \neq 0$,

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}. \quad (3)$$

EXAMPLE 14 Finding d^2y/dx^2 for a Parametrized Curve

Find d^2y/dx^2 as a function of t if $x = t - t^2$, $y = t - t^3$.

Solution

- Express $y' = dy/dx$ in terms of t .

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1 - 3t^2}{1 - 2t}$$

- Differentiate y' with respect to t .

$$\frac{dy'}{dt} = \frac{d}{dt} \left(\frac{1 - 3t^2}{1 - 2t} \right) = \frac{2 - 6t + 6t^2}{(1 - 2t)^2} \quad \text{Quotient Rule}$$

- Divide dy'/dt by dx/dt .

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{(2 - 6t + 6t^2)/(1 - 2t)^2}{1 - 2t} = \frac{2 - 6t + 6t^2}{(1 - 2t)^3} \quad \text{Eq. (3)} \quad \blacksquare$$

EXAMPLE 15 Dropping Emergency Supplies

A Red Cross aircraft is dropping emergency food and medical supplies into a disaster area. If the aircraft releases the supplies immediately above the edge of an open field 700 ft long and if the cargo moves along the path

$$x = 120t \quad \text{and} \quad y = -16t^2 + 500, \quad t \geq 0$$

Finding d^2y/dx^2 in Terms of t

- Express $y' = dy/dx$ in terms of t .
- Find dy'/dt .
- Divide dy'/dt by dx/dt .

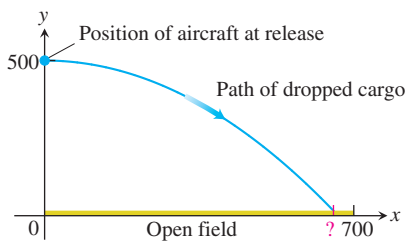


FIGURE 3.32 The path of the dropped cargo of supplies in Example 15.

does the cargo land in the field? The coordinates x and y are measured in feet, and the parameter t (time since release) in seconds. Find a Cartesian equation for the path of the falling cargo (Figure 3.32) and the cargo's rate of descent relative to its forward motion when it hits the ground.

Solution The cargo hits the ground when $y = 0$, which occurs at time t when

$$\begin{aligned} -16t^2 + 500 &= 0 && \text{Set } y = 0. \\ t &= \sqrt{\frac{500}{16}} = \frac{5\sqrt{5}}{2} \text{ sec.} && t \geq 0 \end{aligned}$$

The x -coordinate at the time of the release is $x = 0$. At the time the cargo hits the ground, the x -coordinate is

$$x = 120t = 120\left(\frac{5\sqrt{5}}{2}\right) = 300\sqrt{5} \text{ ft.}$$

Since $300\sqrt{5} \approx 670.8 < 700$, the cargo does land in the field.

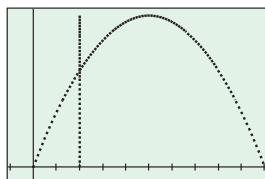
We find a Cartesian equation for the cargo's coordinates by eliminating t between the parametric equations:

$$\begin{aligned} y &= -16t^2 + 500 && \text{Parametric equation for } y \\ &= -16\left(\frac{x}{120}\right)^2 + 500 && \text{Substitute for } t \text{ from the equation } x = 120t. \\ &= -\frac{1}{900}x^2 + 500. && \text{A parabola} \end{aligned}$$

The rate of descent relative to its forward motion when the cargo hits the ground is

$$\begin{aligned} \left.\frac{dy}{dx}\right|_{t=5\sqrt{5}/2} &= \left.\frac{dy/dt}{dx/dt}\right|_{t=5\sqrt{5}/2} \\ &= \left.\frac{-32t}{120}\right|_{t=5\sqrt{5}/2} \\ &= -\frac{2\sqrt{5}}{3} \approx -1.49. \end{aligned}$$

Thus, it is falling about 1.5 feet for every foot of forward motion when it hits the ground. ■



$$\begin{cases} x(t) = 2 \\ y(t) = 160t - 16t^2 \end{cases}$$

and

$$\begin{cases} x(t) = t \\ y(t) = 160t - 16t^2 \end{cases}$$

in dot mode

USING TECHNOLOGY Simulation of Motion on a Vertical Line

The parametric equations

$$x(t) = c, \quad y(t) = f(t)$$

will illuminate pixels along the vertical line $x = c$. If $f(t)$ denotes the height of a moving body at time t , graphing $(x(t), y(t)) = (c, f(t))$ will simulate the actual motion. Try it for the rock in Example 5, Section 3.3 with $x(t) = 2$, say, and $y(t) = 160t - 16t^2$, in dot mode with t Step = 0.1. Why does the spacing of the dots vary? Why does the grapher seem to stop after it reaches the top? (Try the plots for $0 \leq t \leq 5$ and $5 \leq t \leq 10$ separately.)

For a second experiment, plot the parametric equations

$$x(t) = t, \quad y(t) = 160t - 16t^2$$

together with the vertical line simulation of the motion, again in dot mode. Use what you know about the behavior of the rock from the calculations of Example 5 to select a window size that will display all the interesting behavior.

Standard Parametrizations and Derivative Rules

CIRCLE $x^2 + y^2 = a^2$:

$$x = a \cos t$$

$$y = a \sin t$$

$$0 \leq t \leq 2\pi$$

ELLIPSE $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

$$x = a \cos t$$

$$y = b \sin t$$

$$0 \leq t \leq 2\pi$$

FUNCTION $y = f(x)$:

$$x = t$$

$$y = f(t)$$

DERIVATIVES

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}$$

EXERCISES 3.5

Derivative Calculations

In Exercises 1–8, given $y = f(u)$ and $u = g(x)$, find $dy/dx = f'(g(x))g'(x)$.

1. $y = 6u - 9$, $u = (1/2)x^4$
2. $y = 2u^3$, $u = 8x - 1$
3. $y = \sin u$, $u = 3x + 1$
4. $y = \cos u$, $u = -x/3$
5. $y = \cos u$, $u = \sin x$
6. $y = \sin u$, $u = x - \cos x$
7. $y = \tan u$, $u = 10x - 5$
8. $y = -\sec u$, $u = x^2 + 7x$

In Exercises 9–18, write the function in the form $y = f(u)$ and $u = g(x)$. Then find dy/dx as a function of x .

9. $y = (2x + 1)^5$
10. $y = (4 - 3x)^9$
11. $y = \left(1 - \frac{x}{7}\right)^{-7}$
12. $y = \left(\frac{x}{2} - 1\right)^{-10}$
13. $y = \left(\frac{x^2}{8} + x - \frac{1}{x}\right)^4$
14. $y = \left(\frac{x}{5} + \frac{1}{5x}\right)^5$
15. $y = \sec(\tan x)$
16. $y = \cot\left(\pi - \frac{1}{x}\right)$
17. $y = \sin^3 x$
18. $y = 5 \cos^{-4} x$

Find the derivatives of the functions in Exercises 19–38.

19. $p = \sqrt{3 - t}$
20. $q = \sqrt{2r - r^2}$
21. $s = \frac{4}{3\pi} \sin 3t + \frac{4}{5\pi} \cos 5t$

$$22. s = \sin\left(\frac{3\pi t}{2}\right) + \cos\left(\frac{3\pi t}{2}\right)$$

$$23. r = (\csc \theta + \cot \theta)^{-1} \quad 24. r = -(\sec \theta + \tan \theta)^{-1}$$

$$25. y = x^2 \sin^4 x + x \cos^{-2} x \quad 26. y = \frac{1}{x} \sin^{-5} x - \frac{x}{3} \cos^3 x$$

$$27. y = \frac{1}{21} (3x - 2)^7 + \left(4 - \frac{1}{2x^2}\right)^{-1}$$

$$28. y = (5 - 2x)^{-3} + \frac{1}{8} \left(\frac{2}{x} + 1\right)^4$$

$$29. y = (4x + 3)^4 (x + 1)^{-3} \quad 30. y = (2x - 5)^{-1} (x^2 - 5x)^6$$

$$31. h(x) = x \tan(2\sqrt{x}) + 7 \quad 32. k(x) = x^2 \sec\left(\frac{1}{x}\right)$$

$$33. f(\theta) = \left(\frac{\sin \theta}{1 + \cos \theta}\right)^2 \quad 34. g(t) = \left(\frac{1 + \cos t}{\sin t}\right)^{-1}$$

$$35. r = \sin(\theta^2) \cos(2\theta) \quad 36. r = \sec \sqrt{\theta} \tan\left(\frac{1}{\theta}\right)$$

$$37. q = \sin\left(\frac{t}{\sqrt{t+1}}\right) \quad 38. q = \cot\left(\frac{\sin t}{t}\right)$$

In Exercises 39–48, find dy/dt .

39. $y = \sin^2(\pi t - 2)$
40. $y = \sec^2 \pi t$
41. $y = (1 + \cos 2t)^{-4}$
42. $y = (1 + \cot(t/2))^{-2}$

$$\begin{array}{ll}
 43. y = \sin(\cos(2t - 5)) & 44. y = \cos\left(5 \sin\left(\frac{t}{3}\right)\right) \\
 45. y = \left(1 + \tan^4\left(\frac{t}{12}\right)\right)^3 & 46. y = \frac{1}{6}(1 + \cos^2(7t))^3 \\
 47. y = \sqrt{1 + \cos(t^2)} & 48. y = 4 \sin(\sqrt{1 + \sqrt{t}})
 \end{array}$$

Second Derivatives

Find y'' in Exercises 49–52.

$$\begin{array}{ll}
 49. y = \left(1 + \frac{1}{x}\right)^3 & 50. y = (1 - \sqrt{x})^{-1} \\
 51. y = \frac{1}{9} \cot(3x - 1) & 52. y = 9 \tan\left(\frac{x}{3}\right)
 \end{array}$$

Finding Numerical Values of Derivatives

In Exercises 53–58, find the value of $(f \circ g)'$ at the given value of x .

$$\begin{array}{ll}
 53. f(u) = u^5 + 1, \quad u = g(x) = \sqrt{x}, \quad x = 1 & \\
 54. f(u) = 1 - \frac{1}{u}, \quad u = g(x) = \frac{1}{1-x}, \quad x = -1 & \\
 55. f(u) = \cot \frac{\pi u}{10}, \quad u = g(x) = 5\sqrt{x}, \quad x = 1 & \\
 56. f(u) = u + \frac{1}{\cos^2 u}, \quad u = g(x) = \pi x, \quad x = 1/4 & \\
 57. f(u) = \frac{2u}{u^2 + 1}, \quad u = g(x) = 10x^2 + x + 1, \quad x = 0 & \\
 58. f(u) = \left(\frac{u-1}{u+1}\right)^2, \quad u = g(x) = \frac{1}{x^2} - 1, \quad x = -1 &
 \end{array}$$

59. Suppose that functions f and g and their derivatives with respect to x have the following values at $x = 2$ and $x = 3$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	8	2	$1/3$	-3
3	3	-4	2π	5

Find the derivatives with respect to x of the following combinations at the given value of x .

- $$\begin{array}{ll}
 \text{a. } 2f(x), \quad x = 2 & \text{b. } f(x) + g(x), \quad x = 3 \\
 \text{c. } f(x) \cdot g(x), \quad x = 3 & \text{d. } f(x)/g(x), \quad x = 2 \\
 \text{e. } f(g(x)), \quad x = 2 & \text{f. } \sqrt{f(x)}, \quad x = 2 \\
 \text{g. } 1/g^2(x), \quad x = 3 & \text{h. } \sqrt{f^2(x) + g^2(x)}, \quad x = 2
 \end{array}$$
60. Suppose that the functions f and g and their derivatives with respect to x have the following values at $x = 0$ and $x = 1$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	1	1	5	$1/3$
1	3	-4	$-1/3$	$-8/3$

Find the derivatives with respect to x of the following combinations at the given value of x .

- $$\begin{array}{ll}
 \text{a. } 5f(x) - g(x), \quad x = 1 & \text{b. } f(x)g^3(x), \quad x = 0 \\
 \text{c. } \frac{f(x)}{g(x) + 1}, \quad x = 1 & \text{d. } f(g(x)), \quad x = 0 \\
 \text{e. } g(f(x)), \quad x = 0 & \text{f. } (x^{11} + f(x))^{-2}, \quad x = 1 \\
 \text{g. } f(x + g(x)), \quad x = 0 &
 \end{array}$$

61. Find ds/dt when $\theta = 3\pi/2$ if $s = \cos \theta$ and $d\theta/dt = 5$.

62. Find dy/dt when $x = 1$ if $y = x^2 + 7x - 5$ and $dx/dt = 1/3$.

Choices in Composition

What happens if you can write a function as a composite in different ways? Do you get the same derivative each time? The Chain Rule says you should. Try it with the functions in Exercises 63 and 64.

63. Find dy/dx if $y = x$ by using the Chain Rule with y as a composite of

- $$\begin{array}{ll}
 \text{a. } y = (u/5) + 7 \quad \text{and} \quad u = 5x - 35 & \\
 \text{b. } y = 1 + (1/u) \quad \text{and} \quad u = 1/(x - 1). &
 \end{array}$$

64. Find dy/dx if $y = x^{3/2}$ by using the Chain Rule with y as a composite of

- $$\begin{array}{ll}
 \text{a. } y = u^3 \quad \text{and} \quad u = \sqrt{x} & \\
 \text{b. } y = \sqrt{u} \quad \text{and} \quad u = x^3. &
 \end{array}$$

Tangents and Slopes

65. a. Find the tangent to the curve $y = 2 \tan(\pi x/4)$ at $x = 1$.

- b. **Slopes on a tangent curve** What is the smallest value the slope of the curve can ever have on the interval $-2 < x < 2$? Give reasons for your answer.

66. **Slopes on sine curves**

- $$\begin{array}{l}
 \text{a. Find equations for the tangents to the curves } y = \sin 2x \text{ and } y = -\sin(x/2) \text{ at the origin. Is there anything special about how the tangents are related? Give reasons for your answer.} \\
 \text{b. Can anything be said about the tangents to the curves } y = \sin mx \text{ and } y = -\sin(x/m) \text{ at the origin (} m \text{ a constant } \neq 0 \text{)? Give reasons for your answer.} \\
 \text{c. For a given } m, \text{ what are the largest values the slopes of the curves } y = \sin mx \text{ and } y = -\sin(x/m) \text{ can ever have? Give reasons for your answer.} \\
 \text{d. The function } y = \sin x \text{ completes one period on the interval } [0, 2\pi], \text{ the function } y = \sin 2x \text{ completes two periods, the function } y = \sin(x/2) \text{ completes half a period, and so on. Is there any relation between the number of periods } y = \sin mx \text{ completes on } [0, 2\pi] \text{ and the slope of the curve } y = \sin mx \text{ at the origin? Give reasons for your answer.}
 \end{array}$$

Finding Cartesian Equations from Parametric Equations

Exercises 67–78 give parametric equations and parameter intervals for the motion of a particle in the xy -plane. Identify the particle's path by

finding a Cartesian equation for it. Graph the Cartesian equation. (The graphs will vary with the equation used.) Indicate the portion of the graph traced by the particle and the direction of motion.

67. $x = \cos 2t$, $y = \sin 2t$, $0 \leq t \leq \pi$
68. $x = \cos(\pi - t)$, $y = \sin(\pi - t)$, $0 \leq t \leq \pi$
69. $x = 4 \cos t$, $y = 2 \sin t$, $0 \leq t \leq 2\pi$
70. $x = 4 \sin t$, $y = 5 \cos t$, $0 \leq t \leq 2\pi$
71. $x = 3t$, $y = 9t^2$, $-\infty < t < \infty$
72. $x = -\sqrt{t}$, $y = t$, $t \geq 0$
73. $x = 2t - 5$, $y = 4t - 7$, $-\infty < t < \infty$
74. $x = 3 - 3t$, $y = 2t$, $0 \leq t \leq 1$
75. $x = t$, $y = \sqrt{1 - t^2}$, $-1 \leq t \leq 0$
76. $x = \sqrt{t + 1}$, $y = \sqrt{t}$, $t \geq 0$
77. $x = \sec^2 t - 1$, $y = \tan t$, $-\pi/2 < t < \pi/2$
78. $x = -\sec t$, $y = \tan t$, $-\pi/2 < t < \pi/2$

Determining Parametric Equations

79. Find parametric equations and a parameter interval for the motion of a particle that starts at $(a, 0)$ and traces the circle $x^2 + y^2 = a^2$
 - a. once clockwise.
 - b. once counterclockwise.
 - c. twice clockwise.
 - d. twice counterclockwise.

(There are many ways to do these, so your answers may not be the same as the ones in the back of the book.)
80. Find parametric equations and a parameter interval for the motion of a particle that starts at $(a, 0)$ and traces the ellipse $(x^2/a^2) + (y^2/b^2) = 1$
 - a. once clockwise.
 - b. once counterclockwise.
 - c. twice clockwise.
 - d. twice counterclockwise.

(As in Exercise 79, there are many correct answers.)

In Exercises 81–86, find a parametrization for the curve.

81. the line segment with endpoints $(-1, -3)$ and $(4, 1)$
82. the line segment with endpoints $(-1, 3)$ and $(3, -2)$
83. the lower half of the parabola $x - 1 = y^2$
84. the left half of the parabola $y = x^2 + 2x$
85. the ray (half line) with initial point $(2, 3)$ that passes through the point $(-1, -1)$
86. the ray (half line) with initial point $(-1, 2)$ that passes through the point $(0, 0)$

Tangents to Parametrized Curves

In Exercises 87–94, find an equation for the line tangent to the curve at the point defined by the given value of t . Also, find the value of d^2y/dx^2 at this point.

87. $x = 2 \cos t$, $y = 2 \sin t$, $t = \pi/4$
88. $x = \cos t$, $y = \sqrt{3} \cos t$, $t = 2\pi/3$
89. $x = t$, $y = \sqrt{t}$, $t = 1/4$

$$90. x = -\sqrt{t+1}, \quad y = \sqrt{3t}, \quad t = 3$$

$$91. x = 2t^2 + 3, \quad y = t^4, \quad t = -1$$

$$92. x = t - \sin t, \quad y = 1 - \cos t, \quad t = \pi/3$$

$$93. x = \cos t, \quad y = 1 + \sin t, \quad t = \pi/2$$

$$94. x = \sec^2 t - 1, \quad y = \tan t, \quad t = -\pi/4$$

Theory, Examples, and Applications

95. **Running machinery too fast** Suppose that a piston is moving straight up and down and that its position at time t sec is

$$s = A \cos(2\pi bt),$$

with A and b positive. The value of A is the amplitude of the motion, and b is the frequency (number of times the piston moves up and down each second). What effect does doubling the frequency have on the piston's velocity, acceleration, and jerk? (Once you find out, you will know why machinery breaks when you run it too fast.)

96. **Temperatures in Fairbanks, Alaska** The graph in Figure 3.33 shows the average Fahrenheit temperature in Fairbanks, Alaska, during a typical 365-day year. The equation that approximates the temperature on day x is

$$y = 37 \sin \left[\frac{2\pi}{365} (x - 101) \right] + 25.$$

- a. On what day is the temperature increasing the fastest?
- b. About how many degrees per day is the temperature increasing when it is increasing at its fastest?

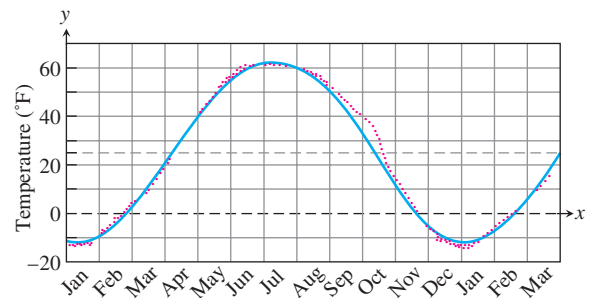


FIGURE 3.33 Normal mean air temperatures at Fairbanks, Alaska, plotted as data points, and the approximating sine function (Exercise 96).

97. **Particle motion** The position of a particle moving along a coordinate line is $s = \sqrt{1 + 4t}$, with s in meters and t in seconds. Find the particle's velocity and acceleration at $t = 6$ sec.
98. **Constant acceleration** Suppose that the velocity of a falling body is $v = k\sqrt{s}$ m/sec (k a constant) at the instant the body has fallen s m from its starting point. Show that the body's acceleration is constant.

- 99. Falling meteorite** The velocity of a heavy meteorite entering Earth's atmosphere is inversely proportional to $\sqrt{s}$ when it is s km from Earth's center. Show that the meteorite's acceleration is inversely proportional to s^2 .
- 100. Particle acceleration** A particle moves along the x -axis with velocity $dx/dt = f(x)$. Show that the particle's acceleration is $f(x)f'(x)$.
- 101. Temperature and the period of a pendulum** For oscillations of small amplitude (short swings), we may safely model the relationship between the period T and the length L of a simple pendulum with the equation

$$T = 2\pi\sqrt{\frac{L}{g}},$$

where g is the constant acceleration of gravity at the pendulum's location. If we measure g in centimeters per second squared, we measure L in centimeters and T in seconds. If the pendulum is made of metal, its length will vary with temperature, either increasing or decreasing at a rate that is roughly proportional to L . In symbols, with u being temperature and k the proportionality constant,

$$\frac{dL}{du} = kL.$$

Assuming this to be the case, show that the rate at which the period changes with respect to temperature is $kT/2$.

- 102. Chain Rule** Suppose that $f(x) = x^2$ and $g(x) = |x|$. Then the composites

$$(f \circ g)(x) = |x|^2 = x^2 \quad \text{and} \quad (g \circ f)(x) = |x^2| = x^2$$

are both differentiable at $x = 0$ even though g itself is not differentiable at $x = 0$. Does this contradict the Chain Rule? Explain.

- 103. Tangents** Suppose that $u = g(x)$ is differentiable at $x = 1$ and that $y = f(u)$ is differentiable at $u = g(1)$. If the graph of $y = f(g(x))$ has a horizontal tangent at $x = 1$, can we conclude anything about the tangent to the graph of g at $x = 1$ or the tangent to the graph of f at $u = g(1)$? Give reasons for your answer.
- 104.** Suppose that $u = g(x)$ is differentiable at $x = -5$, $y = f(u)$ is differentiable at $u = g(-5)$, and $(f \circ g)'(-5)$ is negative. What, if anything, can be said about the values of $g'(-5)$ and $f'(g(-5))$?

- T 105. The derivative of $\sin 2x$** Graph the function $y = 2 \cos 2x$ for $-2 \leq x \leq 3.5$. Then, on the same screen, graph

$$y = \frac{\sin 2(x+h) - \sin 2x}{h}$$

for $h = 1.0, 0.5$, and 0.2 . Experiment with other values of h , including negative values. What do you see happening as $h \rightarrow 0$? Explain this behavior.

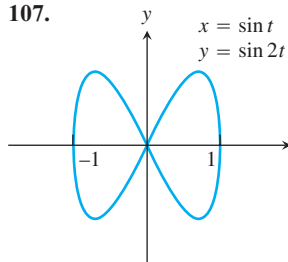
- T 106. The derivative of $\cos(x^2)$** Graph $y = -2x \sin(x^2)$ for $-2 \leq x \leq 3$. Then, on the same screen, graph

$$y = \frac{\cos((x+h)^2) - \cos(x^2)}{h}$$

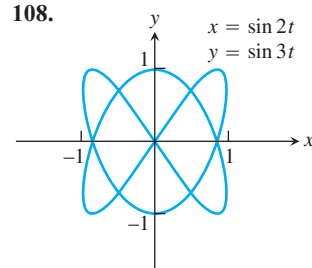
for $h = 1.0, 0.7$, and 0.3 . Experiment with other values of h . What do you see happening as $h \rightarrow 0$? Explain this behavior.

- T** The curves in Exercises 107 and 108 are called *Bowditch curves* or *Lissajous figures*. In each case, find the point in the interior of the first quadrant where the tangent to the curve is horizontal, and find the equations of the two tangents at the origin.

107.



108.



Using the Chain Rule, show that the power rule $(d/dx)x^n = nx^{n-1}$ holds for the functions x^n in Exercises 109 and 110.

109. $x^{1/4} = \sqrt{\sqrt{x}}$

110. $x^{3/4} = \sqrt{x}\sqrt{x}$

COMPUTER EXPLORATIONS

Trigonometric Polynomials

- 111.** As Figure 3.34 shows, the trigonometric “polynomial”

$$s = f(t) = 0.78540 - 0.63662 \cos 2t - 0.07074 \cos 6t - 0.02546 \cos 10t - 0.01299 \cos 14t$$

gives a good approximation of the sawtooth function $s = g(t)$ on the interval $[-\pi, \pi]$. How well does the derivative of f approximate the derivative of g at the points where dg/dt is defined? To find out, carry out the following steps.

- Graph dg/dt (where defined) over $[-\pi, \pi]$.
- Find df/dt .
- Graph df/dt . Where does the approximation of dg/dt by df/dt seem to be best? Least good? Approximations by trigonometric polynomials are important in the theories of heat and oscillation, but we must not expect too much of them, as we see in the next exercise.

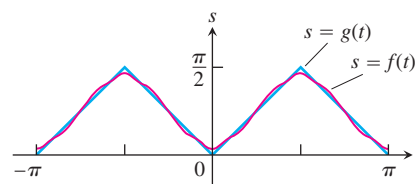


FIGURE 3.34 The approximation of a sawtooth function by a trigonometric “polynomial” (Exercise 111).

- 112.** (Continuation of Exercise 111.) In Exercise 111, the trigonometric polynomial $f(t)$ that approximated the sawtooth function $g(t)$ on $[-\pi, \pi]$ had a derivative that approximated the derivative of the sawtooth function. It is possible, however, for a trigonometric polynomial to approximate a function in a reasonable way without its derivative approximating the function's derivative at all well. As a case in point, the “polynomial”

$$s = h(t) = 1.2732 \sin 2t + 0.4244 \sin 6t + 0.25465 \sin 10t \\ + 0.18189 \sin 14t + 0.14147 \sin 18t$$

graphed in Figure 3.35 approximates the step function $s = k(t)$ shown there. Yet the derivative of h is nothing like the derivative of k .

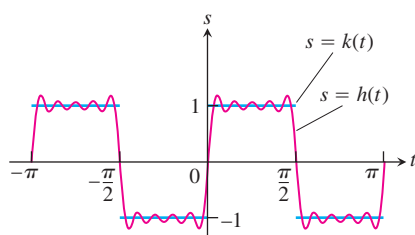


FIGURE 3.35 The approximation of a step function by a trigonometric “polynomial” (Exercise 112).

- Graph dk/dt (where defined) over $[-\pi, \pi]$.
- Find dh/dt .
- Graph dh/dt to see how badly the graph fits the graph of dk/dt . Comment on what you see.

Parametrized Curves

Use a CAS to perform the following steps on the parametrized curves in Exercises 113–116.

- Plot the curve for the given interval of t values.
 - Find dy/dx and d^2y/dx^2 at the point t_0 .
 - Find an equation for the tangent line to the curve at the point defined by the given value t_0 . Plot the curve together with the tangent line on a single graph.
- 113.** $x = \frac{1}{3}t^3$, $y = \frac{1}{2}t^2$, $0 \leq t \leq 1$, $t_0 = 1/2$
- 114.** $x = 2t^3 - 16t^2 + 25t + 5$, $y = t^2 + t - 3$, $0 \leq t \leq 6$, $t_0 = 3/2$
- 115.** $x = t - \cos t$, $y = 1 + \sin t$, $-\pi \leq t \leq \pi$, $t_0 = \pi/4$
- 116.** $x = e^t \cos t$, $y = e^t \sin t$, $0 \leq t \leq \pi$, $t_0 = \pi/2$

3.6

Implicit Differentiation

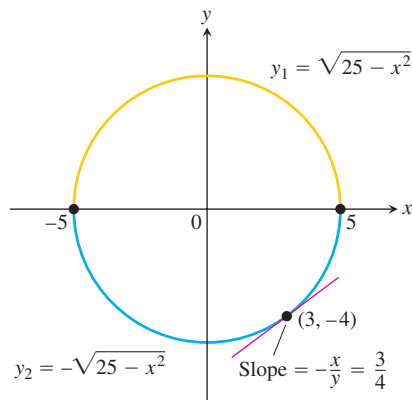


FIGURE 3.36 The circle combines the graphs of two functions. The graph of y_2 is the lower semicircle and passes through $(3, -4)$.

Most of the functions we have dealt with so far have been described by an equation of the form $y = f(x)$ that expresses y explicitly in terms of the variable x . We have learned rules for differentiating functions defined in this way. In Section 3.5 we also learned how to find the derivative dy/dx when a curve is defined parametrically by equations $x = x(t)$ and $y = y(t)$. A third situation occurs when we encounter equations like

$$x^2 + y^2 - 25 = 0, \quad y^2 - x = 0, \quad \text{or} \quad x^3 + y^3 - 9xy = 0.$$

(See Figures 3.36, 3.37, and 3.38.) These equations define an *implicit* relation between the variables x and y . In some cases we may be able to solve such an equation for y as an explicit function (or even several functions) of x . When we cannot put an equation $F(x, y) = 0$ in the form $y = f(x)$ to differentiate it in the usual way, we may still be able to find dy/dx by *implicit differentiation*. This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for y' . This section describes the technique and uses it to extend the Power Rule for differentiation to include rational exponents. In the examples and exercises of this section it is always assumed that the given equation determines y implicitly as a differentiable function of x .

Implicitly Defined Functions

We begin with an example.

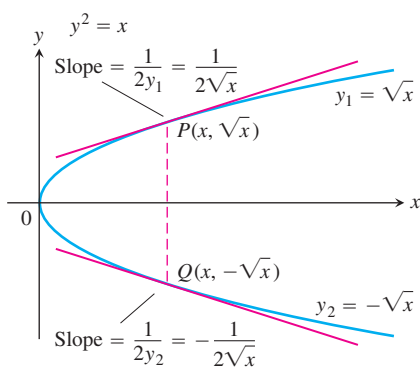


FIGURE 3.37 The equation $y^2 - x = 0$, or $y^2 = x$ as it is usually written, defines two differentiable functions of x on the interval $x \geq 0$. Example 1 shows how to find the derivatives of these functions without solving the equation $y^2 = x$ for y .

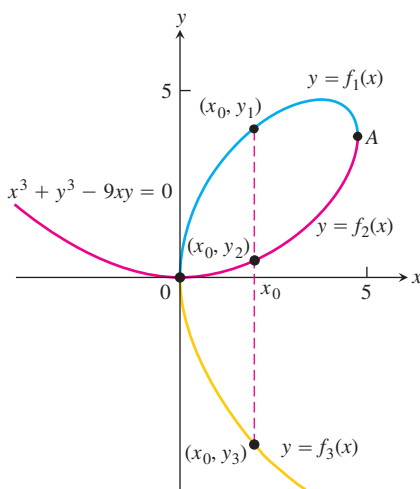


FIGURE 3.38 The curve $x^3 + y^3 - 9xy = 0$ is not the graph of any one function of x . The curve can, however, be divided into separate arcs that are the graphs of functions of x . This particular curve, called a *folium*, dates to Descartes in 1638.

EXAMPLE 1 Differentiating Implicitly

Find dy/dx if $y^2 = x$.

Solution The equation $y^2 = x$ defines two differentiable functions of x that we can actually find, namely $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$ (Figure 3.37). We know how to calculate the derivative of each of these for $x > 0$:

$$\frac{dy_1}{dx} = \frac{1}{2\sqrt{x}} \quad \text{and} \quad \frac{dy_2}{dx} = -\frac{1}{2\sqrt{x}}.$$

But suppose that we knew only that the equation $y^2 = x$ defined y as one or more differentiable functions of x for $x > 0$ without knowing exactly what these functions were. Could we still find dy/dx ?

The answer is yes. To find dy/dx , we simply differentiate both sides of the equation $y^2 = x$ with respect to x , treating $y = f(x)$ as a differentiable function of x :

$$\begin{aligned} y^2 &= x && \text{The Chain Rule gives } \frac{d}{dx}(y^2) = \\ 2y \frac{dy}{dx} &= 1 && \frac{d}{dx}[f(x)]^2 = 2f(x)f'(x) = 2y \frac{dy}{dx}. \\ \frac{dy}{dx} &= \frac{1}{2y}. \end{aligned}$$

This one formula gives the derivatives we calculated for *both* explicit solutions $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$:

$$\frac{dy_1}{dx} = \frac{1}{2y_1} = \frac{1}{2\sqrt{x}} \quad \text{and} \quad \frac{dy_2}{dx} = \frac{1}{2y_2} = \frac{1}{2(-\sqrt{x})} = -\frac{1}{2\sqrt{x}}.$$

EXAMPLE 2 Slope of a Circle at a Point

Find the slope of circle $x^2 + y^2 = 25$ at the point $(3, -4)$.

Solution The circle is not the graph of a single function of x . Rather it is the combined graphs of two differentiable functions, $y_1 = \sqrt{25 - x^2}$ and $y_2 = -\sqrt{25 - x^2}$ (Figure 3.36). The point $(3, -4)$ lies on the graph of y_2 , so we can find the slope by calculating explicitly:

$$\left. \frac{dy_2}{dx} \right|_{x=3} = -\frac{-2x}{2\sqrt{25 - x^2}} \bigg|_{x=3} = -\frac{-6}{2\sqrt{25 - 9}} = \frac{3}{4}.$$

But we can also solve the problem more easily by differentiating the given equation of the circle implicitly with respect to x :

$$\begin{aligned} \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(25) \\ 2x + 2y \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{x}{y}. \end{aligned}$$

$$\text{The slope at } (3, -4) \text{ is } -\frac{x}{y} \bigg|_{(3, -4)} = -\frac{3}{-4} = \frac{3}{4}.$$

Notice that unlike the slope formula for dy_2/dx , which applies only to points below the x -axis, the formula $dy/dx = -x/y$ applies everywhere the circle has a slope. Notice also that the derivative involves *both* variables x and y , not just the independent variable x . ■

To calculate the derivatives of other implicitly defined functions, we proceed as in Examples 1 and 2: We treat y as a differentiable implicit function of x and apply the usual rules to differentiate both sides of the defining equation.

EXAMPLE 3 Differentiating Implicitly

Find dy/dx if $y^2 = x^2 + \sin xy$ (Figure 3.39).

Solution

$$y^2 = x^2 + \sin xy$$

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(x^2) + \frac{d}{dx}(\sin xy)$$

Differentiate both sides with respect to x ...

$$2y \frac{dy}{dx} = 2x + (\cos xy) \frac{d}{dx}(xy)$$

... treating y as a function of x and using the Chain Rule.

$$2y \frac{dy}{dx} = 2x + (\cos xy) \left(y + x \frac{dy}{dx} \right)$$

Treat xy as a product.

$$2y \frac{dy}{dx} - (\cos xy) \left(x \frac{dy}{dx} \right) = 2x + (\cos xy)y$$

Collect terms with dy/dx ...

$$(2y - x \cos xy) \frac{dy}{dx} = 2x + y \cos xy$$

... and factor out dy/dx .

$$\frac{dy}{dx} = \frac{2x + y \cos xy}{2y - x \cos xy}$$

Solve for dy/dx by dividing.

Notice that the formula for dy/dx applies everywhere that the implicitly defined curve has a slope. Notice again that the derivative involves *both* variables x and y , not just the independent variable x . ■

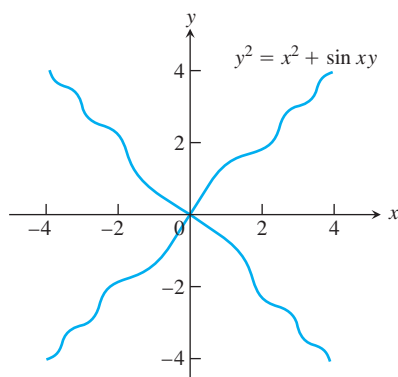


FIGURE 3.39 The graph of $y^2 = x^2 + \sin xy$ in Example 3. The example shows how to find slopes on this implicitly defined curve.

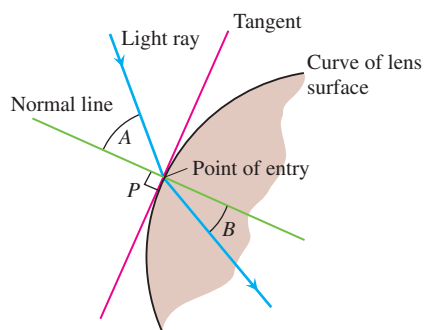


FIGURE 3.40 The profile of a lens, showing the bending (refraction) of a ray of light as it passes through the lens surface.

Implicit Differentiation

1. Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .
2. Collect the terms with dy/dx on one side of the equation.
3. Solve for dy/dx .

Lenses, Tangents, and Normal Lines

In the law that describes how light changes direction as it enters a lens, the important angles are the angles the light makes with the line perpendicular to the surface of the lens at the point of entry (angles A and B in Figure 3.40). This line is called the *normal* to the surface at the point of entry. In a profile view of a lens like the one in Figure 3.40, the **normal** is the line perpendicular to the tangent to the profile curve at the point of entry.

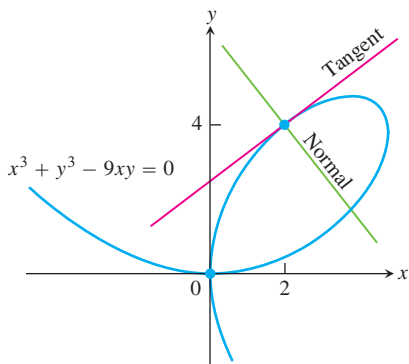


FIGURE 3.41 Example 4 shows how to find equations for the tangent and normal to the folium of Descartes at $(2, 4)$.

EXAMPLE 4 Tangent and Normal to the Folium of Descartes

Show that the point $(2, 4)$ lies on the curve $x^3 + y^3 - 9xy = 0$. Then find the tangent and normal to the curve there (Figure 3.41).

Solution The point $(2, 4)$ lies on the curve because its coordinates satisfy the equation given for the curve: $2^3 + 4^3 - 9(2)(4) = 8 + 64 - 72 = 0$.

To find the slope of the curve at $(2, 4)$, we first use implicit differentiation to find a formula for dy/dx :

$$\begin{aligned}
 x^3 + y^3 - 9xy &= 0 \\
 \frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) - \frac{d}{dx}(9xy) &= \frac{d}{dx}(0) && \text{Differentiate both sides with respect to } x. \\
 3x^2 + 3y^2 \frac{dy}{dx} - 9\left(x \frac{dy}{dx} + y \frac{dx}{dx}\right) &= 0 && \text{Treat } xy \text{ as a product and } y \text{ as a function of } x. \\
 (3y^2 - 9x) \frac{dy}{dx} + 3x^2 - 9y &= 0 \\
 3(y^2 - 3x) \frac{dy}{dx} &= 9y - 3x^2 \\
 \frac{dy}{dx} &= \frac{3y - x^2}{y^2 - 3x}. && \text{Solve for } dy/dx.
 \end{aligned}$$

We then evaluate the derivative at $(x, y) = (2, 4)$:

$$\left. \frac{dy}{dx} \right|_{(2,4)} = \left. \frac{3y - x^2}{y^2 - 3x} \right|_{(2,4)} = \frac{3(4) - 2^2}{4^2 - 3(2)} = \frac{8}{10} = \frac{4}{5}.$$

The tangent at $(2, 4)$ is the line through $(2, 4)$ with slope $4/5$:

$$\begin{aligned}
 y &= 4 + \frac{4}{5}(x - 2) \\
 y &= \frac{4}{5}x + \frac{12}{5}.
 \end{aligned}$$

The normal to the curve at $(2, 4)$ is the line perpendicular to the tangent there, the line through $(2, 4)$ with slope $-5/4$:

$$\begin{aligned}
 y &= 4 - \frac{5}{4}(x - 2) \\
 y &= -\frac{5}{4}x + \frac{13}{2}.
 \end{aligned}$$

The quadratic formula enables us to solve a second-degree equation like $y^2 - 2xy + 3x^2 = 0$ for y in terms of x . There is a formula for the three roots of a cubic equation that is like the quadratic formula but much more complicated. If this formula is used to solve the equation $x^3 + y^3 = 9xy$ for y in terms of x , then three functions determined by the equation are

$$y = f(x) = \sqrt[3]{-\frac{x^3}{2} + \sqrt{\frac{x^6}{4} - 27x^3}} + \sqrt[3]{-\frac{x^3}{2} - \sqrt{\frac{x^6}{4} - 27x^3}}$$

and

$$y = \frac{1}{2} \left[-f(x) \pm \sqrt{-3} \left(\sqrt[3]{-\frac{x^3}{2} + \sqrt{\frac{x^6}{4} - 27x^3}} - \sqrt[3]{-\frac{x^3}{2} - \sqrt{\frac{x^6}{4} - 27x^3}} \right) \right].$$

Using implicit differentiation in Example 4 was much simpler than calculating dy/dx directly from any of the above formulas. Finding slopes on curves defined by higher-degree equations usually requires implicit differentiation.

Derivatives of Higher Order

Implicit differentiation can also be used to find higher derivatives. Here is an example.

EXAMPLE 5 Finding a Second Derivative Implicitly

Find d^2y/dx^2 if $2x^3 - 3y^2 = 8$.

Solution To start, we differentiate both sides of the equation with respect to x in order to find $y' = dy/dx$.

$$\begin{aligned} \frac{d}{dx}(2x^3 - 3y^2) &= \frac{d}{dx}(8) \\ 6x^2 - 6yy' &= 0 && \text{Treat } y \text{ as a function of } x. \\ x^2 - yy' &= 0 \\ y' &= \frac{x^2}{y}, \quad \text{when } y \neq 0 && \text{Solve for } y'. \end{aligned}$$

We now apply the Quotient Rule to find y'' .

$$y'' = \frac{d}{dx} \left(\frac{x^2}{y} \right) = \frac{2xy - x^2y'}{y^2} = \frac{2x}{y} - \frac{x^2}{y^2} \cdot y'$$

Finally, we substitute $y' = x^2/y$ to express y'' in terms of x and y .

$$y'' = \frac{2x}{y} - \frac{x^2}{y^2} \left(\frac{x^2}{y} \right) = \frac{2x}{y} - \frac{x^4}{y^3}, \quad \text{when } y \neq 0$$

Rational Powers of Differentiable Functions

We know that the rule

$$\frac{d}{dx} x^n = nx^{n-1}$$

holds when n is an integer. Using implicit differentiation we can show that it holds when n is any rational number.

THEOREM 4 Power Rule for Rational Powers

If p/q is a rational number, then $x^{p/q}$ is differentiable at every interior point of the domain of $x^{(p/q)-1}$, and

$$\frac{d}{dx} x^{p/q} = \frac{p}{q} x^{(p/q)-1}.$$

EXAMPLE 6 Using the Rational Power Rule

$$(a) \quad \frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}} \quad \text{for } x > 0$$

$$(b) \quad \frac{d}{dx}(x^{2/3}) = \frac{2}{3}x^{-1/3} \quad \text{for } x \neq 0$$

$$(c) \quad \frac{d}{dx}(x^{-4/3}) = -\frac{4}{3}x^{-7/3} \quad \text{for } x \neq 0$$

Proof of Theorem 4 Let p and q be integers with $q > 0$ and suppose that $y = \sqrt[q]{x^p} = x^{p/q}$. Then

$$y^q = x^p.$$

Since p and q are integers (for which we already have the Power Rule), and assuming that y is a differentiable function of x , we can differentiate both sides of the equation with respect to x and get

$$qy^{q-1} \frac{dy}{dx} = px^{p-1}.$$

If $y \neq 0$, we can divide both sides of the equation by qy^{q-1} to solve for dy/dx , obtaining

$$\begin{aligned} \frac{dy}{dx} &= \frac{px^{p-1}}{qy^{q-1}} \\ &= \frac{p}{q} \cdot \frac{x^{p-1}}{(x^{p/q})^{q-1}} && y = x^{p/q} \\ &= \frac{p}{q} \cdot \frac{x^{p-1}}{x^{p-p/q}} && \frac{p}{q}(q-1) = p - \frac{p}{q} \\ &= \frac{p}{q} \cdot x^{(p-1)-(p-p/q)} && \text{A law of exponents} \\ &= \frac{p}{q} \cdot x^{(p/q)-1}, \end{aligned}$$

which proves the rule. ■

We will drop the assumption of differentiability used in the proof of Theorem 4 in Chapter 7, where we prove the Power Rule for any nonzero real exponent. (See Section 7.3.)

By combining the result of Theorem 4 with the Chain Rule, we get an extension of the Power Chain Rule to rational powers of u : If p/q is a rational number and u is a differentiable function of x , then $u^{p/q}$ is a differentiable function of x and

$$\frac{d}{dx} u^{p/q} = \frac{p}{q} u^{(p/q)-1} \frac{du}{dx},$$

provided that $u \neq 0$ if $(p/q) < 1$. This restriction is necessary because 0 might be in the domain of $u^{p/q}$ but not in the domain of $u^{(p/q)-1}$, as we see in the next example.

EXAMPLE 7 Using the Rational Power and Chain Rulesfunction defined on $[-1, 1]$

$$\begin{aligned} \text{(a)} \quad \frac{d}{dx} (1 - x^2)^{1/4} &= \frac{1}{4} (1 - x^2)^{-3/4} (-2x) && \text{Power Chain Rule with } u = 1 - x^2 \\ &= \frac{-x}{2(1 - x^2)^{3/4}} \end{aligned}$$

derivative defined only on $(-1, 1)$

$$\begin{aligned} \text{(b)} \quad \frac{d}{dx} (\cos x)^{-1/5} &= -\frac{1}{5} (\cos x)^{-6/5} \frac{d}{dx} (\cos x) \\ &= -\frac{1}{5} (\cos x)^{-6/5} (-\sin x) \\ &= \frac{1}{5} (\sin x)(\cos x)^{-6/5} \end{aligned}$$

■

EXERCISES 3.6

Derivatives of Rational Powers

Find dy/dx in Exercises 1-10.

1. $y = x^{9/4}$
2. $y = x^{-3/5}$
3. $y = \sqrt[3]{2x}$
4. $y = \sqrt[4]{5x}$
5. $y = 7\sqrt{x+6}$
6. $y = -2\sqrt{x-1}$
7. $y = (2x+5)^{-1/2}$
8. $y = (1-6x)^{2/3}$
9. $y = x(x^2+1)^{1/2}$
10. $y = x(x^2+1)^{-1/2}$

Find the first derivatives of the functions in Exercises 11-18.

11. $s = \sqrt{t^2}$
12. $r = \sqrt[3]{\theta^{-3}}$
13. $y = \sin[(2t+5)^{-2/3}]$
14. $z = \cos[(1-6t)^{2/3}]$
15. $f(x) = \sqrt{1-\sqrt{x}}$
16. $g(x) = 2(2x^{-1/2}+1)^{-1/3}$
17. $h(\theta) = \sqrt[3]{1+\cos(2\theta)}$
18. $k(\theta) = (\sin(\theta+5))^{5/4}$

Differentiating Implicitly

Use implicit differentiation to find dy/dx in Exercises 19-32.

19. $x^2y + xy^2 = 6$
20. $x^3 + y^3 = 18xy$
21. $2xy + y^2 = x + y$
22. $x^3 - xy + y^3 = 1$
23. $x^2(x-y)^2 = x^2 - y^2$
24. $(3xy+7)^2 = 6y$
25. $y^2 = \frac{x-1}{x+1}$
26. $x^2 = \frac{x-y}{x+y}$
27. $x = \tan y$
28. $xy = \cot(xy)$
29. $x + \tan(xy) = 0$
30. $x + \sin y = xy$
31. $y \sin\left(\frac{1}{y}\right) = 1 - xy$
32. $y^2 \cos\left(\frac{1}{y}\right) = 2x + 2y$

Find $dr/d\theta$ in Exercises 33-36.

33. $\theta^{1/2} + r^{1/2} = 1$
34. $r - 2\sqrt{\theta} = \frac{3}{2}\theta^{2/3} + \frac{4}{3}\theta^{3/4}$
35. $\sin(r\theta) = \frac{1}{2}$
36. $\cos r + \cot \theta = r\theta$

Second Derivatives

In Exercises 37-42, use implicit differentiation to find dy/dx and then d^2y/dx^2 .

37. $x^2 + y^2 = 1$
38. $x^{2/3} + y^{2/3} = 1$
39. $y^2 = x^2 + 2x$
40. $y^2 - 2x = 1 - 2y$
41. $2\sqrt{y} = x - y$
42. $xy + y^2 = 1$
43. If $x^3 + y^3 = 16$, find the value of d^2y/dx^2 at the point $(2, 2)$.
44. If $xy + y^2 = 1$, find the value of d^2y/dx^2 at the point $(0, -1)$.

Slopes, Tangents, and Normals

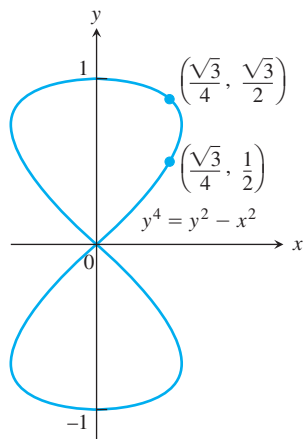
In Exercises 45 and 46, find the slope of the curve at the given points.

45. $y^2 + x^2 = y^4 - 2x$ at $(-2, 1)$ and $(-2, -1)$
46. $(x^2 + y^2)^2 = (x - y)^2$ at $(1, 0)$ and $(1, -1)$

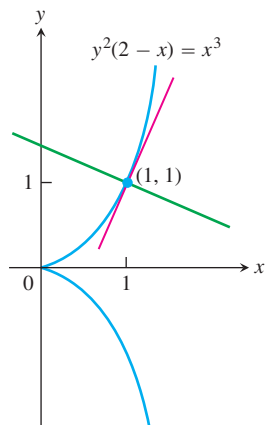
In Exercises 47-56, verify that the given point is on the curve and find the lines that are (a) tangent and (b) normal to the curve at the given point.

47. $x^2 + xy - y^2 = 1$, $(2, 3)$
48. $x^2 + y^2 = 25$, $(3, -4)$
49. $x^2y^2 = 9$, $(-1, 3)$

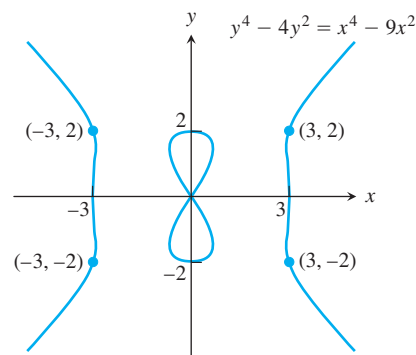
50. $y^2 - 2x - 4y - 1 = 0$, $(-2, 1)$
 51. $6x^2 + 3xy + 2y^2 + 17y - 6 = 0$, $(-1, 0)$
 52. $x^2 - \sqrt{3}xy + 2y^2 = 5$, $(\sqrt{3}, 2)$
 53. $2xy + \pi \sin y = 2\pi$, $(1, \pi/2)$
 54. $x \sin 2y = y \cos 2x$, $(\pi/4, \pi/2)$
 55. $y = 2 \sin(\pi x - y)$, $(1, 0)$
 56. $x^2 \cos^2 y - \sin y = 0$, $(0, \pi)$
 57. **Parallel tangents** Find the two points where the curve $x^2 + xy + y^2 = 7$ crosses the x -axis, and show that the tangents to the curve at these points are parallel. What is the common slope of these tangents?
 58. **Tangents parallel to the coordinate axes** Find points on the curve $x^2 + xy + y^2 = 7$ (a) where the tangent is parallel to the x -axis and (b) where the tangent is parallel to the y -axis. In the latter case, dy/dx is not defined, but dx/dy is. What value does dx/dy have at these points?
 59. **The eight curve** Find the slopes of the curve $y^4 = y^2 - x^2$ at the two points shown here.



60. **The cissoid of Diocles (from about 200 B.C.)** Find equations for the tangent and normal to the cissoid of Diocles $y^2(2-x) = x^3$ at $(1, 1)$.



61. **The devil's curve (Gabriel Cramer [the Cramer of Cramer's rule], 1750)** Find the slopes of the devil's curve $y^4 - 4y^2 = x^4 - 9x^2$ at the four indicated points.



62. **The folium of Descartes** (See Figure 3.38)
 a. Find the slope of the folium of Descartes, $x^3 + y^3 - 9xy = 0$ at the points $(4, 2)$ and $(2, 4)$.
 b. At what point other than the origin does the folium have a horizontal tangent?
 c. Find the coordinates of the point A in Figure 3.38, where the folium has a vertical tangent.

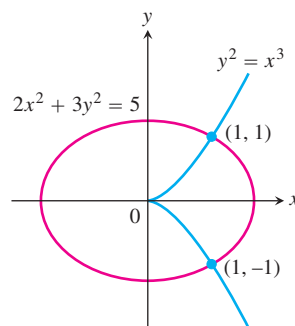
Implicitly Defined Parametrizations

Assuming that the equations in Exercises 63–66 define x and y implicitly as differentiable functions $x = f(t)$, $y = g(t)$, find the slope of the curve $x = f(t)$, $y = g(t)$ at the given value of t .

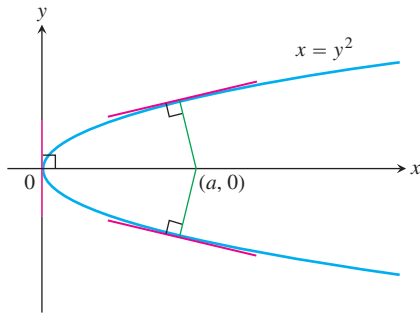
63. $x^2 - 2tx + 2t^2 = 4$, $2y^3 - 3t^2 = 4$, $t = 2$
 64. $x = \sqrt{5 - \sqrt{t}}$, $y(t - 1) = \sqrt{t}$, $t = 4$
 65. $x + 2x^{3/2} = t^2 + t$, $y\sqrt{t + 1} + 2t\sqrt{y} = 4$, $t = 0$
 66. $x \sin t + 2x = t$, $t \sin t - 2t = y$, $t = \pi$

Theory and Examples

67. Which of the following could be true if $f''(x) = x^{-1/3}$?
 a. $f(x) = \frac{3}{2}x^{2/3} - 3$ b. $f(x) = \frac{9}{10}x^{5/3} - 7$
 c. $f'''(x) = -\frac{1}{3}x^{-4/3}$ d. $f'(x) = \frac{3}{2}x^{2/3} + 6$
 68. Is there anything special about the tangents to the curves $y^2 = x^3$ and $2x^2 + 3y^2 = 5$ at the points $(1, \pm 1)$? Give reasons for your answer.



- 69. Intersecting normal** The line that is normal to the curve $x^2 + 2xy - 3y^2 = 0$ at $(1, 1)$ intersects the curve at what other point?
- 70. Normals parallel to a line** Find the normals to the curve $xy + 2x - y = 0$ that are parallel to the line $2x + y = 0$.
- 71. Normals to a parabola** Show that if it is possible to draw three normals from the point $(a, 0)$ to the parabola $x = y^2$ shown here, then a must be greater than $1/2$. One of the normals is the x -axis. For what value of a are the other two normals perpendicular?



- 72.** What is the geometry behind the restrictions on the domains of the derivatives in Example 6(b) and Example 7(a)?

T In Exercises 73 and 74, find both dy/dx (treating y as a differentiable function of x) and dx/dy (treating x as a differentiable function of y). How do dy/dx and dx/dy seem to be related? Explain the relationship geometrically in terms of the graphs.

73. $xy^3 + x^2y = 6$ **74.** $x^3 + y^2 = \sin^2 y$

COMPUTER EXPLORATIONS

- 75. a.** Given that $x^4 + 4y^2 = 1$, find dy/dx two ways: (1) by solving for y and differentiating the resulting functions in the usual way and (2) by implicit differentiation. Do you get the same result each way?
- b.** Solve the equation $x^4 + 4y^2 = 1$ for y and graph the resulting functions together to produce a complete graph of the equation $x^4 + 4y^2 = 1$. Then add the graphs of the first derivatives of these functions to your display. Could you have

predicted the general behavior of the derivative graphs from looking at the graph of $x^4 + 4y^2 = 1$? Could you have predicted the general behavior of the graph of $x^4 + 4y^2 = 1$ by looking at the derivative graphs? Give reasons for your answers.

- 76. a.** Given that $(x - 2)^2 + y^2 = 4$ find dy/dx two ways: (1) by solving for y and differentiating the resulting functions with respect to x and (2) by implicit differentiation. Do you get the same result each way?
- b.** Solve the equation $(x - 2)^2 + y^2 = 4$ for y and graph the resulting functions together to produce a complete graph of the equation $(x - 2)^2 + y^2 = 4$. Then add the graphs of the functions' first derivatives to your picture. Could you have predicted the general behavior of the derivative graphs from looking at the graph of $(x - 2)^2 + y^2 = 4$? Could you have predicted the general behavior of the graph of $(x - 2)^2 + y^2 = 4$ by looking at the derivative graphs? Give reasons for your answers.

Use a CAS to perform the following steps in Exercises 77–84.

- a.** Plot the equation with the implicit plotter of a CAS. Check to see that the given point P satisfies the equation.
- b.** Using implicit differentiation, find a formula for the derivative dy/dx and evaluate it at the given point P .
- c.** Use the slope found in part (b) to find an equation for the tangent line to the curve at P . Then plot the implicit curve and tangent line together on a single graph.
- 77.** $x^3 - xy + y^3 = 7$, $P(2, 1)$
- 78.** $x^5 + y^3x + yx^2 + y^4 = 4$, $P(1, 1)$
- 79.** $y^2 + y = \frac{2+x}{1-x}$, $P(0, 1)$
- 80.** $y^3 + \cos xy = x^2$, $P(1, 0)$
- 81.** $x + \tan\left(\frac{y}{x}\right) = 2$, $P\left(1, \frac{\pi}{4}\right)$
- 82.** $xy^3 + \tan(x + y) = 1$, $P\left(\frac{\pi}{4}, 0\right)$
- 83.** $2y^2 + (xy)^{1/3} = x^2 + 2$, $P(1, 1)$
- 84.** $x\sqrt{1 + 2y} + y = x^2$, $P(1, 0)$

3.7

Related Rates

In this section we look at problems that ask for the rate at which some variable changes. In each case the rate is a derivative that has to be computed from the rate at which some other variable (or perhaps several variables) is known to change. To find it, we write an equation that relates the variables involved and differentiate it to get an equation that relates the rate we seek to the rates we know. The problem of finding a rate you cannot measure easily from some other rates that you can is called a *related rates problem*.

Related Rates Equations

Suppose we are pumping air into a spherical balloon. Both the volume and radius of the balloon are increasing over time. If V is the volume and r is the radius of the balloon at an instant of time, then

$$V = \frac{4}{3} \pi r^3.$$

Using the Chain Rule, we differentiate to find the related rates equation

$$\frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

So if we know the radius r of the balloon and the rate dV/dt at which the volume is increasing at a given instant of time, then we can solve this last equation for dr/dt to find how fast the radius is increasing at that instant. Note that it is easier to measure directly the rate of increase of the volume than it is to measure the increase in the radius. The related rates equation allows us to calculate dr/dt from dV/dt .

Very often the key to relating the variables in a related rates problem is drawing a picture that shows the geometric relations between them, as illustrated in the following example.

EXAMPLE 1 Pumping Out a Tank

How rapidly will the fluid level inside a vertical cylindrical tank drop if we pump the fluid out at the rate of 3000 L/min?

Solution We draw a picture of a partially filled vertical cylindrical tank, calling its radius r and the height of the fluid h (Figure 3.42). Call the volume of the fluid V .

As time passes, the radius remains constant, but V and h change. We think of V and h as differentiable functions of time and use t to represent time. We are told that

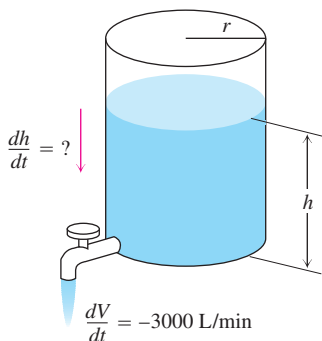


FIGURE 3.42 The rate of change of fluid volume in a cylindrical tank is related to the rate of change of fluid level in the tank (Example 1).

$$\frac{dV}{dt} = -3000.$$

We pump out at the rate of 3000 L/min. The rate is negative because the volume is decreasing.

We are asked to find

$$\frac{dh}{dt}.$$

How fast will the fluid level drop?

To find dh/dt , we first write an equation that relates h to V . The equation depends on the units chosen for V , r , and h . With V in liters and r and h in meters, the appropriate equation for the cylinder's volume is

$$V = 1000 \pi r^2 h$$

because a cubic meter contains 1000 L.

Since V and h are differentiable functions of t , we can differentiate both sides of the equation $V = 1000\pi r^2 h$ with respect to t to get an equation that relates dh/dt to dV/dt :

$$\frac{dV}{dt} = 1000\pi r^2 \frac{dh}{dt} \quad r \text{ is a constant.}$$

We substitute the known value $dV/dt = -3000$ and solve for dh/dt :

$$\frac{dh}{dt} = \frac{-3000}{1000\pi r^2} = -\frac{3}{\pi r^2}.$$

The fluid level will drop at the rate of $3/(\pi r^2)$ m/min.

The equation $dh/dt = -3/\pi r^2$ shows how the rate at which the fluid level drops depends on the tank's radius. If r is small, dh/dt will be large; if r is large, dh/dt will be small.

$$\text{If } r = 1 \text{ m: } \frac{dh}{dt} = -\frac{3}{\pi} \approx -0.95 \text{ m/min} = -95 \text{ cm/min.}$$

$$\text{If } r = 10 \text{ m: } \frac{dh}{dt} = -\frac{3}{100\pi} \approx -0.0095 \text{ m/min} = -0.95 \text{ cm/min.} \quad \blacksquare$$

Related Rates Problem Strategy

1. *Draw a picture and name the variables and constants.* Use t for time. Assume that all variables are differentiable functions of t .
2. *Write down the numerical information* (in terms of the symbols you have chosen).
3. *Write down what you are asked to find* (usually a rate, expressed as a derivative).
4. *Write an equation that relates the variables.* You may have to combine two or more equations to get a single equation that relates the variable whose rate you want to the variables whose rates you know.
5. *Differentiate with respect to t .* Then express the rate you want in terms of the rate and variables whose values you know.
6. *Evaluate.* Use known values to find the unknown rate.

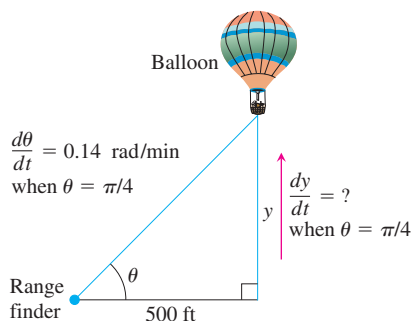


FIGURE 3.43 The rate of change of the balloon's height is related to the rate of change of the angle the range finder makes with the ground (Example 2).

EXAMPLE 2 A Rising Balloon

A hot air balloon rising straight up from a level field is tracked by a range finder 500 ft from the liftoff point. At the moment the range finder's elevation angle is $\pi/4$, the angle is increasing at the rate of 0.14 rad/min. How fast is the balloon rising at that moment?

Solution We answer the question in six steps.

1. *Draw a picture and name the variables and constants* (Figure 3.43). The variables in the picture are
 θ = the angle in radians the range finder makes with the ground.
 y = the height in feet of the balloon.

We let t represent time in minutes and assume that θ and y are differentiable functions of t .

The one constant in the picture is the distance from the range finder to the liftoff point (500 ft). There is no need to give it a special symbol.

2. *Write down the additional numerical information.*

$$\frac{d\theta}{dt} = 0.14 \text{ rad/min} \quad \text{when} \quad \theta = \frac{\pi}{4}$$

3. *Write down what we are to find.* We want dy/dt when $\theta = \pi/4$.

4. Write an equation that relates the variables y and θ .

$$\frac{y}{500} = \tan \theta \quad \text{or} \quad y = 500 \tan \theta$$

5. Differentiate with respect to t using the Chain Rule. The result tells how dy/dt (which we want) is related to $d\theta/dt$ (which we know).

$$\frac{dy}{dt} = 500 (\sec^2 \theta) \frac{d\theta}{dt}$$

6. Evaluate with $\theta = \pi/4$ and $d\theta/dt = 0.14$ to find dy/dt .

$$\frac{dy}{dt} = 500(\sqrt{2})^2(0.14) = 140 \quad \sec \frac{\pi}{4} = \sqrt{2}$$

At the moment in question, the balloon is rising at the rate of 140 ft/min. ■

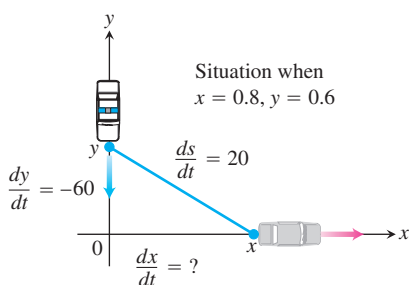


FIGURE 3.44 The speed of the car is related to the speed of the police cruiser and the rate of change of the distance between them (Example 3).

EXAMPLE 3 A Highway Chase

A police cruiser, approaching a right-angled intersection from the north, is chasing a speeding car that has turned the corner and is now moving straight east. When the cruiser is 0.6 mi north of the intersection and the car is 0.8 mi to the east, the police determine with radar that the distance between them and the car is increasing at 20 mph. If the cruiser is moving at 60 mph at the instant of measurement, what is the speed of the car?

Solution We picture the car and cruiser in the coordinate plane, using the positive x -axis as the eastbound highway and the positive y -axis as the southbound highway (Figure 3.44). We let t represent time and set

$$\begin{aligned} x &= \text{position of car at time } t \\ y &= \text{position of cruiser at time } t \\ s &= \text{distance between car and cruiser at time } t. \end{aligned}$$

We assume that x , y , and s are differentiable functions of t .

We want to find dx/dt when

$$x = 0.8 \text{ mi}, \quad y = 0.6 \text{ mi}, \quad \frac{dy}{dt} = -60 \text{ mph}, \quad \frac{ds}{dt} = 20 \text{ mph}.$$

Note that dy/dt is negative because y is decreasing.

We differentiate the distance equation

$$s^2 = x^2 + y^2$$

(we could also use $s = \sqrt{x^2 + y^2}$), and obtain

$$\begin{aligned} 2s \frac{ds}{dt} &= 2x \frac{dx}{dt} + 2y \frac{dy}{dt} \\ \frac{ds}{dt} &= \frac{1}{s} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) \\ &= \frac{1}{\sqrt{x^2 + y^2}} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right). \end{aligned}$$

Finally, use $x = 0.8$, $y = 0.6$, $dy/dt = -60$, $ds/dt = 20$, and solve for dx/dt .

$$20 = \frac{1}{\sqrt{(0.8)^2 + (0.6)^2}} \left(0.8 \frac{dx}{dt} + (0.6)(-60) \right)$$

$$\frac{dx}{dt} = \frac{20\sqrt{(0.8)^2 + (0.6)^2} + (0.6)(60)}{0.8} = 70$$

At the moment in question, the car's speed is 70 mph. ■

EXAMPLE 4 Filling a Conical Tank

Water runs into a conical tank at the rate of $9 \text{ ft}^3/\text{min}$. The tank stands point down and has a height of 10 ft and a base radius of 5 ft. How fast is the water level rising when the water is 6 ft deep?

Solution Figure 3.45 shows a partially filled conical tank. The variables in the problem are

V = volume (ft^3) of the water in the tank at time t (min)

x = radius (ft) of the surface of the water at time t

y = depth (ft) of water in tank at time t .

We assume that V , x , and y are differentiable functions of t . The constants are the dimensions of the tank. We are asked for dy/dt when

$$y = 6 \text{ ft} \quad \text{and} \quad \frac{dV}{dt} = 9 \text{ ft}^3/\text{min}.$$

The water forms a cone with volume

$$V = \frac{1}{3} \pi x^2 y.$$

This equation involves x as well as V and y . Because no information is given about x and dx/dt at the time in question, we need to eliminate x . The similar triangles in Figure 3.45 give us a way to express x in terms of y :

$$\frac{x}{y} = \frac{5}{10} \quad \text{or} \quad x = \frac{y}{2}.$$

Therefore,

$$V = \frac{1}{3} \pi \left(\frac{y}{2} \right)^2 y = \frac{\pi}{12} y^3$$

to give the derivative

$$\frac{dV}{dt} = \frac{\pi}{12} \cdot 3y^2 \frac{dy}{dt} = \frac{\pi}{4} y^2 \frac{dy}{dt}.$$

Finally, use $y = 6$ and $dV/dt = 9$ to solve for dy/dt .

$$9 = \frac{\pi}{4} (6)^2 \frac{dy}{dt}$$

$$\frac{dy}{dt} = \frac{1}{\pi} \approx 0.32$$

At the moment in question, the water level is rising at about 0.32 ft/min. ■

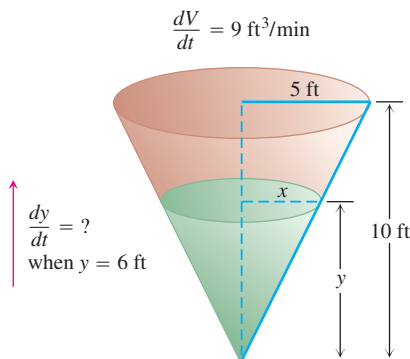
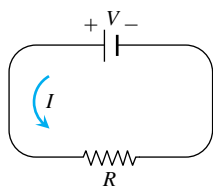


FIGURE 3.45 The geometry of the conical tank and the rate at which water fills the tank determine how fast the water level rises (Example 4).

EXERCISES 3.7

- Area** Suppose that the radius r and area $A = \pi r^2$ of a circle are differentiable functions of t . Write an equation that relates dA/dt to dr/dt .
- Surface area** Suppose that the radius r and surface area $S = 4\pi r^2$ of a sphere are differentiable functions of t . Write an equation that relates dS/dt to dr/dt .
- Volume** The radius r and height h of a right circular cylinder are related to the cylinder's volume V by the formula $V = \pi r^2 h$.
 - How is dV/dt related to dh/dt if r is constant?
 - How is dV/dt related to dr/dt if h is constant?
 - How is dV/dt related to dr/dt and dh/dt if neither r nor h is constant?
- Volume** The radius r and height h of a right circular cone are related to the cone's volume V by the equation $V = (1/3)\pi r^2 h$.
 - How is dV/dt related to dh/dt if r is constant?
 - How is dV/dt related to dr/dt if h is constant?
 - How is dV/dt related to dr/dt and dh/dt if neither r nor h is constant?
- Changing voltage** The voltage V (volts), current I (amperes), and resistance R (ohms) of an electric circuit like the one shown here are related by the equation $V = IR$. Suppose that V is increasing at the rate of 1 volt/sec while I is decreasing at the rate of $1/3$ amp/sec. Let t denote time in seconds.



- What is the value of dV/dt ?
 - What is the value of dI/dt ?
 - What equation relates dR/dt to dV/dt and dI/dt ?
 - Find the rate at which R is changing when $V = 12$ volts and $I = 2$ amp. Is R increasing, or decreasing?
- Electrical power** The power P (watts) of an electric circuit is related to the circuit's resistance R (ohms) and current I (amperes) by the equation $P = RI^2$.
 - How are dP/dt , dR/dt , and dI/dt related if none of P , R , and I are constant?
 - How is dR/dt related to dI/dt if P is constant?
 - Distance** Let x and y be differentiable functions of t and let $s = \sqrt{x^2 + y^2}$ be the distance between the points $(x, 0)$ and $(0, y)$ in the xy -plane.
 - How is ds/dt related to dx/dt if y is constant?

- How is ds/dt related to dx/dt and dy/dt if neither x nor y is constant?
 - How is dx/dt related to dy/dt if s is constant?
- Diagonals** If x , y , and z are lengths of the edges of a rectangular box, the common length of the box's diagonals is $s = \sqrt{x^2 + y^2 + z^2}$.
 - Assuming that x , y , and z are differentiable functions of t , how is ds/dt related to dx/dt , dy/dt , and dz/dt ?
 - How is ds/dt related to dy/dt and dz/dt if x is constant?
 - How are dx/dt , dy/dt , and dz/dt related if s is constant?
 - Area** The area A of a triangle with sides of lengths a and b enclosing an angle of measure θ is

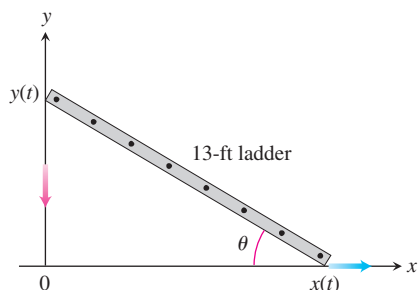
$$A = \frac{1}{2} ab \sin \theta.$$

- How is dA/dt related to $d\theta/dt$ if a and b are constant?
 - How is dA/dt related to $d\theta/dt$ and da/dt if only b is constant?
 - How is dA/dt related to $d\theta/dt$, da/dt , and db/dt if none of a , b , and θ are constant?
- Heating a plate** When a circular plate of metal is heated in an oven, its radius increases at the rate of 0.01 cm/min. At what rate is the plate's area increasing when the radius is 50 cm?
 - Changing dimensions in a rectangle** The length l of a rectangle is decreasing at the rate of 2 cm/sec while the width w is increasing at the rate of 2 cm/sec. When $l = 12$ cm and $w = 5$ cm, find the rates of change of (a) the area, (b) the perimeter, and (c) the lengths of the diagonals of the rectangle. Which of these quantities are decreasing, and which are increasing?
 - Changing dimensions in a rectangular box** Suppose that the edge lengths x , y , and z of a closed rectangular box are changing at the following rates:

$$\frac{dx}{dt} = 1 \text{ m/sec}, \quad \frac{dy}{dt} = -2 \text{ m/sec}, \quad \frac{dz}{dt} = 1 \text{ m/sec}.$$

Find the rates at which the box's (a) volume, (b) surface area, and (c) diagonal length $s = \sqrt{x^2 + y^2 + z^2}$ are changing at the instant when $x = 4$, $y = 3$, and $z = 2$.

- A sliding ladder** A 13-ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec.
 - How fast is the top of the ladder sliding down the wall then?
 - At what rate is the area of the triangle formed by the ladder, wall, and ground changing then?
 - At what rate is the angle θ between the ladder and the ground changing then?



14. Commercial air traffic Two commercial airplanes are flying at 40,000 ft along straight-line courses that intersect at right angles. Plane A is approaching the intersection point at a speed of 442 knots (nautical miles per hour; a nautical mile is 2000 yd). Plane B is approaching the intersection at 481 knots. At what rate is the distance between the planes changing when A is 5 nautical miles from the intersection point and B is 12 nautical miles from the intersection point?

15. Flying a kite A girl flies a kite at a height of 300 ft, the wind carrying the kite horizontally away from her at a rate of 25 ft/sec. How fast must she let out the string when the kite is 500 ft away from her?

16. Boring a cylinder The mechanics at Lincoln Automotive are reboring a 6-in.-deep cylinder to fit a new piston. The machine they are using increases the cylinder's radius one-thousandth of an inch every 3 min. How rapidly is the cylinder volume increasing when the bore (diameter) is 3.800 in.?

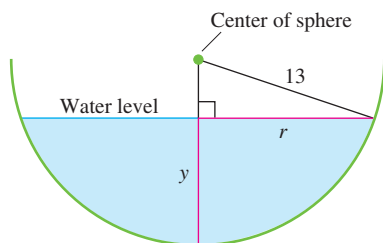
17. A growing sand pile Sand falls from a conveyor belt at the rate of $10 \text{ m}^3/\text{min}$ onto the top of a conical pile. The height of the pile is always three-eighths of the base diameter. How fast are the (a) height and (b) radius changing when the pile is 4 m high? Answer in centimeters per minute.

18. A draining conical reservoir Water is flowing at the rate of $50 \text{ m}^3/\text{min}$ from a shallow concrete conical reservoir (vertex down) of base radius 45 m and height 6 m.

a. How fast (centimeters per minute) is the water level falling when the water is 5 m deep?

b. How fast is the radius of the water's surface changing then? Answer in centimeters per minute.

19. A draining hemispherical reservoir Water is flowing at the rate of $6 \text{ m}^3/\text{min}$ from a reservoir shaped like a hemispherical bowl of radius 13 m, shown here in profile. Answer the following questions, given that the volume of water in a hemispherical bowl of radius R is $V = (\pi/3)y^2(3R - y)$ when the water is y meters deep.



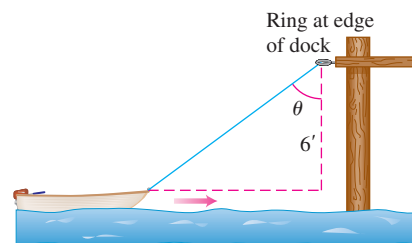
- At what rate is the water level changing when the water is 8 m deep?
- What is the radius r of the water's surface when the water is y m deep?
- At what rate is the radius r changing when the water is 8 m deep?

20. A growing raindrop Suppose that a drop of mist is a perfect sphere and that, through condensation, the drop picks up moisture at a rate proportional to its surface area. Show that under these circumstances the drop's radius increases at a constant rate.

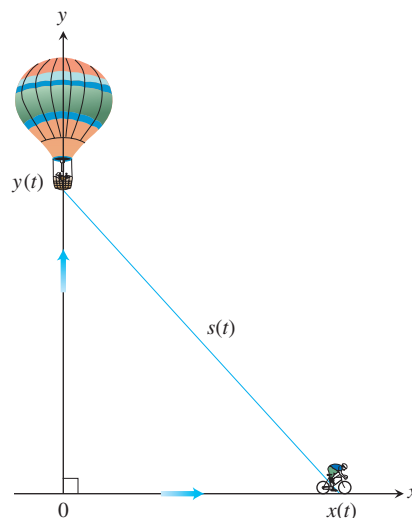
21. The radius of an inflating balloon A spherical balloon is inflated with helium at the rate of $100\pi \text{ ft}^3/\text{min}$. How fast is the balloon's radius increasing at the instant the radius is 5 ft? How fast is the surface area increasing?

22. Hauling in a dinghy A dinghy is pulled toward a dock by a rope from the bow through a ring on the dock 6 ft above the bow. The rope is hauled in at the rate of 2 ft/sec.

- How fast is the boat approaching the dock when 10 ft of rope are out?
- At what rate is the angle θ changing then (see the figure)?

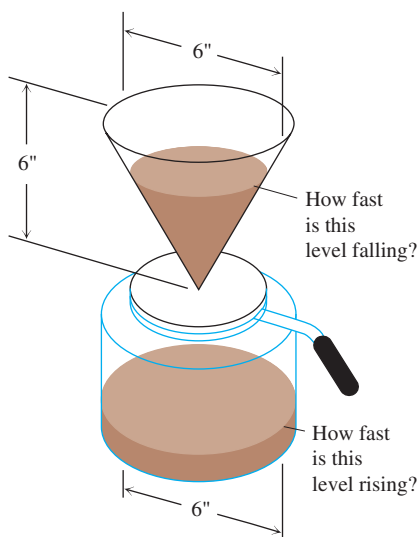


23. A balloon and a bicycle A balloon is rising vertically above a level, straight road at a constant rate of 1 ft/sec. Just when the balloon is 65 ft above the ground, a bicycle moving at a constant rate of 17 ft/sec passes under it. How fast is the distance $s(t)$ between the bicycle and balloon increasing 3 sec later?



- 24. Making coffee** Coffee is draining from a conical filter into a cylindrical coffeepot at the rate of $10 \text{ in}^3/\text{min}$.

- How fast is the level in the pot rising when the coffee in the cone is 5 in. deep?
- How fast is the level in the cone falling then?



- 25. Cardiac output** In the late 1860s, Adolf Fick, a professor of physiology in the Faculty of Medicine in Würzburg, Germany, developed one of the methods we use today for measuring how much blood your heart pumps in a minute. Your cardiac output as you read this sentence is probably about 7 L/min. At rest it is likely to be a bit under 6 L/min. If you are a trained marathon runner running a marathon, your cardiac output can be as high as 30 L/min.

Your cardiac output can be calculated with the formula

$$y = \frac{Q}{D},$$

where Q is the number of milliliters of CO_2 you exhale in a minute and D is the difference between the CO_2 concentration (ml/L) in the blood pumped to the lungs and the CO_2 concentration in the blood returning from the lungs. With $Q = 233 \text{ ml/min}$ and $D = 97 - 56 = 41 \text{ ml/L}$,

$$y = \frac{233 \text{ ml/min}}{41 \text{ ml/L}} \approx 5.68 \text{ L/min},$$

fairly close to the 6 L/min that most people have at basal (resting) conditions. (Data courtesy of J. Kenneth Herd, M.D., Quillan College of Medicine, East Tennessee State University.)

Suppose that when $Q = 233$ and $D = 41$, we also know that D is decreasing at the rate of 2 units a minute but that Q remains unchanged. What is happening to the cardiac output?

- 26. Cost, revenue, and profit** A company can manufacture x items at a cost of $c(x)$ thousand dollars, a sales revenue of $r(x)$ thousand dollars, and a profit of $p(x) = r(x) - c(x)$ thousand dollars. Find dc/dt , dr/dt , and dp/dt for the following values of x and dx/dt .

a. $r(x) = 9x$, $c(x) = x^3 - 6x^2 + 15x$, and $dx/dt = 0.1$ when $x = 2$

b. $r(x) = 70x$, $c(x) = x^3 - 6x^2 + 45/x$, and $dx/dt = 0.05$ when $x = 1.5$

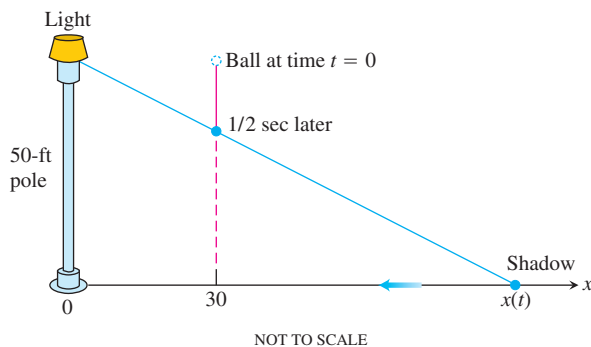
- 27. Moving along a parabola** A particle moves along the parabola $y = x^2$ in the first quadrant in such a way that its x -coordinate (measured in meters) increases at a steady 10 m/sec. How fast is the angle of inclination θ of the line joining the particle to the origin changing when $x = 3 \text{ m}$?

- 28. Moving along another parabola** A particle moves from right to left along the parabolic curve $y = \sqrt{-x}$ in such a way that its x -coordinate (measured in meters) decreases at the rate of 8 m/sec. How fast is the angle of inclination θ of the line joining the particle to the origin changing when $x = -4$?

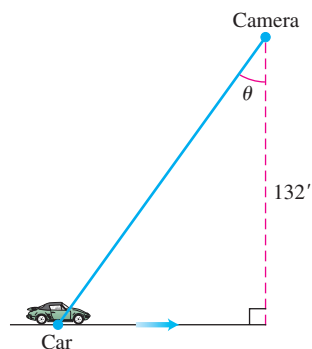
- 29. Motion in the plane** The coordinates of a particle in the metric xy -plane are differentiable functions of time t with $dx/dt = -1 \text{ m/sec}$ and $dy/dt = -5 \text{ m/sec}$. How fast is the particle's distance from the origin changing as it passes through the point $(5, 12)$?

- 30. A moving shadow** A man 6 ft tall walks at the rate of 5 ft/sec toward a streetlight that is 16 ft above the ground. At what rate is the tip of his shadow moving? At what rate is the length of his shadow changing when he is 10 ft from the base of the light?

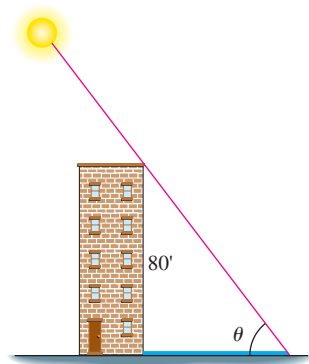
- 31. Another moving shadow** A light shines from the top of a pole 50 ft high. A ball is dropped from the same height from a point 30 ft away from the light. (See accompanying figure.) How fast is the shadow of the ball moving along the ground $1/2 \text{ sec}$ later? (Assume the ball falls a distance $s = 16t^2 \text{ ft}$ in $t \text{ sec}$.)



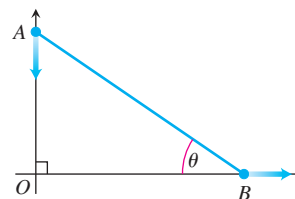
- 32. Videotaping a moving car** You are videotaping a race from a stand 132 ft from the track, following a car that is moving at 180 mi/h (264 ft/sec). How fast will your camera angle θ be changing when the car is right in front of you? A half second later?



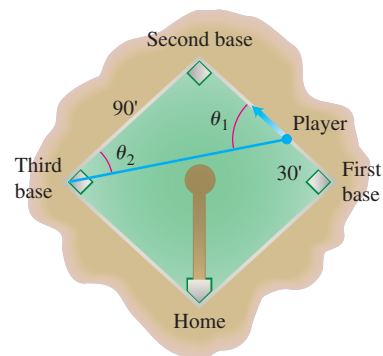
- 33. A melting ice layer** A spherical iron ball 8 in. in diameter is coated with a layer of ice of uniform thickness. If the ice melts at the rate of $10 \text{ in}^3/\text{min}$, how fast is the thickness of the ice decreasing when it is 2 in. thick? How fast is the outer surface area of ice decreasing?
- 34. Highway patrol** A highway patrol plane flies 3 mi above a level, straight road at a steady 120 mi/h. The pilot sees an oncoming car and with radar determines that at the instant the line-of-sight distance from plane to car is 5 mi, the line-of-sight distance is decreasing at the rate of 160 mi/h. Find the car's speed along the highway.
- 35. A building's shadow** On a morning of a day when the sun will pass directly overhead, the shadow of an 80-ft building on level ground is 60 ft long. At the moment in question, the angle θ the sun makes with the ground is increasing at the rate of $0.27^\circ/\text{min}$. At what rate is the shadow decreasing? (Remember to use radians. Express your answer in inches per minute, to the nearest tenth.)



- 36. Walkers** A and B are walking on straight streets that meet at right angles. A approaches the intersection at 2 m/sec; B moves away from the intersection 1 m/sec. At what rate is the angle θ changing when A is 10 m from the intersection and B is 20 m from the intersection? Express your answer in degrees per second to the nearest degree.



- 37. Baseball players** A baseball diamond is a square 90 ft on a side. A player runs from first base to second at a rate of 16 ft/sec.
- At what rate is the player's distance from third base changing when the player is 30 ft from first base?
 - At what rates are angles θ_1 and θ_2 (see the figure) changing at that time?
 - The player slides into second base at the rate of 15 ft/sec. At what rates are angles θ_1 and θ_2 changing as the player touches base?



- 38. Ships** Two ships are steaming straight away from a point O along routes that make a 120° angle. Ship A moves at 14 knots (nautical miles per hour; a nautical mile is 2000 yd). Ship B moves at 21 knots. How fast are the ships moving apart when $OA = 5$ and $OB = 3$ nautical miles?

3.8

Linearization and Differentials

Sometimes we can approximate complicated functions with simpler ones that give the accuracy we want for specific applications and are easier to work with. The approximating functions discussed in this section are called *linearizations*, and they are based on tangent lines. Other approximating functions, such as polynomials, are discussed in Chapter 11.

We introduce new variables dx and dy , called *differentials*, and define them in a way that makes Leibniz’s notation for the derivative dy/dx a true ratio. We use dy to estimate error in measurement and sensitivity of a function to change. Application of these ideas then provides for a precise proof of the Chain Rule (Section 3.5).

Linearization

As you can see in Figure 3.46, the tangent to the curve $y = x^2$ lies close to the curve near the point of tangency. For a brief interval to either side, the y -values along the tangent line give good approximations to the y -values on the curve. We observe this phenomenon by zooming in on the two graphs at the point of tangency or by looking at tables of values for the difference between $f(x)$ and its tangent line near the x -coordinate of the point of tangency. Locally, every differentiable curve behaves like a straight line.

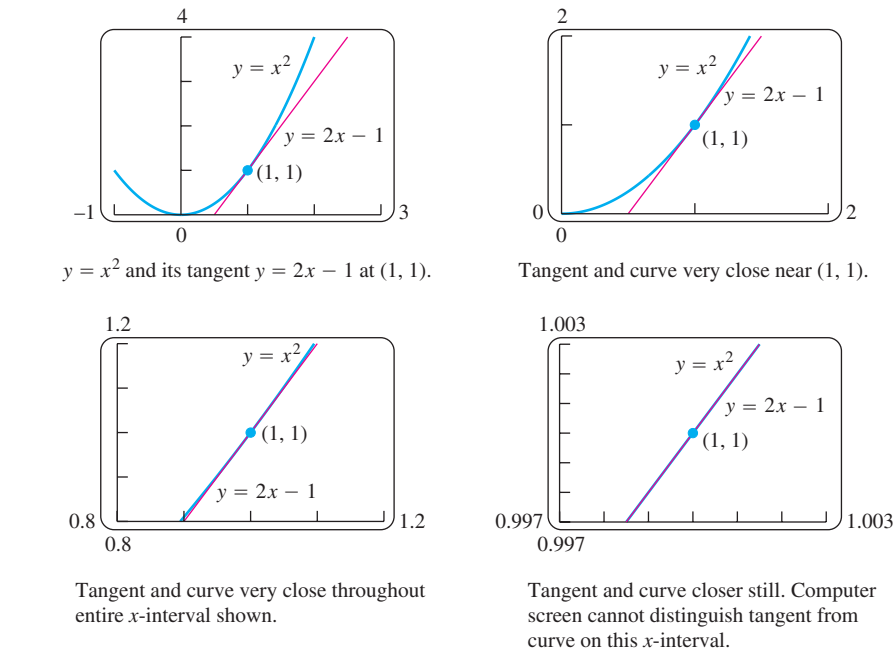


FIGURE 3.46 The more we magnify the graph of a function near a point where the function is differentiable, the flatter the graph becomes and the more it resembles its tangent.

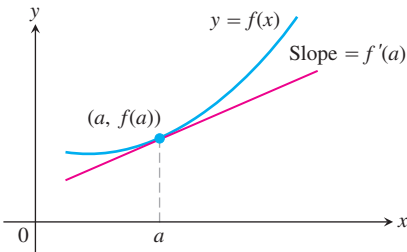


FIGURE 3.47 The tangent to the curve $y = f(x)$ at $x = a$ is the line $L(x) = f(a) + f'(a)(x - a)$.

In general, the tangent to $y = f(x)$ at a point $x = a$, where f is differentiable (Figure 3.47), passes through the point $(a, f(a))$, so its point-slope equation is

$$y = f(a) + f'(a)(x - a).$$

Thus, this tangent line is the graph of the linear function

$$L(x) = f(a) + f'(a)(x - a).$$

For as long as this line remains close to the graph of f , $L(x)$ gives a good approximation to $f(x)$.

DEFINITIONS Linearization, Standard Linear Approximation

If f is differentiable at $x = a$, then the approximating function

$$L(x) = f(a) + f'(a)(x - a)$$

is the **linearization** of f at a . The approximation

$$f(x) \approx L(x)$$

of f by L is the **standard linear approximation** of f at a . The point $x = a$ is the **center** of the approximation.

EXAMPLE 1 Finding a Linearization

Find the linearization of $f(x) = \sqrt{1+x}$ at $x = 0$ (Figure 3.48).

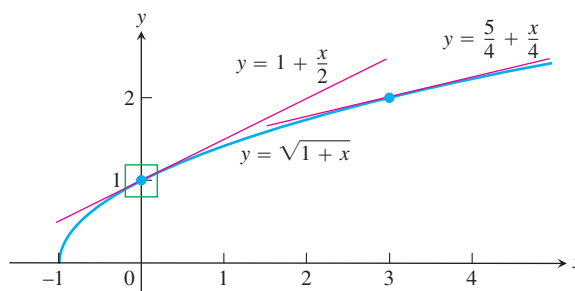


FIGURE 3.48 The graph of $y = \sqrt{1+x}$ and its linearizations at $x = 0$ and $x = 3$. Figure 3.49 shows a magnified view of the small window about 1 on the y -axis.

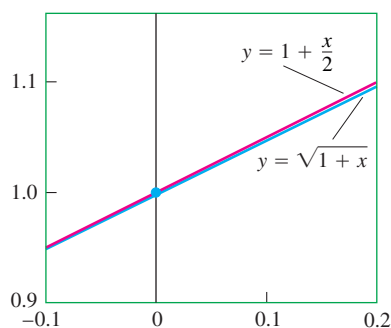


FIGURE 3.49 Magnified view of the window in Figure 3.48.

Solution Since

$$f'(x) = \frac{1}{2}(1+x)^{-1/2},$$

we have $f(0) = 1$ and $f'(0) = 1/2$, giving the linearization

$$L(x) = f(a) + f'(a)(x - a) = 1 + \frac{1}{2}(x - 0) = 1 + \frac{x}{2}.$$

See Figure 3.49. ■

Look at how accurate the approximation $\sqrt{1+x} \approx 1 + (x/2)$ from Example 1 is for values of x near 0.

As we move away from zero, we lose accuracy. For example, for $x = 2$, the linearization gives 2 as the approximation for $\sqrt{3}$, which is not even accurate to one decimal place.

Do not be misled by the preceding calculations into thinking that whatever we do with a linearization is better done with a calculator. In practice, we would never use a linearization to find a particular square root. The utility of a linearization is its ability to replace a complicated formula by a simpler one over an entire interval of values. If we have to work with $\sqrt{1+x}$ for x close to 0 and can tolerate the small amount of error involved, we can

Approximation	True value	True value – approximation
$\sqrt{1.2} \approx 1 + \frac{0.2}{2} = 1.10$	1.095445	$< 10^{-2}$
$\sqrt{1.05} \approx 1 + \frac{0.05}{2} = 1.025$	1.024695	$< 10^{-3}$
$\sqrt{1.005} \approx 1 + \frac{0.005}{2} = 1.00250$	1.002497	$< 10^{-5}$

work with $1 + (x/2)$ instead. Of course, we then need to know how much error there is. We have more to say on the estimation of error in Chapter 11.

A linear approximation normally loses accuracy away from its center. As Figure 3.48 suggests, the approximation $\sqrt{1+x} \approx 1 + (x/2)$ will probably be too crude to be useful near $x = 3$. There, we need the linearization at $x = 3$.

EXAMPLE 2 Finding a Linearization at Another Point

Find the linearization of $f(x) = \sqrt{1+x}$ at $x = 3$.

Solution We evaluate the equation defining $L(x)$ at $a = 3$. With

$$f(3) = 2, \quad f'(3) = \frac{1}{2}(1+x)^{-1/2} \Big|_{x=3} = \frac{1}{4},$$

we have

$$L(x) = 2 + \frac{1}{4}(x - 3) = \frac{5}{4} + \frac{x}{4}.$$

At $x = 3.2$, the linearization in Example 2 gives

$$\sqrt{1+x} = \sqrt{1+3.2} \approx \frac{5}{4} + \frac{3.2}{4} = 1.250 + 0.800 = 2.050,$$

which differs from the true value $\sqrt{4.2} \approx 2.04939$ by less than one one-thousandth. The linearization in Example 1 gives

$$\sqrt{1+x} = \sqrt{1+3.2} \approx 1 + \frac{3.2}{2} = 1 + 1.6 = 2.6,$$

a result that is off by more than 25%.

EXAMPLE 3 Finding a Linearization for the Cosine Function

Find the linearization of $f(x) = \cos x$ at $x = \pi/2$ (Figure 3.50).

Solution Since $f(\pi/2) = \cos(\pi/2) = 0$, $f'(x) = -\sin x$, and $f'(\pi/2) = -\sin(\pi/2) = -1$, we have

$$\begin{aligned} L(x) &= f(a) + f'(a)(x - a) \\ &= 0 + (-1)\left(x - \frac{\pi}{2}\right) \\ &= -x + \frac{\pi}{2}. \end{aligned}$$

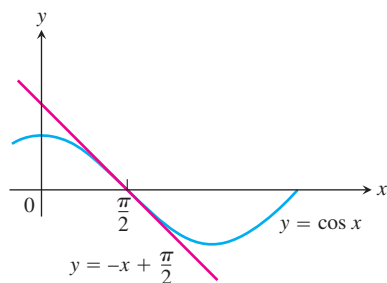


FIGURE 3.50 The graph of $f(x) = \cos x$ and its linearization at $x = \pi/2$. Near $x = \pi/2$, $\cos x \approx -x + (\pi/2)$ (Example 3).

An important linear approximation for roots and powers is

$$(1 + x)^k \approx 1 + kx \quad (x \text{ near } 0; \text{ any number } k)$$

(Exercise 15). This approximation, good for values of x sufficiently close to zero, has broad application. For example, when x is small,

$$\sqrt{1 + x} \approx 1 + \frac{1}{2}x \quad k = 1/2$$

$$\frac{1}{1 - x} = (1 - x)^{-1} \approx 1 + (-1)(-x) = 1 + x \quad k = -1; \text{ replace } x \text{ by } -x.$$

$$\sqrt[3]{1 + 5x^4} = (1 + 5x^4)^{1/3} \approx 1 + \frac{1}{3}(5x^4) = 1 + \frac{5}{3}x^4 \quad k = 1/3; \text{ replace } x \text{ by } 5x^4.$$

$$\frac{1}{\sqrt{1 - x^2}} = (1 - x^2)^{-1/2} \approx 1 + \left(-\frac{1}{2}\right)(-x^2) = 1 + \frac{1}{2}x^2 \quad k = -1/2; \text{ replace } x \text{ by } -x^2.$$

Differentials

We sometimes use the Leibniz notation dy/dx to represent the derivative of y with respect to x . Contrary to its appearance, it is not a ratio. We now introduce two new variables dx and dy with the property that if their ratio exists, it will be equal to the derivative.

DEFINITION Differential

Let $y = f(x)$ be a differentiable function. The **differential dx** is an independent variable. The **differential dy** is

$$dy = f'(x) dx.$$

Unlike the independent variable dx , the variable dy is always a dependent variable. It depends on both x and dx . If dx is given a specific value and x is a particular number in the domain of the function f , then the numerical value of dy is determined.

EXAMPLE 4 Finding the Differential dy

- (a) Find dy if $y = x^5 + 37x$.
 (b) Find the value of dy when $x = 1$ and $dx = 0.2$.

Solution

- (a) $dy = (5x^4 + 37) dx$
 (b) Substituting $x = 1$ and $dx = 0.2$ in the expression for dy , we have

$$dy = (5 \cdot 1^4 + 37)0.2 = 8.4. \quad \blacksquare$$

The geometric meaning of differentials is shown in Figure 3.51. Let $x = a$ and set $dx = \Delta x$. The corresponding change in $y = f(x)$ is

$$\Delta y = f(a + dx) - f(a).$$

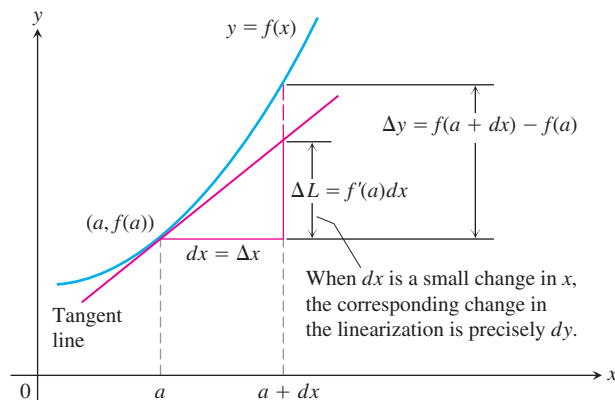


FIGURE 3.51 Geometrically, the differential dy is the change ΔL in the linearization of f when $x = a$ changes by an amount $dx = \Delta x$.

The corresponding change in the tangent line L is

$$\begin{aligned}\Delta L &= L(a + dx) - L(a) \\ &= \underbrace{f(a) + f'(a)[(a + dx) - a]}_{L(a + dx)} - \underbrace{f(a)}_{L(a)} \\ &= f'(a) dx.\end{aligned}$$

That is, the change in the linearization of f is precisely the value of the differential dy when $x = a$ and $dx = \Delta x$. Therefore, dy represents the amount the tangent line rises or falls when x changes by an amount $dx = \Delta x$.

If $dx \neq 0$, then the quotient of the differential dy by the differential dx is equal to the derivative $f'(x)$ because

$$dy \div dx = \frac{f'(x) dx}{dx} = f'(x) = \frac{dy}{dx}.$$

We sometimes write

$$df = f'(x) dx$$

in place of $dy = f'(x) dx$, calling df the **differential of f** . For instance, if $f(x) = 3x^2 - 6$, then

$$df = d(3x^2 - 6) = 6x dx.$$

Every differentiation formula like

$$\frac{d(u + v)}{dx} = \frac{du}{dx} + \frac{dv}{dx} \quad \text{or} \quad \frac{d(\sin u)}{dx} = \cos u \frac{du}{dx}$$

has a corresponding differential form like

$$d(u + v) = du + dv \quad \text{or} \quad d(\sin u) = \cos u du.$$

EXAMPLE 5 Finding Differentials of Functions

$$(a) \quad d(\tan 2x) = \sec^2(2x) d(2x) = 2 \sec^2 2x \, dx$$

$$(b) \quad d\left(\frac{x}{x+1}\right) = \frac{(x+1) \, dx - x \, d(x+1)}{(x+1)^2} = \frac{x \, dx + dx - x \, dx}{(x+1)^2} = \frac{dx}{(x+1)^2}$$

Estimating with Differentials

Suppose we know the value of a differentiable function $f(x)$ at a point a and want to predict how much this value will change if we move to a nearby point $a + dx$. If dx is small, then we can see from Figure 3.51 that Δy is approximately equal to the differential dy . Since

$$f(a + dx) = f(a) + \Delta y,$$

the differential approximation gives

$$f(a + dx) \approx f(a) + dy$$

where $dx = \Delta x$. Thus the approximation $\Delta y \approx dy$ can be used to calculate $f(a + dx)$ when $f(a)$ is known and dx is small.

EXAMPLE 6 Estimating with Differentials

The radius r of a circle increases from $a = 10$ m to 10.1 m (Figure 3.52). Use dA to estimate the increase in the circle's area A . Estimate the area of the enlarged circle and compare your estimate to the true area.

Solution Since $A = \pi r^2$, the estimated increase is

$$dA = A'(a) \, dr = 2\pi a \, dr = 2\pi(10)(0.1) = 2\pi \, \text{m}^2.$$

Thus,

$$\begin{aligned} A(10 + 0.1) &\approx A(10) + 2\pi \\ &= \pi(10)^2 + 2\pi = 102\pi. \end{aligned}$$

The area of a circle of radius 10.1 m is approximately $102\pi \, \text{m}^2$.

The true area is

$$\begin{aligned} A(10.1) &= \pi(10.1)^2 \\ &= 102.01\pi \, \text{m}^2. \end{aligned}$$

The error in our estimate is $0.01\pi \, \text{m}^2$, which is the difference $\Delta A - dA$.

Error in Differential Approximation

Let $f(x)$ be differentiable at $x = a$ and suppose that $dx = \Delta x$ is an increment of x . We have two ways to describe the change in f as x changes from a to $a + \Delta x$:

$$\text{The true change:} \quad \Delta f = f(a + \Delta x) - f(a)$$

$$\text{The differential estimate:} \quad df = f'(a) \, \Delta x.$$

How well does df approximate Δf ?

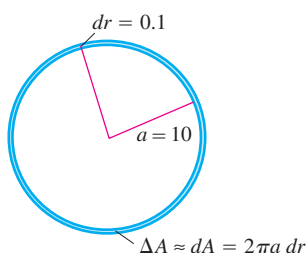


FIGURE 3.52 When dr is small compared with a , as it is when $dr = 0.1$ and $a = 10$, the differential $dA = 2\pi a \, dr$ gives a way to estimate the area of the circle with radius $r = a + dr$ (Example 6).

We measure the approximation error by subtracting df from Δf :

$$\begin{aligned}
 \text{Approximation error} &= \Delta f - df \\
 &= \Delta f - f'(a)\Delta x \\
 &= \underbrace{f(a + \Delta x) - f(a)}_{\Delta f} - f'(a)\Delta x \\
 &= \underbrace{\left(\frac{f(a + \Delta x) - f(a)}{\Delta x} - f'(a) \right)}_{\text{Call this part } \epsilon} \cdot \Delta x \\
 &= \epsilon \cdot \Delta x.
 \end{aligned}$$

As $\Delta x \rightarrow 0$, the difference quotient

$$\frac{f(a + \Delta x) - f(a)}{\Delta x}$$

approaches $f'(a)$ (remember the definition of $f'(a)$), so the quantity in parentheses becomes a very small number (which is why we called it ϵ). In fact, $\epsilon \rightarrow 0$ as $\Delta x \rightarrow 0$. When Δx is small, the approximation error $\epsilon \Delta x$ is smaller still.

$$\underbrace{\Delta f}_{\text{true change}} = \underbrace{f'(a)\Delta x}_{\text{estimated change}} + \underbrace{\epsilon \Delta x}_{\text{error}}$$

Although we do not know exactly how small the error is and will not be able to make much progress on this front until Chapter 11, there is something worth noting here, namely the *form* taken by the equation.

Change in $y = f(x)$ near $x = a$

If $y = f(x)$ is differentiable at $x = a$ and x changes from a to $a + \Delta x$, the change Δy in f is given by an equation of the form

$$\Delta y = f'(a) \Delta x + \epsilon \Delta x \quad (1)$$

in which $\epsilon \rightarrow 0$ as $\Delta x \rightarrow 0$.

In Example 6 we found that

$$\Delta A = \pi(10.1)^2 - \pi(10)^2 = (102.01 - 100)\pi = \underbrace{(2\pi)}_{dA} + \underbrace{(0.01\pi)}_{\text{error}} \text{ m}^2$$

so the approximation error is $\Delta A - dA = \epsilon \Delta r = 0.01\pi$ and $\epsilon = 0.01\pi/\Delta r = 0.01\pi/0.1 = 0.1\pi$ m.

Equation (1) enables us to bring the proof of the Chain Rule to a successful conclusion.

Proof of the Chain Rule

Our goal is to show that if $f(u)$ is a differentiable function of u and $u = g(x)$ is a differentiable function of x , then the composite $y = f(g(x))$ is a differentiable function of x .

More precisely, if g is differentiable at x_0 and f is differentiable at $g(x_0)$, then the composite is differentiable at x_0 and

$$\left. \frac{dy}{dx} \right|_{x=x_0} = f'(g(x_0)) \cdot g'(x_0).$$

Let Δx be an increment in x and let Δu and Δy be the corresponding increments in u and y . Applying Equation (1) we have,

$$\Delta u = g'(x_0)\Delta x + \epsilon_1 \Delta x = (g'(x_0) + \epsilon_1)\Delta x,$$

where $\epsilon_1 \rightarrow 0$ as $\Delta x \rightarrow 0$. Similarly,

$$\Delta y = f'(u_0)\Delta u + \epsilon_2 \Delta u = (f'(u_0) + \epsilon_2)\Delta u,$$

where $\epsilon_2 \rightarrow 0$ as $\Delta u \rightarrow 0$. Notice also that $\Delta u \rightarrow 0$ as $\Delta x \rightarrow 0$. Combining the equations for Δu and Δy gives

$$\Delta y = (f'(u_0) + \epsilon_2)(g'(x_0) + \epsilon_1)\Delta x,$$

so

$$\frac{\Delta y}{\Delta x} = f'(u_0)g'(x_0) + \epsilon_2 g'(x_0) + f'(u_0)\epsilon_1 + \epsilon_2\epsilon_1.$$

Since ϵ_1 and ϵ_2 go to zero as Δx goes to zero, three of the four terms on the right vanish in the limit, leaving

$$\left. \frac{dy}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = f'(u_0)g'(x_0) = f'(g(x_0)) \cdot g'(x_0).$$

This concludes the proof. ■

Sensitivity to Change

The equation $df = f'(x) dx$ tells how *sensitive* the output of f is to a change in input at different values of x . The larger the value of f' at x , the greater the effect of a given change dx . As we move from a to a nearby point $a + dx$, we can describe the change in f in three ways:

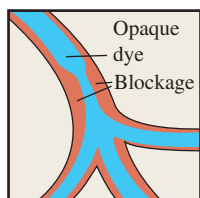
	True	Estimated
Absolute change	$\Delta f = f(a + dx) - f(a)$	$df = f'(a) dx$
Relative change	$\frac{\Delta f}{f(a)}$	$\frac{df}{f(a)}$
Percentage change	$\frac{\Delta f}{f(a)} \times 100$	$\frac{df}{f(a)} \times 100$

EXAMPLE 7 Finding the Depth of a Well

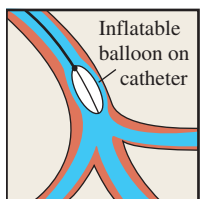
You want to calculate the depth of a well from the equation $s = 16t^2$ by timing how long it takes a heavy stone you drop to splash into the water below. How sensitive will your calculations be to a 0.1-sec error in measuring the time?

Solution The size of ds in the equation

$$ds = 32t dt$$

**Angiography**

An opaque dye is injected into a partially blocked artery to make the inside visible under X-rays. This reveals the location and severity of the blockage.

**Angioplasty**

A balloon-tipped catheter is inflated inside the artery to widen it at the blockage site.

depends on how big t is. If $t = 2$ sec, the change caused by $dt = 0.1$ is about

$$ds = 32(2)(0.1) = 6.4 \text{ ft.}$$

Three seconds later at $t = 5$ sec, the change caused by the same dt is

$$ds = 32(5)(0.1) = 16 \text{ ft.}$$

The estimated depth of the well differs from its true depth by a greater distance the longer the time it takes the stone to splash into the water below, for a given error in measuring the time. ■

EXAMPLE 8 Unclogging Arteries

In the late 1830s, French physiologist Jean Poiseuille (“pwa-ZOY”) discovered the formula we use today to predict how much the radius of a partially clogged artery has to be expanded to restore normal flow. His formula,

$$V = kr^4,$$

says that the volume V of fluid flowing through a small pipe or tube in a unit of time at a fixed pressure is a constant times the fourth power of the tube’s radius r . How will a 10% increase in r affect V ?

Solution The differentials of r and V are related by the equation

$$dV = \frac{dV}{dr} dr = 4kr^3 dr.$$

The relative change in V is

$$\frac{dV}{V} = \frac{4kr^3 dr}{kr^4} = 4 \frac{dr}{r}.$$

The relative change in V is 4 times the relative change in r ; so a 10% increase in r will produce a 40% increase in the flow. ■

EXAMPLE 9 Converting Mass to Energy

Newton’s second law,

$$F = \frac{d}{dt}(mv) = m \frac{dv}{dt} = ma,$$

is stated with the assumption that mass is constant, but we know this is not strictly true because the mass of a body increases with velocity. In Einstein’s corrected formula, mass has the value

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}},$$

where the “rest mass” m_0 represents the mass of a body that is not moving and c is the speed of light, which is about 300,000 km/sec. Use the approximation

$$\frac{1}{\sqrt{1 - x^2}} \approx 1 + \frac{1}{2}x^2 \quad (2)$$

to estimate the increase Δm in mass resulting from the added velocity v .

Solution When v is very small compared with c , v^2/c^2 is close to zero and it is safe to use the approximation

$$\frac{1}{\sqrt{1 - v^2/c^2}} \approx 1 + \frac{1}{2} \left(\frac{v^2}{c^2} \right) \quad \text{Eq. (2) with } x = \frac{v}{c}$$

to obtain

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}} \approx m_0 \left[1 + \frac{1}{2} \left(\frac{v^2}{c^2} \right) \right] = m_0 + \frac{1}{2} m_0 v^2 \left(\frac{1}{c^2} \right),$$

or

$$m \approx m_0 + \frac{1}{2} m_0 v^2 \left(\frac{1}{c^2} \right). \quad (3)$$

Equation (3) expresses the increase in mass that results from the added velocity v .

Energy Interpretation

In Newtonian physics, $(1/2)m_0 v^2$ is the kinetic energy (KE) of the body, and if we rewrite Equation (3) in the form

$$(m - m_0)c^2 \approx \frac{1}{2} m_0 v^2,$$

we see that

$$(m - m_0)c^2 \approx \frac{1}{2} m_0 v^2 = \frac{1}{2} m_0 v^2 - \frac{1}{2} m_0 (0)^2 = \Delta(\text{KE}),$$

or

$$(\Delta m)c^2 \approx \Delta(\text{KE}).$$

So the change in kinetic energy $\Delta(\text{KE})$ in going from velocity 0 to velocity v is approximately equal to $(\Delta m)c^2$, the change in mass times the square of the speed of light. Using $c \approx 3 \times 10^8$ m/sec, we see that a small change in mass can create a large change in energy. ■

EXERCISES 3.8

Finding Linearizations

In Exercises 1–4, find the linearization $L(x)$ of $f(x)$ at $x = a$.

1. $f(x) = x^3 - 2x + 3, \quad a = 2$

2. $f(x) = \sqrt{x^2 + 9}, \quad a = -4$

3. $f(x) = x + \frac{1}{x}, \quad a = 1$

4. $f(x) = \sqrt[3]{x}, \quad a = -8$

Linearization for Approximation

You want linearizations that will replace the functions in Exercises 5–10 over intervals that include the given points x_0 . To make your

subsequent work as simple as possible, you want to center each linearization not at x_0 but at a nearby integer $x = a$ at which the given function and its derivative are easy to evaluate. What linearization do you use in each case?

5. $f(x) = x^2 + 2x, \quad x_0 = 0.1$

6. $f(x) = x^{-1}, \quad x_0 = 0.9$

7. $f(x) = 2x^2 + 4x - 3, \quad x_0 = -0.9$

8. $f(x) = 1 + x, \quad x_0 = 8.1$

9. $f(x) = \sqrt[3]{x}, \quad x_0 = 8.5$

10. $f(x) = \frac{x}{x+1}, \quad x_0 = 1.3$

Linearizing Trigonometric Functions

In Exercises 11–14, find the linearization of f at $x = a$. Then graph the linearization and f together.

11. $f(x) = \sin x$ at (a) $x = 0$, (b) $x = \pi$
12. $f(x) = \cos x$ at (a) $x = 0$, (b) $x = -\pi/2$
13. $f(x) = \sec x$ at (a) $x = 0$, (b) $x = -\pi/3$
14. $f(x) = \tan x$ at (a) $x = 0$, (b) $x = \pi/4$

The Approximation $(1 + x)^k \approx 1 + kx$

15. Show that the linearization of $f(x) = (1 + x)^k$ at $x = 0$ is $L(x) = 1 + kx$.
16. Use the linear approximation $(1 + x)^k \approx 1 + kx$ to find an approximation for the function $f(x)$ for values of x near zero.
 - a. $f(x) = (1 - x)^6$
 - b. $f(x) = \frac{2}{1 - x}$
 - c. $f(x) = \frac{1}{\sqrt{1 + x}}$
 - d. $f(x) = \sqrt{2 + x^2}$
 - e. $f(x) = (4 + 3x)^{1/3}$
 - f. $f(x) = \sqrt[3]{\left(1 - \frac{1}{2 + x}\right)^2}$
17. **Faster than a calculator** Use the approximation $(1 + x)^k \approx 1 + kx$ to estimate the following.
 - a. $(1.0002)^{50}$
 - b. $\sqrt[3]{1.009}$
18. Find the linearization of $f(x) = \sqrt{x + 1} + \sin x$ at $x = 0$. How is it related to the individual linearizations of $\sqrt{x + 1}$ and $\sin x$ at $x = 0$?

Derivatives in Differential Form

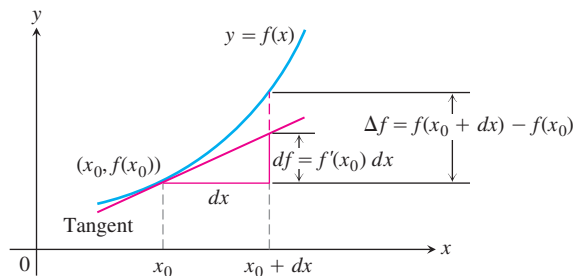
In Exercises 19–30, find dy .

19. $y = x^3 - 3\sqrt{x}$
20. $y = x\sqrt{1 - x^2}$
21. $y = \frac{2x}{1 + x^2}$
22. $y = \frac{2\sqrt{x}}{3(1 + \sqrt{x})}$
23. $2y^{3/2} + xy - x = 0$
24. $xy^2 - 4x^{3/2} - y = 0$
25. $y = \sin(5\sqrt{x})$
26. $y = \cos(x^2)$
27. $y = 4 \tan(x^3/3)$
28. $y = \sec(x^2 - 1)$
29. $y = 3 \csc(1 - 2\sqrt{x})$
30. $y = 2 \cot\left(\frac{1}{\sqrt{x}}\right)$

Approximation Error

In Exercises 31–36, each function $f(x)$ changes value when x changes from x_0 to $x_0 + dx$. Find

- a. the change $\Delta f = f(x_0 + dx) - f(x_0)$;
- b. the value of the estimate $df = f'(x_0) dx$; and
- c. the approximation error $|\Delta f - df|$.



31. $f(x) = x^2 + 2x$, $x_0 = 1$, $dx = 0.1$
32. $f(x) = 2x^2 + 4x - 3$, $x_0 = -1$, $dx = 0.1$
33. $f(x) = x^3 - x$, $x_0 = 1$, $dx = 0.1$
34. $f(x) = x^4$, $x_0 = 1$, $dx = 0.1$
35. $f(x) = x^{-1}$, $x_0 = 0.5$, $dx = 0.1$
36. $f(x) = x^3 - 2x + 3$, $x_0 = 2$, $dx = 0.1$

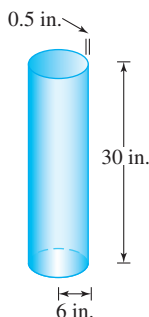
Differential Estimates of Change

In Exercises 37–42, write a differential formula that estimates the given change in volume or surface area.

37. The change in the volume $V = (4/3)\pi r^3$ of a sphere when the radius changes from r_0 to $r_0 + dr$
38. The change in the volume $V = x^3$ of a cube when the edge lengths change from x_0 to $x_0 + dx$
39. The change in the surface area $S = 6x^2$ of a cube when the edge lengths change from x_0 to $x_0 + dx$
40. The change in the lateral surface area $S = \pi r \sqrt{r^2 + h^2}$ of a right circular cone when the radius changes from r_0 to $r_0 + dr$ and the height does not change
41. The change in the volume $V = \pi r^2 h$ of a right circular cylinder when the radius changes from r_0 to $r_0 + dr$ and the height does not change
42. The change in the lateral surface area $S = 2\pi r h$ of a right circular cylinder when the height changes from h_0 to $h_0 + dh$ and the radius does not change

Applications

43. The radius of a circle is increased from 2.00 to 2.02 m.
 - a. Estimate the resulting change in area.
 - b. Express the estimate as a percentage of the circle's original area.
44. The diameter of a tree was 10 in. During the following year, the circumference increased 2 in. About how much did the tree's diameter increase? The tree's cross-section area?
45. **Estimating volume** Estimate the volume of material in a cylindrical shell with height 30 in., radius 6 in., and shell thickness 0.5 in.



- 46. Estimating height of a building** A surveyor, standing 30 ft from the base of a building, measures the angle of elevation to the top of the building to be 75° . How accurately must the angle be measured for the percentage error in estimating the height of the building to be less than 4%?
- 47. Tolerance** The height and radius of a right circular cylinder are equal, so the cylinder's volume is $V = \pi h^3$. The volume is to be calculated with an error of no more than 1% of the true value. Find approximately the greatest error that can be tolerated in the measurement of h , expressed as a percentage of h .
- 48. Tolerance**
- About how accurately must the interior diameter of a 10-m-high cylindrical storage tank be measured to calculate the tank's volume to within 1% of its true value?
 - About how accurately must the tank's exterior diameter be measured to calculate the amount of paint it will take to paint the side of the tank to within 5% of the true amount?
- 49. Minting coins** A manufacturer contracts to mint coins for the federal government. How much variation dr in the radius of the coins can be tolerated if the coins are to weigh within 1/1000 of their ideal weight? Assume that the thickness does not vary.
- 50. Sketching the change in a cube's volume** The volume $V = x^3$ of a cube with edges of length x increases by an amount ΔV when x increases by an amount Δx . Show with a sketch how to represent ΔV geometrically as the sum of the volumes of
- three slabs of dimensions x by x by Δx
 - three bars of dimensions x by Δx by Δx
 - one cube of dimensions Δx by Δx by Δx .
- The differential formula $dV = 3x^2 dx$ estimates the change in V with the three slabs.
- 51. The effect of flight maneuvers on the heart** The amount of work done by the heart's main pumping chamber, the left ventricle, is given by the equation

$$W = PV + \frac{V\delta v^2}{2g},$$

where W is the work per unit time, P is the average blood pressure, V is the volume of blood pumped out during the unit of time,

δ ("delta") is the weight density of the blood, v is the average velocity of the exiting blood, and g is the acceleration of gravity.

When P , V , δ , and v remain constant, W becomes a function of g , and the equation takes the simplified form

$$W = a + \frac{b}{g} \quad (a, b \text{ constant}).$$

As a member of NASA's medical team, you want to know how sensitive W is to apparent changes in g caused by flight maneuvers, and this depends on the initial value of g . As part of your investigation, you decide to compare the effect on W of a given change dg on the moon, where $g = 5.2 \text{ ft/sec}^2$, with the effect the same change dg would have on Earth, where $g = 32 \text{ ft/sec}^2$. Use the simplified equation above to find the ratio of dW_{moon} to dW_{Earth} .

- 52. Measuring acceleration of gravity** When the length L of a clock pendulum is held constant by controlling its temperature, the pendulum's period T depends on the acceleration of gravity g . The period will therefore vary slightly as the clock is moved from place to place on the earth's surface, depending on the change in g . By keeping track of ΔT , we can estimate the variation in g from the equation $T = 2\pi(L/g)^{1/2}$ that relates T , g , and L .
- With L held constant and g as the independent variable, calculate dT and use it to answer parts (b) and (c).
 - If g increases, will T increase or decrease? Will a pendulum clock speed up or slow down? Explain.
 - A clock with a 100-cm pendulum is moved from a location where $g = 980 \text{ cm/sec}^2$ to a new location. This increases the period by $dT = 0.001 \text{ sec}$. Find dg and estimate the value of g at the new location.
- 53.** The edge of a cube is measured as 10 cm with an error of 1%. The cube's volume is to be calculated from this measurement. Estimate the percentage error in the volume calculation.
- 54.** About how accurately should you measure the side of a square to be sure of calculating the area within 2% of its true value?
- 55.** The diameter of a sphere is measured as $100 \pm 1 \text{ cm}$ and the volume is calculated from this measurement. Estimate the percentage error in the volume calculation.
- 56.** Estimate the allowable percentage error in measuring the diameter D of a sphere if the volume is to be calculated correctly to within 3%.
- 57. (Continuation of Example 7.)** Show that a 5% error in measuring t will cause about a 10% error in calculating s from the equation $s = 16t^2$.
- 58. (Continuation of Example 8.)** By what percentage should r be increased to increase V by 50%?

Theory and Examples

- 59.** Show that the approximation of $\sqrt{1+x}$ by its linearization at the origin must improve as $x \rightarrow 0$ by showing that

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x}}{1 + (x/2)} = 1.$$

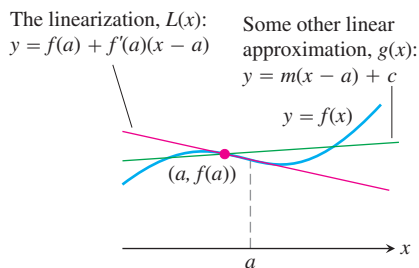
60. Show that the approximation of $\tan x$ by its linearization at the origin must improve as $x \rightarrow 0$ by showing that

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1.$$

61. **The linearization is the best linear approximation** (This is why we use the linearization.) Suppose that $y = f(x)$ is differentiable at $x = a$ and that $g(x) = m(x - a) + c$ is a linear function in which m and c are constants. If the error $E(x) = f(x) - g(x)$ were small enough near $x = a$, we might think of using g as a linear approximation of f instead of the linearization $L(x) = f(a) + f'(a)(x - a)$. Show that if we impose on g the conditions

1. $E(a) = 0$ The approximation error is zero at $x = a$.
2. $\lim_{x \rightarrow a} \frac{E(x)}{x - a} = 0$ The error is negligible when compared with $x - a$.

then $g(x) = f(a) + f'(a)(x - a)$. Thus, the linearization $L(x)$ gives the only linear approximation whose error is both zero at $x = a$ and negligible in comparison with $x - a$.



62. Quadratic approximations

- a. Let $Q(x) = b_0 + b_1(x - a) + b_2(x - a)^2$ be a quadratic approximation to $f(x)$ at $x = a$ with the properties:
 - i. $Q(a) = f(a)$
 - ii. $Q'(a) = f'(a)$
 - iii. $Q''(a) = f''(a)$

Determine the coefficients b_0 , b_1 , and b_2 .

- b. Find the quadratic approximation to $f(x) = 1/(1 - x)$ at $x = 0$.

- T** c. Graph $f(x) = 1/(1 - x)$ and its quadratic approximation at $x = 0$. Then zoom in on the two graphs at the point $(0, 1)$. Comment on what you see.

- T** d. Find the quadratic approximation to $g(x) = 1/x$ at $x = 1$. Graph g and its quadratic approximation together. Comment on what you see.

- T** e. Find the quadratic approximation to $h(x) = \sqrt{1 + x}$ at $x = 0$. Graph h and its quadratic approximation together. Comment on what you see.

- f. What are the linearizations of f , g , and h at the respective points in parts (b), (d), and (e)?

- T** 63. **Reading derivatives from graphs** The idea that differentiable curves flatten out when magnified can be used to estimate the values of the derivatives of functions at particular points. We magnify the curve until the portion we see looks like a straight line through the point in question, and then we use the screen's coordinate grid to read the slope of the curve as the slope of the line it resembles.

- a. To see how the process works, try it first with the function $y = x^2$ at $x = 1$. The slope you read should be 2.
- b. Then try it with the curve $y = e^x$ at $x = 1$, $x = 0$, and $x = -1$. In each case, compare your estimate of the derivative with the value of e^x at the point. What pattern do you see? Test it with other values of x . Chapter 7 will explain what is going on.

64. Suppose that the graph of a differentiable function $f(x)$ has a horizontal tangent at $x = a$. Can anything be said about the linearization of f at $x = a$? Give reasons for your answer.

65. To what relative speed should a body at rest be accelerated to increase its mass by 1%?

T 66. Repeated root-taking

- a. Enter 2 in your calculator and take successive square roots by pressing the square root key repeatedly (or raising the displayed number repeatedly to the 0.5 power). What pattern do you see emerging? Explain what is going on. What happens if you take successive tenth roots instead?
- b. Repeat the procedure with 0.5 in place of 2 as the original entry. What happens now? Can you use any positive number x in place of 2? Explain what is going on.

COMPUTER EXPLORATIONS

Comparing Functions with Their Linearizations

In Exercises 67–70, use a CAS to estimate the magnitude of the error in using the linearization in place of the function over a specified interval I . Perform the following steps:

- a. Plot the function f over I .
- b. Find the linearization L of the function at the point a .
- c. Plot f and L together on a single graph.
- d. Plot the absolute error $|f(x) - L(x)|$ over I and find its maximum value.
- e. From your graph in part (d), estimate as large a $\delta > 0$ as you can, satisfying

$$|x - a| < \delta \quad \Rightarrow \quad |f(x) - L(x)| < \epsilon$$

for $\epsilon = 0.5, 0.1$, and 0.01 . Then check graphically to see if your δ -estimate holds true.

67. $f(x) = x^3 + x^2 - 2x$, $[-1, 2]$, $a = 1$

68. $f(x) = \frac{x-1}{4x^2+1}$, $\left[-\frac{3}{4}, 1\right]$, $a = \frac{1}{2}$

69. $f(x) = x^{2/3}(x-2)$, $[-2, 3]$, $a = 2$

70. $f(x) = \sqrt{x} - \sin x$, $[0, 2\pi]$, $a = 2$

Chapter 3

Additional and Advanced Exercises

1. An equation like $\sin^2 \theta + \cos^2 \theta = 1$ is called an **identity** because it holds for all values of θ . An equation like $\sin \theta = 0.5$ is not an identity because it holds only for selected values of θ , not all. If you differentiate both sides of a trigonometric identity in θ with respect to θ , the resulting new equation will also be an identity.

Differentiate the following to show that the resulting equations hold for all θ .

- a. $\sin 2\theta = 2 \sin \theta \cos \theta$
- b. $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

2. If the identity $\sin(x + a) = \sin x \cos a + \cos x \sin a$ is differentiated with respect to x , is the resulting equation also an identity? Does this principle apply to the equation $x^2 - 2x - 8 = 0$? Explain.

3. a. Find values for the constants a , b , and c that will make

$$f(x) = \cos x \quad \text{and} \quad g(x) = a + bx + cx^2$$

satisfy the conditions

$$f(0) = g(0), \quad f'(0) = g'(0), \quad \text{and} \quad f''(0) = g''(0).$$

- b. Find values for b and c that will make

$$f(x) = \sin(x + a) \quad \text{and} \quad g(x) = b \sin x + c \cos x$$

satisfy the conditions

$$f(0) = g(0) \quad \text{and} \quad f'(0) = g'(0).$$

- c. For the determined values of a , b , and c , what happens for the third and fourth derivatives of f and g in each of parts (a) and (b)?

4. Solutions to differential equations

- a. Show that $y = \sin x$, $y = \cos x$, and $y = a \cos x + b \sin x$ (a and b constants) all satisfy the equation

$$y'' + y = 0.$$

- b. How would you modify the functions in part (a) to satisfy the equation

$$y'' + 4y = 0?$$

Generalize this result.

5. **An osculating circle** Find the values of h , k , and a that make the circle $(x - h)^2 + (y - k)^2 = a^2$ tangent to the parabola $y = x^2 + 1$ at the point $(1, 2)$ and that also make the second derivatives d^2y/dx^2 have the same value on both curves there. Circles like this one that are tangent to a curve and have the same second derivative as the curve at the point of tangency are called *osculating circles* (from the Latin *osculari*, meaning “to kiss”). We encounter them again in Chapter 13.

6. **Marginal revenue** A bus will hold 60 people. The number x of people per trip who use the bus is related to the fare charged (p dollars) by the law $p = [3 - (x/40)]^2$. Write an expression for the total revenue $r(x)$ per trip received by the bus company. What number of people per trip will make the marginal revenue dr/dx equal to zero? What is the corresponding fare? (This fare is the one that maximizes the revenue, so the bus company should probably rethink its fare policy.)

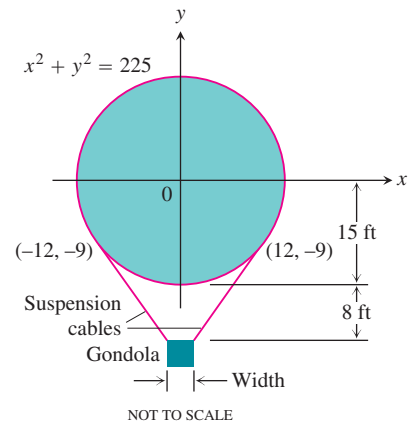
7. Industrial production

- a. Economists often use the expression “rate of growth” in relative rather than absolute terms. For example, let $u = f(t)$ be the number of people in the labor force at time t in a given industry. (We treat this function as though it were differentiable even though it is an integer-valued step function.)

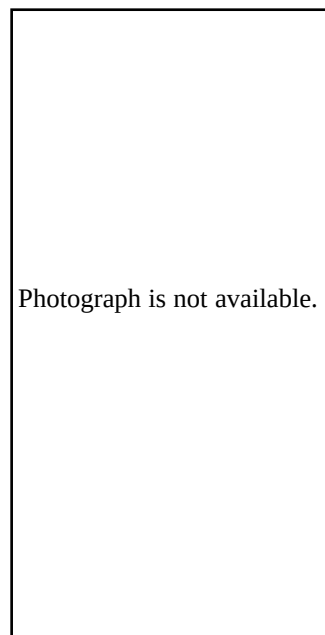
Let $v = g(t)$ be the average production per person in the labor force at time t . The total production is then $y = uv$. If the labor force is growing at the rate of 4% per year ($du/dt = 0.04u$) and the production per worker is growing at the rate of 5% per year ($dv/dt = 0.05v$), find the rate of growth of the total production, y .

- b. Suppose that the labor force in part (a) is decreasing at the rate of 2% per year while the production per person is increasing at the rate of 3% per year. Is the total production increasing, or is it decreasing, and at what rate?

8. **Designing a gondola** The designer of a 30-ft-diameter spherical hot air balloon wants to suspend the gondola 8 ft below the bottom of the balloon with cables tangent to the surface of the balloon, as shown. Two of the cables are shown running from the top edges of the gondola to their points of tangency, $(-12, -9)$ and $(12, -9)$. How wide should the gondola be?



9. **Pisa by parachute** The photograph shows Mike McCarthy parachuting from the top of the Tower of Pisa on August 5, 1988. Make a rough sketch to show the shape of the graph of his speed during the jump.



Mike McCarthy of London jumped from the Tower of Pisa and then opened his parachute in what he said was a world record low-level parachute jump of 179 ft. (Source: *Boston Globe*, Aug. 6, 1988.)

- 10. Motion of a particle** The position at time $t \geq 0$ of a particle moving along a coordinate line is

$$s = 10 \cos(t + \pi/4).$$

- What is the particle's starting position ($t = 0$)?
 - What are the points farthest to the left and right of the origin reached by the particle?
 - Find the particle's velocity and acceleration at the points in part (b).
 - When does the particle first reach the origin? What are its velocity, speed, and acceleration then?
- 11. Shooting a paper clip** On Earth, you can easily shoot a paper clip 64 ft straight up into the air with a rubber band. In t sec after firing, the paper clip is $s = 64t - 16t^2$ ft above your hand.
- How long does it take the paper clip to reach its maximum height? With what velocity does it leave your hand?
 - On the moon, the same acceleration will send the paper clip to a height of $s = 64t - 2.6t^2$ ft in t sec. About how long will it take the paper clip to reach its maximum height, and how high will it go?
- 12. Velocities of two particles** At time t sec, the positions of two particles on a coordinate line are $s_1 = 3t^3 - 12t^2 + 18t + 5$ m and $s_2 = -t^3 + 9t^2 - 12t$ m. When do the particles have the same velocities?
- 13. Velocity of a particle** A particle of constant mass m moves along the x -axis. Its velocity v and position x satisfy the equation

$$\frac{1}{2} m(v^2 - v_0^2) = \frac{1}{2} k(x_0^2 - x^2),$$

where k , v_0 , and x_0 are constants. Show that whenever $v \neq 0$,

$$m \frac{dv}{dt} = -kx.$$

14. Average and instantaneous velocity

- Show that if the position x of a moving point is given by a quadratic function of t , $x = At^2 + Bt + C$, then the average velocity over any time interval $[t_1, t_2]$ is equal to the instantaneous velocity at the midpoint of the time interval.
 - What is the geometric significance of the result in part (a)?
- 15.** Find all values of the constants m and b for which the function

$$y = \begin{cases} \sin x, & x < \pi \\ mx + b, & x \geq \pi \end{cases}$$

is

- continuous at $x = \pi$.
 - differentiable at $x = \pi$.
- 16.** Does the function

$$f(x) = \begin{cases} \frac{1 - \cos x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

have a derivative at $x = 0$? Explain.

- 17. a.** For what values of a and b will

$$f(x) = \begin{cases} ax, & x < 2 \\ ax^2 - bx + 3, & x \geq 2 \end{cases}$$

be differentiable for all values of x ?

- b.** Discuss the geometry of the resulting graph of f .

- 18. a.** For what values of a and b will

$$g(x) = \begin{cases} ax + b, & x \leq -1 \\ ax^3 + x + 2b, & x > -1 \end{cases}$$

be differentiable for all values of x ?

- b.** Discuss the geometry of the resulting graph of g .

- 19. Odd differentiable functions** Is there anything special about the derivative of an odd differentiable function of x ? Give reasons for your answer.

- 20. Even differentiable functions** Is there anything special about the derivative of an even differentiable function of x ? Give reasons for your answer.

- 21.** Suppose that the functions f and g are defined throughout an open interval containing the point x_0 , that f is differentiable at x_0 , that $f(x_0) = 0$, and that g is continuous at x_0 . Show that the product fg is differentiable at x_0 . This process shows, for example, that although $|x|$ is not differentiable at $x = 0$, the product $x|x|$ is differentiable at $x = 0$.

- 22. (Continuation of Exercise 21.)** Use the result of Exercise 21 to show that the following functions are differentiable at $x = 0$.

a. $|x| \sin x$ **b.** $x^{2/3} \sin x$ **c.** $\sqrt[3]{x}(1 - \cos x)$

d. $h(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$

- 23.** Is the derivative of

$$h(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

continuous at $x = 0$? How about the derivative of $k(x) = xh(x)$? Give reasons for your answers.

- 24.** Suppose that a function f satisfies the following conditions for all real values of x and y :

- $f(x + y) = f(x) \cdot f(y)$.
- $f(x) = 1 + xg(x)$, where $\lim_{x \rightarrow 0} g(x) = 1$.

Show that the derivative $f'(x)$ exists at every value of x and that $f'(x) = f(x)$.

- 25. The generalized product rule** Use mathematical induction to prove that if $y = u_1 u_2 \cdots u_n$ is a finite product of differentiable functions, then y is differentiable on their common domain and

$$\frac{dy}{dx} = \frac{du_1}{dx} u_2 \cdots u_n + u_1 \frac{du_2}{dx} \cdots u_n + \cdots + u_1 u_2 \cdots u_{n-1} \frac{du_n}{dx}.$$

- 26. Leibniz's rule for higher-order derivatives of products** Leibniz's rule for higher-order derivatives of products of differentiable functions says that

$$\begin{aligned} \text{a. } \frac{d^2(uv)}{dx^2} &= \frac{d^2u}{dx^2}v + 2\frac{du}{dx}\frac{dv}{dx} + u\frac{d^2v}{dx^2} \\ \text{b. } \frac{d^3(uv)}{dx^3} &= \frac{d^3u}{dx^3}v + 3\frac{d^2u}{dx^2}\frac{dv}{dx} + 3\frac{du}{dx}\frac{d^2v}{dx^2} + u\frac{d^3v}{dx^3} \\ \text{c. } \frac{d^n(uv)}{dx^n} &= \frac{d^nu}{dx^n}v + n\frac{d^{n-1}u}{dx^{n-1}}\frac{dv}{dx} + \cdots \\ &\quad + \frac{n(n-1)\cdots(n-k+1)}{k!}\frac{d^{n-k}u}{dx^{n-k}}\frac{d^kv}{dx^k} \\ &\quad + \cdots + u\frac{d^nv}{dx^n}. \end{aligned}$$

The equations in parts (a) and (b) are special cases of the equation in part (c). Derive the equation in part (c) by mathematical induction, using

$$\binom{m}{k} + \binom{m}{k+1} = \frac{m!}{k!(m-k)!} + \frac{m!}{(k+1)!(m-k-1)!}.$$

- 27. The period of a clock pendulum** The period T of a clock pendulum (time for one full swing and back) is given by the formula $T^2 = 4\pi^2 L/g$, where T is measured in seconds, $g = 32.2$ ft/sec², and L , the length of the pendulum, is measured in feet. Find approximately

- the length of a clock pendulum whose period is $T = 1$ sec.
- the change dT in T if the pendulum in part (a) is lengthened 0.01 ft.
- the amount the clock gains or loses in a day as a result of the period's changing by the amount dT found in part (b).

- 28. The melting ice cube** Assume an ice cube retains its cubical shape as it melts. If we call its edge length s , its volume is $V = s^3$ and its surface area is $6s^2$. We assume that V and s are differentiable functions of time t . We assume also that the cube's volume decreases at a rate that is proportional to its surface area. (This latter assumption seems reasonable enough when we think that the melting takes place at the surface: Changing the amount of surface changes the amount of ice exposed to melt.) In mathematical terms,

$$\frac{dV}{dt} = -k(6s^2), \quad k > 0.$$

The minus sign indicates that the volume is decreasing. We assume that the proportionality factor k is constant. (It probably depends on many things, such as the relative humidity of the surrounding air, the air temperature, and the incidence or absence of sunlight, to name only a few.) Assume a particular set of conditions in which the cube lost 1/4 of its volume during the first hour, and that the volume is V_0 when $t = 0$. How long will it take the ice cube to melt?

Chapter 3

Practice Exercises

Derivatives of Functions

Find the derivatives of the functions in Exercises 1-40.

1. $y = x^5 - 0.125x^2 + 0.25x$ 2. $y = 3 - 0.7x^3 + 0.3x^7$
3. $y = x^3 - 3(x^2 + \pi^2)$ 4. $y = x^7 + \sqrt{7}x - \frac{1}{\pi + 1}$

5. $y = (x + 1)^2(x^2 + 2x)$

6. $y = (2x - 5)(4 - x)^{-1}$

7. $y = (\theta^2 + \sec \theta + 1)^3$

8. $y = \left(-1 - \frac{\csc \theta}{2} - \frac{\theta^2}{4}\right)^2$

9. $s = \frac{\sqrt{t}}{1 + \sqrt{t}}$

10. $s = \frac{1}{\sqrt{t} - 1}$

11. $y = 2 \tan^2 x - \sec^2 x$ 12. $y = \frac{1}{\sin^2 x} - \frac{2}{\sin x}$
13. $s = \cos^4(1 - 2t)$ 14. $s = \cot^3\left(\frac{2}{t}\right)$
15. $s = (\sec t + \tan t)^5$ 16. $s = \csc^5(1 - t + 3t^2)$
17. $r = \sqrt{2\theta \sin \theta}$ 18. $r = 2\theta\sqrt{\cos \theta}$
19. $r = \sin \sqrt{2\theta}$ 20. $r = \sin(\theta + \sqrt{\theta + 1})$
21. $y = \frac{1}{2}x^2 \csc \frac{2}{x}$ 22. $y = 2\sqrt{x} \sin \sqrt{x}$
23. $y = x^{-1/2} \sec(2x)^2$ 24. $y = \sqrt{x} \csc(x + 1)^3$
25. $y = 5 \cot x^2$ 26. $y = x^2 \cot 5x$
27. $y = x^2 \sin^2(2x^2)$ 28. $y = x^{-2} \sin^2(x^3)$
29. $s = \left(\frac{4t}{t+1}\right)^{-2}$ 30. $s = \frac{-1}{15(15t-1)^3}$
31. $y = \left(\frac{\sqrt{x}}{1+x}\right)^2$ 32. $y = \left(\frac{2\sqrt{x}}{2\sqrt{x}+1}\right)^2$
33. $y = \sqrt{\frac{x^2+x}{x^2}}$ 34. $y = 4x\sqrt{x+\sqrt{x}}$
35. $r = \left(\frac{\sin \theta}{\cos \theta - 1}\right)^2$ 36. $r = \left(\frac{1 + \sin \theta}{1 - \cos \theta}\right)^2$
37. $y = (2x + 1)\sqrt{2x + 1}$ 38. $y = 20(3x - 4)^{1/4}(3x - 4)^{-1/5}$
39. $y = \frac{3}{(5x^2 + \sin 2x)^{3/2}}$ 40. $y = (3 + \cos^3 3x)^{-1/3}$

Implicit Differentiation

In Exercises 41–48, find dy/dx .

41. $xy + 2x + 3y = 1$ 42. $x^2 + xy + y^2 - 5x = 2$
43. $x^3 + 4xy - 3y^{4/3} = 2x$ 44. $5x^{4/5} + 10y^{6/5} = 15$
45. $\sqrt{xy} = 1$ 46. $x^2y^2 = 1$
47. $y^2 = \frac{x}{x+1}$ 48. $y^2 = \sqrt{\frac{1+x}{1-x}}$

In Exercises 49 and 50, find dp/dq .

49. $p^3 + 4pq - 3q^2 = 2$ 50. $q = (5p^2 + 2p)^{-3/2}$

In Exercises 51 and 52, find dr/ds .

51. $r \cos 2s + \sin^2 s = \pi$ 52. $2rs - r - s + s^2 = -3$

53. Find d^2y/dx^2 by implicit differentiation:

- a. $x^3 + y^3 = 1$ b. $y^2 = 1 - \frac{2}{x}$
54. a. By differentiating $x^2 - y^2 = 1$ implicitly, show that $dy/dx = x/y$.
b. Then show that $d^2y/dx^2 = -1/y^3$.

Numerical Values of Derivatives

55. Suppose that functions $f(x)$ and $g(x)$ and their first derivatives have the following values at $x = 0$ and $x = 1$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	1	1	-3	1/2
1	3	5	1/2	-4

Find the first derivatives of the following combinations at the given value of x .

- a. $6f(x) - g(x)$, $x = 1$ b. $f(x)g^2(x)$, $x = 0$
- c. $\frac{f(x)}{g(x) + 1}$, $x = 1$ d. $f(g(x))$, $x = 0$
- e. $g(f(x))$, $x = 0$ f. $(x + f(x))^{3/2}$, $x = 1$
- g. $f(x + g(x))$, $x = 0$
56. Suppose that the function $f(x)$ and its first derivative have the following values at $x = 0$ and $x = 1$.

x	$f(x)$	$f'(x)$
0	9	-2
1	-3	1/5

Find the first derivatives of the following combinations at the given value of x .

- a. $\sqrt{x} f(x)$, $x = 1$ b. $\sqrt{f(x)}$, $x = 0$
- c. $f(\sqrt{x})$, $x = 1$ d. $f(1 - 5 \tan x)$, $x = 0$
- e. $\frac{f(x)}{2 + \cos x}$, $x = 0$ f. $10 \sin\left(\frac{\pi x}{2}\right) f^2(x)$, $x = 1$
57. Find the value of dy/dt at $t = 0$ if $y = 3 \sin 2x$ and $x = t^2 + \pi$.
58. Find the value of ds/du at $u = 2$ if $s = t^2 + 5t$ and $t = (u^2 + 2u)^{1/3}$.
59. Find the value of dw/ds at $s = 0$ if $w = \sin(\sqrt{r} - 2)$ and $r = 8 \sin(s + \pi/6)$.
60. Find the value of dr/dt at $t = 0$ if $r = (\theta^2 + 7)^{1/3}$ and $\theta^2 t + \theta = 1$.
61. If $y^3 + y = 2 \cos x$, find the value of d^2y/dx^2 at the point $(0, 1)$.
62. If $x^{1/3} + y^{1/3} = 4$, find d^2y/dx^2 at the point $(8, 8)$.

Derivative Definition

In Exercises 63 and 64, find the derivative using the definition.

63. $f(t) = \frac{1}{2t + 1}$ 64. $g(x) = 2x^2 + 1$

65. a. Graph the function

$$f(x) = \begin{cases} x^2, & -1 \leq x < 0 \\ -x^2, & 0 \leq x \leq 1. \end{cases}$$

- b. Is f continuous at $x = 0$?
c. Is f differentiable at $x = 0$?
Give reasons for your answers.

66. a. Graph the function

$$f(x) = \begin{cases} x, & -1 \leq x < 0 \\ \tan x, & 0 \leq x \leq \pi/4. \end{cases}$$

- b. Is f continuous at $x = 0$?
c. Is f differentiable at $x = 0$?

Give reasons for your answers.

67. a. Graph the function

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2 - x, & 1 < x \leq 2. \end{cases}$$

- b. Is f continuous at $x = 1$?
c. Is f differentiable at $x = 1$?

Give reasons for your answers.

68. For what value or values of the constant m , if any, is

$$f(x) = \begin{cases} \sin 2x, & x \leq 0 \\ mx, & x > 0 \end{cases}$$

- a. continuous at $x = 0$?
b. differentiable at $x = 0$?

Give reasons for your answers.

Slopes, Tangents, and Normals

69. **Tangents with specified slope** Are there any points on the curve $y = (x/2) + 1/(2x - 4)$ where the slope is $-3/2$? If so, find them.
70. **Tangents with specified slope** Are there any points on the curve $y = x - 1/(2x)$ where the slope is 3? If so, find them.
71. **Horizontal tangents** Find the points on the curve $y = 2x^3 - 3x^2 - 12x + 20$ where the tangent is parallel to the x -axis.
72. **Tangent intercepts** Find the x - and y -intercepts of the line that is tangent to the curve $y = x^3$ at the point $(-2, -8)$.
73. **Tangents perpendicular or parallel to lines** Find the points on the curve $y = 2x^3 - 3x^2 - 12x + 20$ where the tangent is
- perpendicular to the line $y = 1 - (x/24)$.
 - parallel to the line $y = \sqrt{2} - 12x$.
74. **Intersecting tangents** Show that the tangents to the curve $y = (\pi \sin x)/x$ at $x = \pi$ and $x = -\pi$ intersect at right angles.
75. **Normals parallel to a line** Find the points on the curve $y = \tan x$, $-\pi/2 < x < \pi/2$, where the normal is parallel to the line $y = -x/2$. Sketch the curve and normals together, labeling each with its equation.
76. **Tangent and normal lines** Find equations for the tangent and normal to the curve $y = 1 + \cos x$ at the point $(\pi/2, 1)$. Sketch the curve, tangent, and normal together, labeling each with its equation.

77. **Tangent parabola** The parabola $y = x^2 + C$ is to be tangent to the line $y = x$. Find C .

78. **Slope of tangent** Show that the tangent to the curve $y = x^3$ at any point (a, a^3) meets the curve again at a point where the slope is four times the slope at (a, a^3) .

79. **Tangent curve** For what value of c is the curve $y = c/(x + 1)$ tangent to the line through the points $(0, 3)$ and $(5, -2)$?

80. **Normal to a circle** Show that the normal line at any point of the circle $x^2 + y^2 = a^2$ passes through the origin.

Tangents and Normals to Implicitly Defined Curves

In Exercises 81–86, find equations for the lines that are tangent and normal to the curve at the given point.

81. $x^2 + 2y^2 = 9$, $(1, 2)$

82. $x^3 + y^2 = 2$, $(1, 1)$

83. $xy + 2x - 5y = 2$, $(3, 2)$

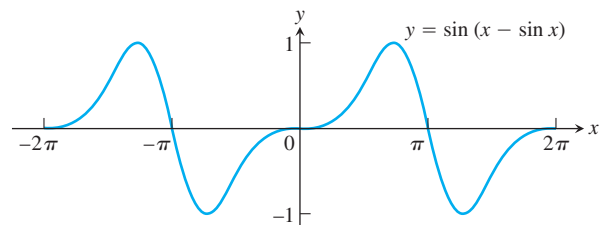
84. $(y - x)^2 = 2x + 4$, $(6, 2)$

85. $x + \sqrt{xy} = 6$, $(4, 1)$

86. $x^{3/2} + 2y^{3/2} = 17$, $(1, 4)$

87. Find the slope of the curve $x^3y^3 + y^2 = x + y$ at the points $(1, 1)$ and $(1, -1)$.

88. The graph shown suggests that the curve $y = \sin(x - \sin x)$ might have horizontal tangents at the x -axis. Does it? Give reasons for your answer.



Tangents to Parametrized Curves

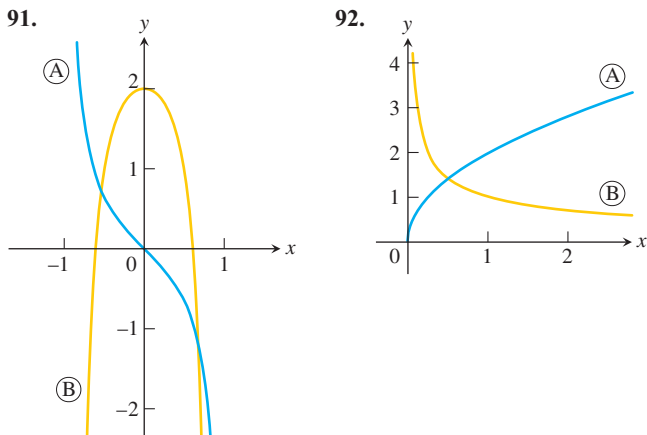
In Exercises 89 and 90, find an equation for the line in the xy -plane that is tangent to the curve at the point corresponding to the given value of t . Also, find the value of d^2y/dx^2 at this point.

89. $x = (1/2) \tan t$, $y = (1/2) \sec t$, $t = \pi/3$

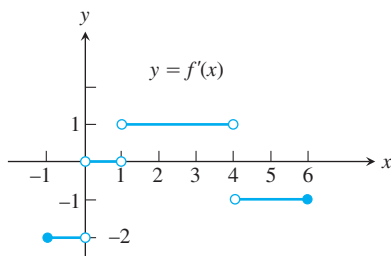
90. $x = 1 + 1/t^2$, $y = 1 - 3/t$, $t = 2$

Analyzing Graphs

Each of the figures in Exercises 91 and 92 shows two graphs, the graph of a function $y = f(x)$ together with the graph of its derivative $f'(x)$. Which graph is which? How do you know?



93. Use the following information to graph the function $y = f(x)$ for $-1 \leq x \leq 6$.
- The graph of f is made of line segments joined end to end.
 - The graph starts at the point $(-1, 2)$.
 - The derivative of f , where defined, agrees with the step function shown here.



94. Repeat Exercise 93, supposing that the graph starts at $(-1, 0)$ instead of $(-1, 2)$.

Exercises 95 and 96 are about the graphs in Figure 3.53 (right-hand column). The graphs in part (a) show the numbers of rabbits and foxes in a small arctic population. They are plotted as functions of time for 200 days. The number of rabbits increases at first, as the rabbits reproduce. But the foxes prey on rabbits and, as the number of foxes increases, the rabbit population levels off and then drops. Figure 3.53b shows the graph of the derivative of the rabbit population. We made it by plotting slopes.

95. a. What is the value of the derivative of the rabbit population in Figure 3.53 when the number of rabbits is largest? Smallest?
 b. What is the size of the rabbit population in Figure 3.53 when its derivative is largest? Smallest (negative value)?
96. In what units should the slopes of the rabbit and fox population curves be measured?

Trigonometric Limits

97. $\lim_{x \rightarrow 0} \frac{\sin x}{2x^2 - x}$ 98. $\lim_{x \rightarrow 0} \frac{3x - \tan 7x}{2x}$

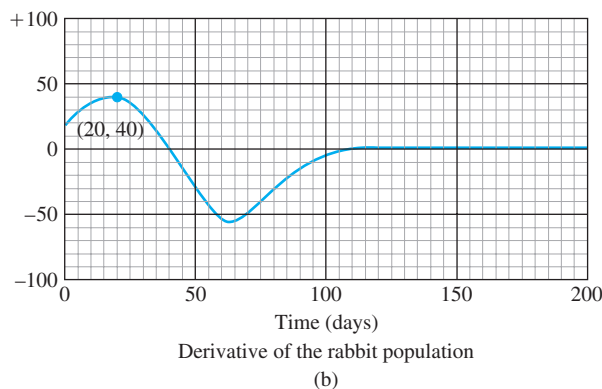
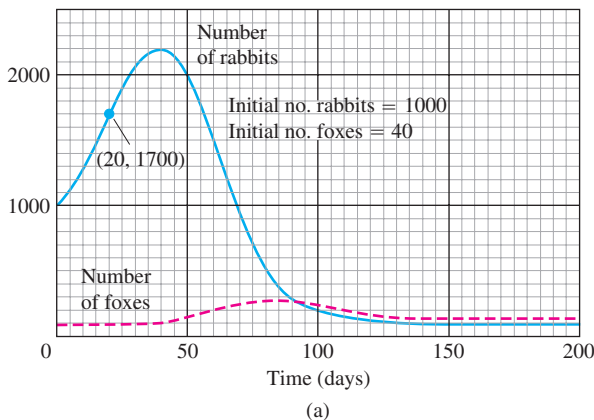


FIGURE 3.53 Rabbits and foxes in an arctic predator-prey food chain.

99. $\lim_{r \rightarrow 0} \frac{\sin r}{\tan 2r}$ 100. $\lim_{\theta \rightarrow 0} \frac{\sin(\sin \theta)}{\theta}$

101. $\lim_{\theta \rightarrow (\pi/2)^-} \frac{4 \tan^2 \theta + \tan \theta + 1}{\tan^2 \theta + 5}$

102. $\lim_{\theta \rightarrow 0^+} \frac{1 - 2 \cot^2 \theta}{5 \cot^2 \theta - 7 \cot \theta - 8}$

103. $\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x}$ 104. $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$

Show how to extend the functions in Exercises 105 and 106 to be continuous at the origin.

105. $g(x) = \frac{\tan(\tan x)}{\tan x}$ 106. $f(x) = \frac{\tan(\tan x)}{\sin(\sin x)}$

Related Rates

107. **Right circular cylinder** The total surface area S of a right circular cylinder is related to the base radius r and height h by the equation $S = 2\pi r^2 + 2\pi rh$.

- How is dS/dt related to dr/dt if h is constant?
- How is dS/dt related to dh/dt if r is constant?

c. How is dS/dt related to dr/dt and dh/dt if neither r nor h is constant?

d. How is dr/dt related to dh/dt if S is constant?

108. Right circular cone The lateral surface area S of a right circular cone is related to the base radius r and height h by the equation $S = \pi r \sqrt{r^2 + h^2}$.

a. How is dS/dt related to dr/dt if h is constant?

b. How is dS/dt related to dh/dt if r is constant?

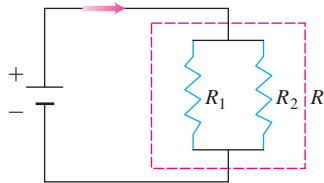
c. How is dS/dt related to dr/dt and dh/dt if neither r nor h is constant?

109. Circle's changing area The radius of a circle is changing at the rate of $-2/\pi$ m/sec. At what rate is the circle's area changing when $r = 10$ m?

110. Cube's changing edges The volume of a cube is increasing at the rate of $1200 \text{ cm}^3/\text{min}$ at the instant its edges are 20 cm long. At what rate are the lengths of the edges changing at that instant?

111. Resistors connected in parallel If two resistors of R_1 and R_2 ohms are connected in parallel in an electric circuit to make an R -ohm resistor, the value of R can be found from the equation

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$



If R_1 is decreasing at the rate of 1 ohm/sec and R_2 is increasing at the rate of 0.5 ohm/sec, at what rate is R changing when $R_1 = 75$ ohms and $R_2 = 50$ ohms?

112. Impedance in a series circuit The impedance Z (ohms) in a series circuit is related to the resistance R (ohms) and reactance X (ohms) by the equation $Z = \sqrt{R^2 + X^2}$. If R is increasing at 3 ohms/sec and X is decreasing at 2 ohms/sec, at what rate is Z changing when $R = 10$ ohms and $X = 20$ ohms?

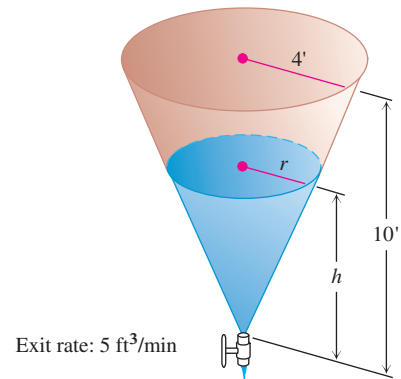
113. Speed of moving particle The coordinates of a particle moving in the metric xy -plane are differentiable functions of time t with $dx/dt = 10$ m/sec and $dy/dt = 5$ m/sec. How fast is the particle moving away from the origin as it passes through the point $(3, -4)$?

114. Motion of a particle A particle moves along the curve $y = x^{3/2}$ in the first quadrant in such a way that its distance from the origin increases at the rate of 11 units per second. Find dx/dt when $x = 3$.

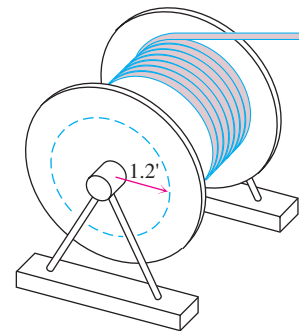
115. Draining a tank Water drains from the conical tank shown in the accompanying figure at the rate of $5 \text{ ft}^3/\text{min}$.

a. What is the relation between the variables h and r in the figure?

b. How fast is the water level dropping when $h = 6$ ft?



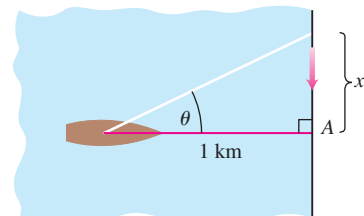
116. Rotating spool As television cable is pulled from a large spool to be strung from the telephone poles along a street, it unwinds from the spool in layers of constant radius (see accompanying figure). If the truck pulling the cable moves at a steady 6 ft/sec (a touch over 4 mph), use the equation $s = r\theta$ to find how fast (radians per second) the spool is turning when the layer of radius 1.2 ft is being unwound.



117. Moving searchlight beam The figure shows a boat 1 km offshore, sweeping the shore with a searchlight. The light turns at a constant rate, $d\theta/dt = -0.6$ rad/sec.

a. How fast is the light moving along the shore when it reaches point A ?

b. How many revolutions per minute is 0.6 rad/sec?



118. Points moving on coordinate axes Points A and B move along the x - and y -axes, respectively, in such a way that the distance r (meters) along the perpendicular from the origin to the line AB remains constant. How fast is OA changing, and is it increasing, or decreasing, when $OB = 2r$ and B is moving toward O at the rate of $0.3r$ m/sec?

Linearization

119. Find the linearizations of

a. $\tan x$ at $x = -\pi/4$ b. $\sec x$ at $x = -\pi/4$.

Graph the curves and linearizations together.

120. We can obtain a useful linear approximation of the function $f(x) = 1/(1 + \tan x)$ at $x = 0$ by combining the approximations

$$\frac{1}{1+x} \approx 1-x \quad \text{and} \quad \tan x \approx x$$

to get

$$\frac{1}{1+\tan x} \approx 1-x.$$

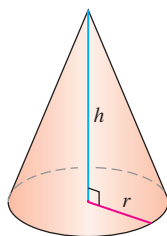
Show that this result is the standard linear approximation of $1/(1 + \tan x)$ at $x = 0$.

121. Find the linearization of $f(x) = \sqrt{1+x} + \sin x - 0.5$ at $x = 0$.

122. Find the linearization of $f(x) = 2/(1-x) + \sqrt{1+x} - 3.1$ at $x = 0$.

Differential Estimates of Change

123. **Surface area of a cone** Write a formula that estimates the change that occurs in the lateral surface area of a right circular cone when the height changes from h_0 to $h_0 + dh$ and the radius does not change.



$$V = \frac{1}{3}\pi r^2 h$$

$$S = \pi r \sqrt{r^2 + h^2}$$

(Lateral surface area)

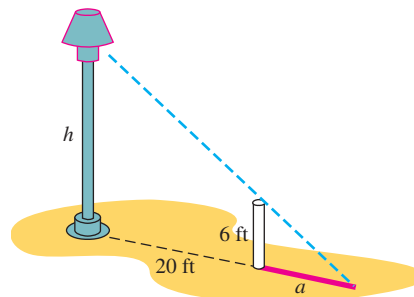
124. Controlling error

- How accurately should you measure the edge of a cube to be reasonably sure of calculating the cube's surface area with an error of no more than 2%?
- Suppose that the edge is measured with the accuracy required in part (a). About how accurately can the cube's volume be calculated from the edge measurement? To find out, estimate the percentage error in the volume calculation that might result from using the edge measurement.

125. **Compounding error** The circumference of the equator of a sphere is measured as 10 cm with a possible error of 0.4 cm. This measurement is then used to calculate the radius. The radius is then used to calculate the surface area and volume of the sphere. Estimate the percentage errors in the calculated values of

- the radius.
- the surface area.
- the volume.

126. **Finding height** To find the height of a lamppost (see accompanying figure), you stand a 6 ft pole 20 ft from the lamp and measure the length a of its shadow, finding it to be 15 ft, give or take an inch. Calculate the height of the lamppost using the value $a = 15$ and estimate the possible error in the result.



Chapter 3

Questions to Guide Your Review

1. What is the derivative of a function f ? How is its domain related to the domain of f ? Give examples.
2. What role does the derivative play in defining slopes, tangents, and rates of change?
3. How can you sometimes graph the derivative of a function when all you have is a table of the function's values?
4. What does it mean for a function to be differentiable on an open interval? On a closed interval?
5. How are derivatives and one-sided derivatives related?
6. Describe geometrically when a function typically does *not* have a derivative at a point.
7. How is a function's differentiability at a point related to its continuity there, if at all?
8. Could the unit step function

$$U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

possibly be the derivative of some other function on $[-1, 1]$? Explain.

9. What rules do you know for calculating derivatives? Give some examples.
10. Explain how the three formulas
 - a. $\frac{d}{dx}(x^n) = nx^{n-1}$
 - b. $\frac{d}{dx}(cu) = c \frac{du}{dx}$
 - c. $\frac{d}{dx}(u_1 + u_2 + \cdots + u_n) = \frac{du_1}{dx} + \frac{du_2}{dx} + \cdots + \frac{du_n}{dx}$
 enable us to differentiate any polynomial.
11. What formula do we need, in addition to the three listed in Question 10, to differentiate rational functions?
12. What is a second derivative? A third derivative? How many derivatives do the functions you know have? Give examples.
13. What is the relationship between a function's average and instantaneous rates of change? Give an example.
14. How do derivatives arise in the study of motion? What can you learn about a body's motion along a line by examining the derivatives of the body's position function? Give examples.
15. How can derivatives arise in economics?
16. Give examples of still other applications of derivatives.
17. What do the limits $\lim_{h \rightarrow 0} ((\sin h)/h)$ and $\lim_{h \rightarrow 0} ((\cos h - 1)/h)$ have to do with the derivatives of the sine and cosine functions? What *are* the derivatives of these functions?
18. Once you know the derivatives of $\sin x$ and $\cos x$, how can you find the derivatives of $\tan x$, $\cot x$, $\sec x$, and $\csc x$? What *are* the derivatives of these functions?
19. At what points are the six basic trigonometric functions continuous? How do you know?
20. What is the rule for calculating the derivative of a composite of two differentiable functions? How is such a derivative evaluated? Give examples.
21. What is the formula for the slope dy/dx of a parametrized curve $x = f(t)$, $y = g(t)$? When does the formula apply? When can you expect to be able to find d^2y/dx^2 as well? Give examples.
22. If u is a differentiable function of x , how do you find $(d/dx)(u^n)$ if n is an integer? If n is a rational number? Give examples.
23. What is implicit differentiation? When do you need it? Give examples.
24. How do related rates problems arise? Give examples.
25. Outline a strategy for solving related rates problems. Illustrate with an example.
26. What is the linearization $L(x)$ of a function $f(x)$ at a point $x = a$? What is required of f at a for the linearization to exist? How are linearizations used? Give examples.
27. If x moves from a to a nearby value $a + dx$, how do you estimate the corresponding change in the value of a differentiable function $f(x)$? How do you estimate the relative change? The percentage change? Give an example.

Chapter 3 Technology Application Projects

Mathematica/Maple Module

Convergence of Secant Slopes to the Derivative Function

You will visualize the secant line between successive points on a curve and observe what happens as the distance between them becomes small. The function, sample points, and secant lines are plotted on a single graph, while a second graph compares the slopes of the secant lines with the derivative function.

Mathematica/Maple Module

Derivatives, Slopes, Tangent Lines, and Making Movies

Parts I–III. You will visualize the derivative at a point, the linearization of a function, and the derivative of a function. You learn how to plot the function and selected tangents on the same graph.

Part IV (Plotting Many Tangents)

Part V (Making Movies). Parts IV and V of the module can be used to animate tangent lines as one moves along the graph of a function.

Mathematica/Maple Module

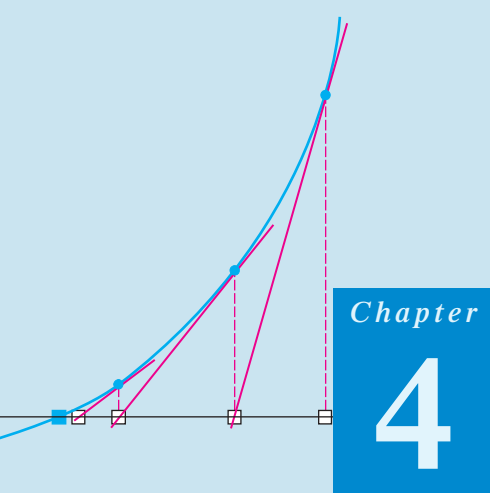
Convergence of Secant Slopes to the Derivative Function

You will visualize right-hand and left-hand derivatives.

Mathematica/Maple Module

Motion Along a Straight Line: Position $\rightarrow$ Velocity $\rightarrow$ Acceleration

Observe dramatic animated visualizations of the derivative relations among the position, velocity, and acceleration functions. Figures in the text can be animated.



Chapter

4

APPLICATIONS OF DERIVATIVES

OVERVIEW This chapter studies some of the important applications of derivatives. We learn how derivatives are used to find extreme values of functions, to determine and analyze the shapes of graphs, to calculate limits of fractions whose numerators and denominators both approach zero or infinity, and to find numerically where a function equals zero. We also consider the process of recovering a function from its derivative. The key to many of these accomplishments is the Mean Value Theorem, a theorem whose corollaries provide the gateway to integral calculus in Chapter 5.

4.1

Extreme Values of Functions

This section shows how to locate and identify extreme values of a continuous function from its derivative. Once we can do this, we can solve a variety of *optimization problems* in which we find the optimal (best) way to do something in a given situation.

DEFINITIONS Absolute Maximum, Absolute Minimum

Let f be a function with domain D . Then f has an **absolute maximum** value on D at a point c if

$$f(x) \leq f(c) \quad \text{for all } x \text{ in } D$$

and an **absolute minimum** value on D at c if

$$f(x) \geq f(c) \quad \text{for all } x \text{ in } D.$$

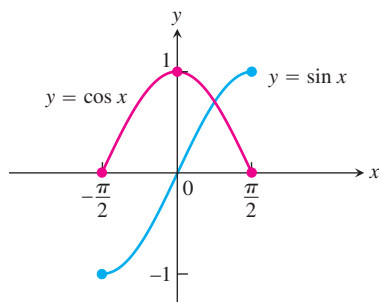


FIGURE 4.1 Absolute extrema for the sine and cosine functions on $[-\pi/2, \pi/2]$. These values can depend on the domain of a function.

Absolute maximum and minimum values are called absolute **extrema** (plural of the Latin *extremum*). Absolute extrema are also called **global** extrema, to distinguish them from *local extrema* defined below.

For example, on the closed interval $[-\pi/2, \pi/2]$ the function $f(x) = \cos x$ takes on an absolute maximum value of 1 (once) and an absolute minimum value of 0 (twice). On the same interval, the function $g(x) = \sin x$ takes on a maximum value of 1 and a minimum value of -1 (Figure 4.1).

Functions with the same defining rule can have different extrema, depending on the domain.

EXAMPLE 1 Exploring Absolute Extrema

The absolute extrema of the following functions on their domains can be seen in Figure 4.2.

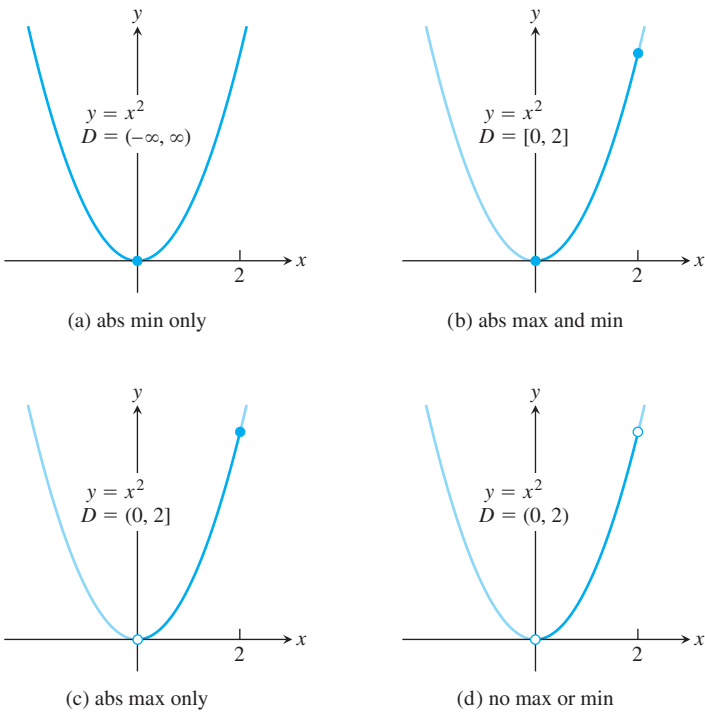


FIGURE 4.2 Graphs for Example 1.

Function rule	Domain D	Absolute extrema on D
(a) $y = x^2$	$(-\infty, \infty)$	No absolute maximum. Absolute minimum of 0 at $x = 0$.
(b) $y = x^2$	$[0, 2]$	Absolute maximum of 4 at $x = 2$. Absolute minimum of 0 at $x = 0$.
(c) $y = x^2$	$(0, 2]$	Absolute maximum of 4 at $x = 2$. No absolute minimum.
(d) $y = x^2$	$(0, 2)$	No absolute extrema.



HISTORICAL BIOGRAPHY

Daniel Bernoulli
(1700–1789)

The following theorem asserts that a function which is continuous at every point of a closed interval $[a, b]$ has an absolute maximum and an absolute minimum value on the interval. We always look for these values when we graph a function.

THEOREM 1 The Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$. That is, there are numbers x_1 and x_2 in $[a, b]$ with $f(x_1) = m$, $f(x_2) = M$, and $m \leq f(x) \leq M$ for every other x in $[a, b]$ (Figure 4.3).

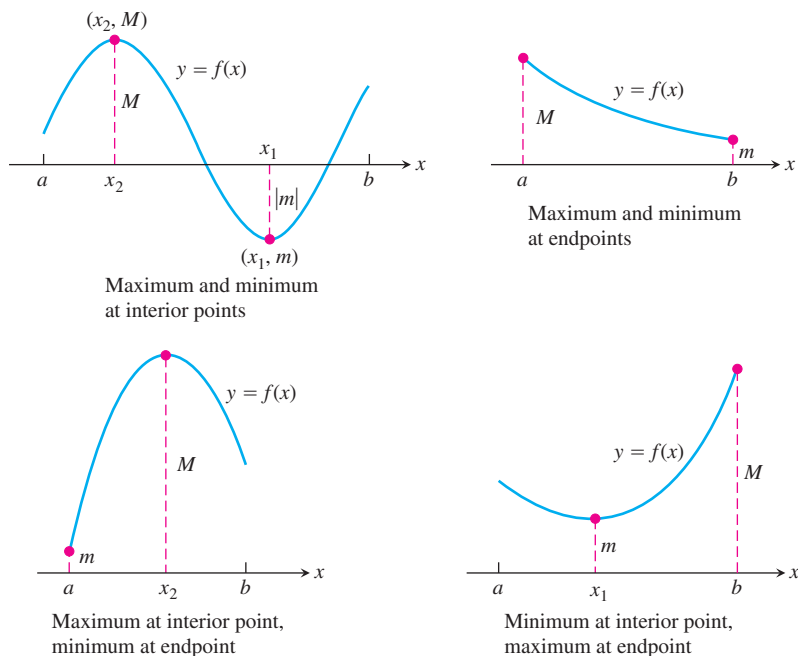


FIGURE 4.3 Some possibilities for a continuous function's maximum and minimum on a closed interval $[a, b]$.

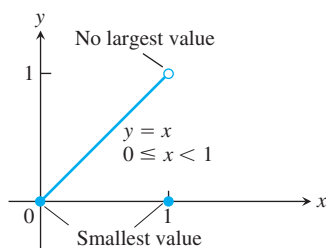


FIGURE 4.4 Even a single point of discontinuity can keep a function from having either a maximum or minimum value on a closed interval. The function

$$y = \begin{cases} x, & 0 \leq x < 1 \\ 0, & x = 1 \end{cases}$$

is continuous at every point of $[0, 1]$ except $x = 1$, yet its graph over $[0, 1]$ does not have a highest point.

The proof of The Extreme Value Theorem requires a detailed knowledge of the real number system (see Appendix 4) and we will not give it here. Figure 4.3 illustrates possible locations for the absolute extrema of a continuous function on a closed interval $[a, b]$. As we observed for the function $y = \cos x$, it is possible that an absolute minimum (or absolute maximum) may occur at two or more different points of the interval.

The requirements in Theorem 1 that the interval be closed and finite, and that the function be continuous, are key ingredients. Without them, the conclusion of the theorem need not hold. Example 1 shows that an absolute extreme value may not exist if the interval fails to be both closed and finite. Figure 4.4 shows that the continuity requirement cannot be omitted.

Local (Relative) Extreme Values

Figure 4.5 shows a graph with five points where a function has extreme values on its domain $[a, b]$. The function's absolute minimum occurs at a even though at e the function's value is

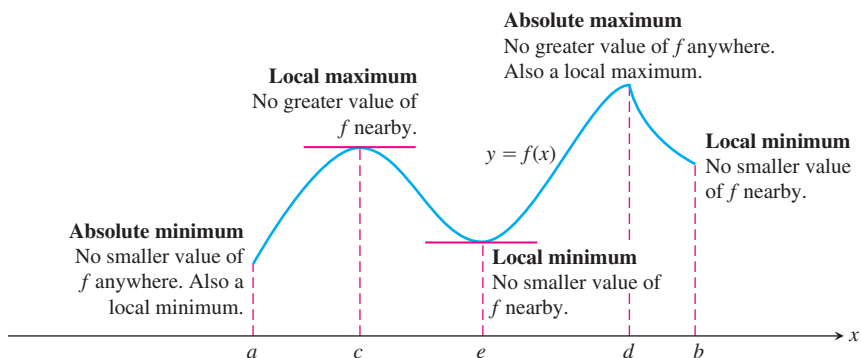


FIGURE 4.5 How to classify maxima and minima.

smaller than at any other point *nearby*. The curve rises to the left and falls to the right around c , making $f(c)$ a maximum locally. The function attains its absolute maximum at d .

DEFINITIONS Local Maximum, Local Minimum

A function f has a **local maximum** value at an interior point c of its domain if

$$f(x) \leq f(c) \quad \text{for all } x \text{ in some open interval containing } c.$$

A function f has a **local minimum** value at an interior point c of its domain if

$$f(x) \geq f(c) \quad \text{for all } x \text{ in some open interval containing } c.$$

We can extend the definitions of local extrema to the endpoints of intervals by defining f to have a **local maximum** or **local minimum** value *at an endpoint* c if the appropriate inequality holds for all x in some half-open interval in its domain containing c . In Figure 4.5, the function f has local maxima at c and d and local minima at a , e , and b . Local extrema are also called **relative extrema**.

An absolute maximum is also a local maximum. Being the largest value overall, it is also the largest value in its immediate neighborhood. Hence, *a list of all local maxima will automatically include the absolute maximum if there is one*. Similarly, *a list of all local minima will include the absolute minimum if there is one*.

Finding Extrema

The next theorem explains why we usually need to investigate only a few values to find a function's extrema.

THEOREM 2 The First Derivative Theorem for Local Extreme Values

If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c , then

$$f'(c) = 0.$$

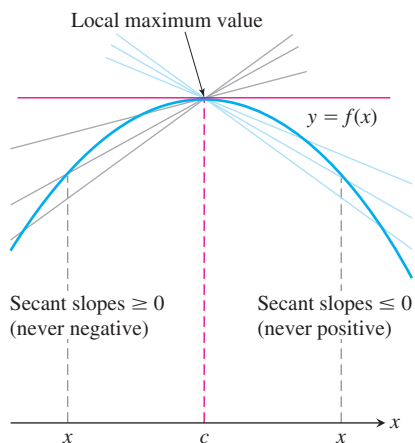


FIGURE 4.6 A curve with a local maximum value. The slope at c , simultaneously the limit of nonpositive numbers and nonnegative numbers, is zero.

Proof To prove that $f'(c)$ is zero at a local extremum, we show first that $f'(c)$ cannot be positive and second that $f'(c)$ cannot be negative. The only number that is neither positive nor negative is zero, so that is what $f'(c)$ must be.

To begin, suppose that f has a local maximum value at $x = c$ (Figure 4.6) so that $f(x) - f(c) \leq 0$ for all values of x near enough to c . Since c is an interior point of f 's domain, $f'(c)$ is defined by the two-sided limit

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

This means that the right-hand and left-hand limits both exist at $x = c$ and equal $f'(c)$. When we examine these limits separately, we find that

$$f'(c) = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} \leq 0. \quad \begin{array}{l} \text{Because } (x - c) > 0 \\ \text{and } f(x) \leq f(c) \end{array} \quad (1)$$

Similarly,

$$f'(c) = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} \geq 0. \quad \begin{array}{l} \text{Because } (x - c) < 0 \\ \text{and } f(x) \leq f(c) \end{array} \quad (2)$$

Together, Equations (1) and (2) imply $f'(c) = 0$.

This proves the theorem for local maximum values. To prove it for local minimum values, we simply use $f(x) \geq f(c)$, which reverses the inequalities in Equations (1) and (2). ■

Theorem 2 says that a function's first derivative is always zero at an interior point where the function has a local extreme value and the derivative is defined. Hence the only places where a function f can possibly have an extreme value (local or global) are

1. interior points where $f' = 0$,
2. interior points where f' is undefined,
3. endpoints of the domain of f .

The following definition helps us to summarize.

DEFINITION Critical Point

An interior point of the domain of a function f where f' is zero or undefined is a **critical point** of f .

Thus the only domain points where a function can assume extreme values are critical points and endpoints.

Be careful not to misinterpret Theorem 2 because its converse is false. A differentiable function may have a critical point at $x = c$ without having a local extreme value there. For instance, the function $f(x) = x^3$ has a critical point at the origin and zero value there, but is positive to the right of the origin and negative to the left. So it cannot have a local extreme value at the origin. Instead, it has a *point of inflection* there. This idea is defined and discussed further in Section 4.4.

Most quests for extreme values call for finding the absolute extrema of a continuous function on a closed and finite interval. Theorem 1 assures us that such values exist; Theorem 2 tells us that they are taken on only at critical points and endpoints. Often we can

simply list these points and calculate the corresponding function values to find what the largest and smallest values are, and where they are located.

How to Find the Absolute Extrema of a Continuous Function f on a Finite Closed Interval

1. Evaluate f at all critical points and endpoints.
2. Take the largest and smallest of these values.

EXAMPLE 2 Finding Absolute Extrema

Find the absolute maximum and minimum values of $f(x) = x^2$ on $[-2, 1]$.

Solution The function is differentiable over its entire domain, so the only critical point is where $f'(x) = 2x = 0$, namely $x = 0$. We need to check the function's values at $x = 0$ and at the endpoints $x = -2$ and $x = 1$:

$$\begin{aligned}\text{Critical point value: } f(0) &= 0 \\ \text{Endpoint values: } f(-2) &= 4 \\ f(1) &= 1\end{aligned}$$

The function has an absolute maximum value of 4 at $x = -2$ and an absolute minimum value of 0 at $x = 0$. ■

EXAMPLE 3 Absolute Extrema at Endpoints

Find the absolute extrema values of $g(t) = 8t - t^4$ on $[-2, 1]$.

Solution The function is differentiable on its entire domain, so the only critical points occur where $g'(t) = 0$. Solving this equation gives

$$8 - 4t^3 = 0 \quad \text{or} \quad t = \sqrt[3]{2} > 1,$$

a point not in the given domain. The function's absolute extrema therefore occur at the endpoints, $g(-2) = -32$ (absolute minimum), and $g(1) = 7$ (absolute maximum). See Figure 4.7. ■

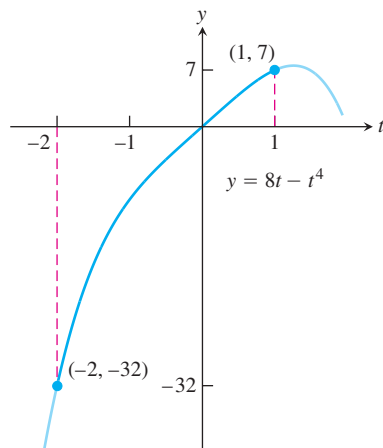


FIGURE 4.7 The extreme values of $g(t) = 8t - t^4$ on $[-2, 1]$ (Example 3).

EXAMPLE 4 Finding Absolute Extrema on a Closed Interval

Find the absolute maximum and minimum values of $f(x) = x^{2/3}$ on the interval $[-2, 3]$.

Solution We evaluate the function at the critical points and endpoints and take the largest and smallest of the resulting values.

The first derivative

$$f'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3\sqrt[3]{x}}$$

has no zeros but is undefined at the interior point $x = 0$. The values of f at this one critical point and at the endpoints are

$$\begin{aligned}\text{Critical point value: } f(0) &= 0 \\ \text{Endpoint values: } f(-2) &= (-2)^{2/3} = \sqrt[3]{4} \\ f(3) &= (3)^{2/3} = \sqrt[3]{9}.\end{aligned}$$

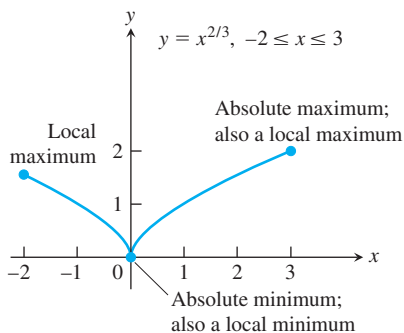


FIGURE 4.8 The extreme values of $f(x) = x^{2/3}$ on $[-2, 3]$ occur at $x = 0$ and $x = 3$ (Example 4).

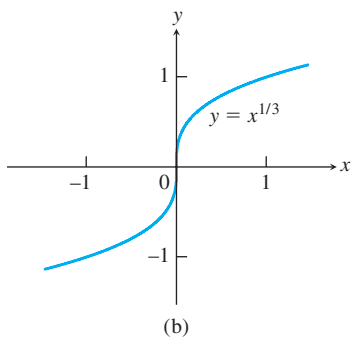
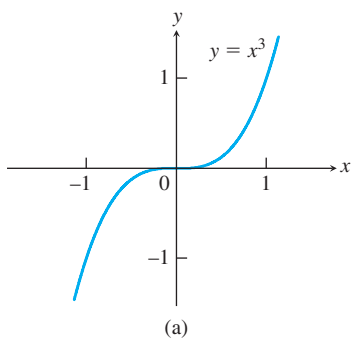


FIGURE 4.9 Critical points without extreme values. (a) $y' = 3x^2$ is 0 at $x = 0$, but $y = x^3$ has no extremum there. (b) $y' = (1/3)x^{-2/3}$ is undefined at $x = 0$, but $y = x^{1/3}$ has no extremum there.

We can see from this list that the function's absolute maximum value is $\sqrt[3]{9} \approx 2.08$, and it occurs at the right endpoint $x = 3$. The absolute minimum value is 0, and it occurs at the interior point $x = 0$. (Figure 4.8).

While a function's extrema can occur only at critical points and endpoints, not every critical point or endpoint signals the presence of an extreme value. Figure 4.9 illustrates this for interior points.

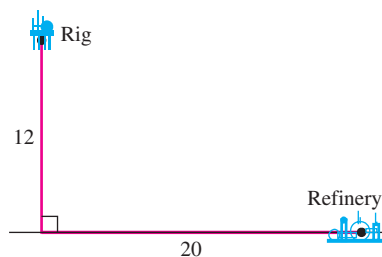
We complete this section with an example illustrating how the concepts we studied are used to solve a real-world optimization problem.

EXAMPLE 5 Piping Oil from a Drilling Rig to a Refinery

A drilling rig 12 mi offshore is to be connected by pipe to a refinery onshore, 20 mi straight down the coast from the rig. If underwater pipe costs \$500,000 per mile and land-based pipe costs \$300,000 per mile, what combination of the two will give the least expensive connection?

Solution We try a few possibilities to get a feel for the problem:

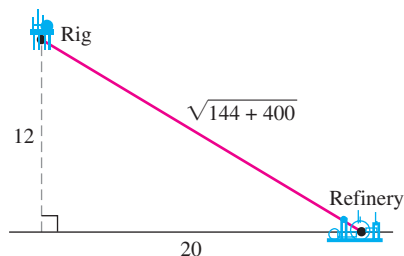
(a) *Smallest amount of underwater pipe*



Underwater pipe is more expensive, so we use as little as we can. We run straight to shore (12 mi) and use land pipe for 20 mi to the refinery.

$$\begin{aligned}\text{Dollar cost} &= 12(500,000) + 20(300,000) \\ &= 12,000,000\end{aligned}$$

(b) *All pipe underwater (most direct route)*

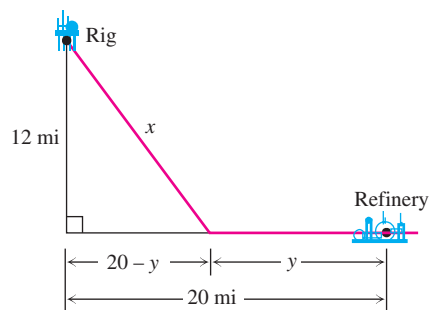


We go straight to the refinery underwater.

$$\begin{aligned}\text{Dollar cost} &= \sqrt{544} (500,000) \\ &\approx 11,661,900\end{aligned}$$

This is less expensive than plan (a).

(c) *Something in between*



Now we introduce the length x of underwater pipe and the length y of land-based pipe as variables. The right angle opposite the rig is the key to expressing the relationship between x and y , for the Pythagorean theorem gives

$$\begin{aligned}x^2 &= 12^2 + (20 - y)^2 \\x &= \sqrt{144 + (20 - y)^2}.\end{aligned}\tag{3}$$

Only the positive root has meaning in this model.

The dollar cost of the pipeline is

$$c = 500,000x + 300,000y.$$

To express c as a function of a single variable, we can substitute for x , using Equation (3):

$$c(y) = 500,000\sqrt{144 + (20 - y)^2} + 300,000y.$$

Our goal now is to find the minimum value of $c(y)$ on the interval $0 \leq y \leq 20$. The first derivative of $c(y)$ with respect to y according to the Chain Rule is

$$\begin{aligned}c'(y) &= 500,000 \cdot \frac{1}{2} \cdot \frac{2(20 - y)(-1)}{\sqrt{144 + (20 - y)^2}} + 300,000 \\&= -500,000 \frac{20 - y}{\sqrt{144 + (20 - y)^2}} + 300,000.\end{aligned}$$

Setting c' equal to zero gives

$$500,000(20 - y) = 300,000\sqrt{144 + (20 - y)^2}$$

$$\frac{5}{3}(20 - y) = \sqrt{144 + (20 - y)^2}$$

$$\frac{25}{9}(20 - y)^2 = 144 + (20 - y)^2$$

$$\frac{16}{9}(20 - y)^2 = 144$$

$$(20 - y) = \pm \frac{3}{4} \cdot 12 = \pm 9$$

$$y = 20 \pm 9$$

$$y = 11 \quad \text{or} \quad y = 29.$$

Only $y = 11$ lies in the interval of interest. The values of c at this one critical point and at the endpoints are

$$c(11) = 10,800,000$$

$$c(0) = 11,661,900$$

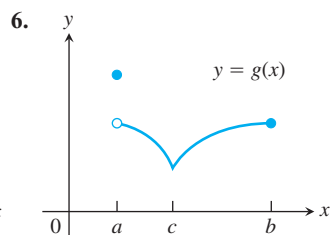
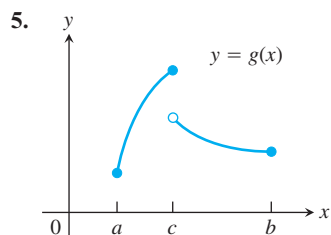
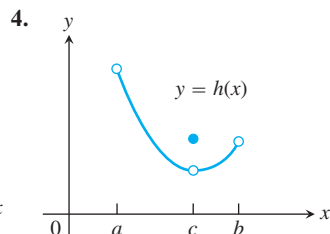
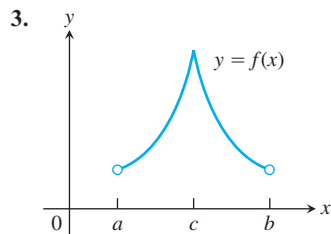
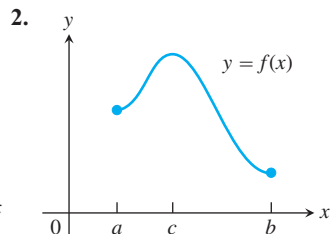
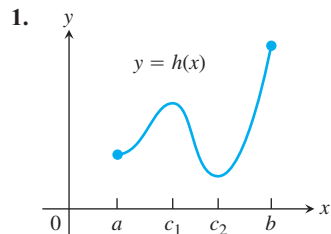
$$c(20) = 12,000,000$$

The least expensive connection costs \$10,800,000, and we achieve it by running the line underwater to the point on shore 11 mi from the refinery. ■

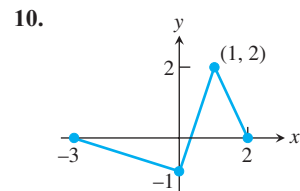
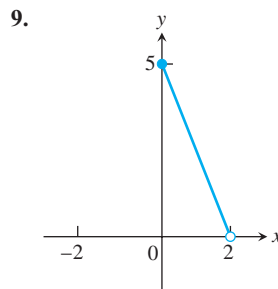
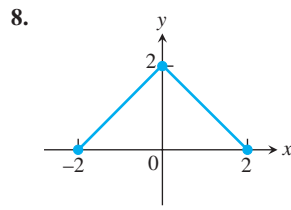
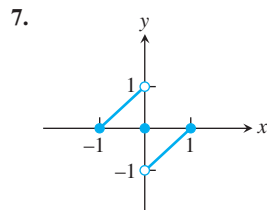
EXERCISES 4.1

Finding Extrema from Graphs

In Exercises 1–6, determine from the graph whether the function has any absolute extreme values on $[a, b]$. Then explain how your answer is consistent with Theorem 1.



In Exercises 7–10, find the extreme values and where they occur.



In Exercises 11–14, match the table with a graph.

11.

x	$f'(x)$
a	0
b	0
c	5

12.

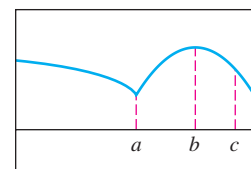
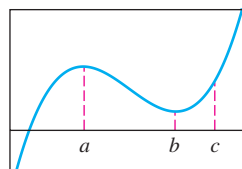
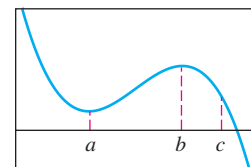
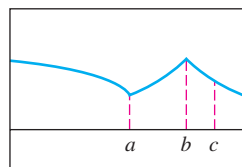
x	$f'(x)$
a	0
b	0
c	-5

13.

x	$f'(x)$
a	does not exist
b	0
c	-2

14.

x	$f'(x)$
a	does not exist
b	does not exist
c	-1.7



(a)

(b)

(c)

(d)

Absolute Extrema on Finite Closed Intervals

In Exercises 15–30, find the absolute maximum and minimum values of each function on the given interval. Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

15. $f(x) = \frac{2}{3}x - 5, \quad -2 \leq x \leq 3$
16. $f(x) = -x - 4, \quad -4 \leq x \leq 1$
17. $f(x) = x^2 - 1, \quad -1 \leq x \leq 2$
18. $f(x) = 4 - x^2, \quad -3 \leq x \leq 1$
19. $F(x) = -\frac{1}{x^2}, \quad 0.5 \leq x \leq 2$
20. $F(x) = -\frac{1}{x}, \quad -2 \leq x \leq -1$
21. $h(x) = \sqrt[3]{x}, \quad -1 \leq x \leq 8$
22. $h(x) = -3x^{2/3}, \quad -1 \leq x \leq 1$
23. $g(x) = \sqrt{4 - x^2}, \quad -2 \leq x \leq 1$
24. $g(x) = -\sqrt{5 - x^2}, \quad -\sqrt{5} \leq x \leq 0$
25. $f(\theta) = \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{5\pi}{6}$
26. $f(\theta) = \tan \theta, \quad -\frac{\pi}{3} \leq \theta \leq \frac{\pi}{4}$
27. $g(x) = \csc x, \quad \frac{\pi}{3} \leq x \leq \frac{2\pi}{3}$
28. $g(x) = \sec x, \quad -\frac{\pi}{3} \leq x \leq \frac{\pi}{6}$
29. $f(t) = 2 - |t|, \quad -1 \leq t \leq 3$
30. $f(t) = |t - 5|, \quad 4 \leq t \leq 7$

In Exercises 31–34, find the function's absolute maximum and minimum values and say where they are assumed.

31. $f(x) = x^{4/3}, \quad -1 \leq x \leq 8$
32. $f(x) = x^{5/3}, \quad -1 \leq x \leq 8$
33. $g(\theta) = \theta^{3/5}, \quad -32 \leq \theta \leq 1$
34. $h(\theta) = 3\theta^{2/3}, \quad -27 \leq \theta \leq 8$

Finding Extreme Values

In Exercises 35–44, find the extreme values of the function and where they occur.

35. $y = 2x^2 - 8x + 9$
36. $y = x^3 - 2x + 4$
37. $y = x^3 + x^2 - 8x + 5$
38. $y = x^3 - 3x^2 + 3x - 2$
39. $y = \sqrt{x^2 - 1}$
40. $y = \frac{1}{\sqrt{1 - x^2}}$
41. $y = \frac{1}{\sqrt[3]{1 - x^2}}$
42. $y = \sqrt{3 + 2x - x^2}$
43. $y = \frac{x}{x^2 + 1}$
44. $y = \frac{x + 1}{x^2 + 2x + 2}$

Local Extrema and Critical Points

In Exercises 45–52, find the derivative at each critical point and determine the local extreme values.

45. $y = x^{2/3}(x + 2)$
46. $y = x^{2/3}(x^2 - 4)$
47. $y = x\sqrt{4 - x^2}$
48. $y = x^2\sqrt{3 - x}$
49. $y = \begin{cases} 4 - 2x, & x \leq 1 \\ x + 1, & x > 1 \end{cases}$
50. $y = \begin{cases} 3 - x, & x < 0 \\ 3 + 2x - x^2, & x \geq 0 \end{cases}$
51. $y = \begin{cases} -x^2 - 2x + 4, & x \leq 1 \\ -x^2 + 6x - 4, & x > 1 \end{cases}$
52. $y = \begin{cases} -\frac{1}{4}x^2 - \frac{1}{2}x + \frac{15}{4}, & x \leq 1 \\ x^3 - 6x^2 + 8x, & x > 1 \end{cases}$

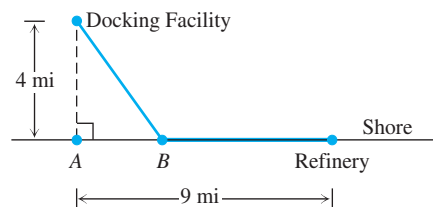
In Exercises 53 and 54, give reasons for your answers.

53. Let $f(x) = (x - 2)^{2/3}$.
 - a. Does $f'(2)$ exist?
 - b. Show that the only local extreme value of f occurs at $x = 2$.
 - c. Does the result in part (b) contradict the Extreme Value Theorem?
 - d. Repeat parts (a) and (b) for $f(x) = (x - a)^{2/3}$, replacing 2 by a .
54. Let $f(x) = |x^3 - 9x|$.
 - a. Does $f'(0)$ exist?
 - b. Does $f'(3)$ exist?
 - c. Does $f'(-3)$ exist?
 - d. Determine all extrema of f .

Optimization Applications

Whenever you are maximizing or minimizing a function of a single variable, we urge you to graph the function over the domain that is appropriate to the problem you are solving. The graph will provide insight before you begin to calculate and will furnish a visual context for understanding your answer.

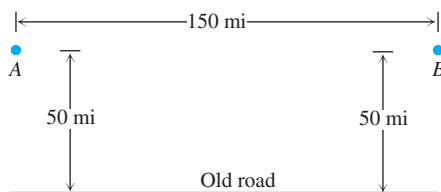
55. **Constructing a pipeline** Supertankers off-load oil at a docking facility 4 mi offshore. The nearest refinery is 9 mi east of the shore point nearest the docking facility. A pipeline must be constructed connecting the docking facility with the refinery. The pipeline costs \$300,000 per mile if constructed underwater and \$200,000 per mile if overland.



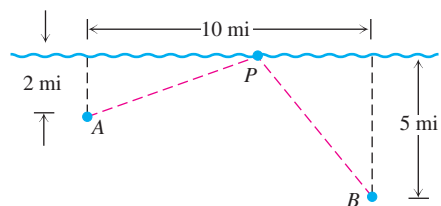
- a. Locate Point B to minimize the cost of the construction.

- b. The cost of underwater construction is expected to increase, whereas the cost of overland construction is expected to stay constant. At what cost does it become optimal to construct the pipeline directly to Point A ?

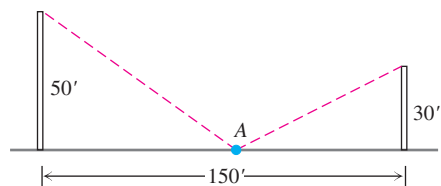
- 56. Upgrading a highway** A highway must be constructed to connect Village A with Village B . There is a rudimentary roadway that can be upgraded 50 mi south of the line connecting the two villages. The cost of upgrading the existing roadway is \$300,000 per mile, whereas the cost of constructing a new highway is \$500,000 per mile. Find the combination of upgrading and new construction that minimizes the cost of connecting the two villages. Clearly define the location of the proposed highway.



- 57. Locating a pumping station** Two towns lie on the south side of a river. A pumping station is to be located to serve the two towns. A pipeline will be constructed from the pumping station to each of the towns along the line connecting the town and the pumping station. Locate the pumping station to minimize the amount of pipeline that must be constructed.



- 58. Length of a guy wire** One tower is 50 ft high and another tower is 30 ft high. The towers are 150 ft apart. A guy wire is to run from Point A to the top of each tower.



- Locate Point A so that the total length of guy wire is minimal.
 - Show in general that regardless of the height of the towers, the length of guy wire is minimized if the angles at A are equal.
- 59. The function**

$$V(x) = x(10 - 2x)(16 - 2x), \quad 0 < x < 5,$$

models the volume of a box.

- Find the extreme values of V .

- Interpret any values found in part (a) in terms of volume of the box.

- 60. The function**

$$P(x) = 2x + \frac{200}{x}, \quad 0 < x < \infty,$$

models the perimeter of a rectangle of dimensions x by $100/x$.

- Find any extreme values of P .
 - Give an interpretation in terms of perimeter of the rectangle for any values found in part (a).
- 61. Area of a right triangle** What is the largest possible area for a right triangle whose hypotenuse is 5 cm long?
- 62. Area of an athletic field** An athletic field is to be built in the shape of a rectangle x units long capped by semicircular regions of radius r at the two ends. The field is to be bounded by a 400-m racetrack.
- Express the area of the rectangular portion of the field as a function of x alone or r alone (your choice).
 - What values of x and r give the rectangular portion the largest possible area?
- 63. Maximum height of a vertically moving body** The height of a body moving vertically is given by

$$s = -\frac{1}{2}gt^2 + v_0t + s_0, \quad g > 0,$$

with s in meters and t in seconds. Find the body's maximum height.

- 64. Peak alternating current** Suppose that at any given time t (in seconds) the current i (in amperes) in an alternating current circuit is $i = 2 \cos t + 2 \sin t$. What is the peak current for this circuit (largest magnitude)?

Theory and Examples

- 65. A minimum with no derivative** The function $f(x) = |x|$ has an absolute minimum value at $x = 0$ even though f is not differentiable at $x = 0$. Is this consistent with Theorem 2? Give reasons for your answer.
- 66. Even functions** If an even function $f(x)$ has a local maximum value at $x = c$, can anything be said about the value of f at $x = -c$? Give reasons for your answer.
- 67. Odd functions** If an odd function $g(x)$ has a local minimum value at $x = c$, can anything be said about the value of g at $x = -c$? Give reasons for your answer.
- 68.** We know how to find the extreme values of a continuous function $f(x)$ by investigating its values at critical points and endpoints. But what if there *are* no critical points or endpoints? What happens then? Do such functions really exist? Give reasons for your answers.
- 69. Cubic functions** Consider the cubic function

$$f(x) = ax^3 + bx^2 + cx + d.$$

- Show that f can have 0, 1, or 2 critical points. Give examples and graphs to support your argument.
- How many local extreme values can f have?

T 70. Functions with no extreme values at endpoints

- a. Graph the function

$$f(x) = \begin{cases} \sin \frac{1}{x}, & x > 0 \\ 0, & x = 0. \end{cases}$$

Explain why $f(0) = 0$ is not a local extreme value of f .

- b. Construct a function of your own that fails to have an extreme value at a domain endpoint.

T Graph the functions in Exercises 71–74. Then find the extreme values of the function on the interval and say where they occur.

71. $f(x) = |x - 2| + |x + 3|, \quad -5 \leq x \leq 5$

72. $g(x) = |x - 1| - |x - 5|, \quad -2 \leq x \leq 7$

73. $h(x) = |x + 2| - |x - 3|, \quad -\infty < x < \infty$

74. $k(x) = |x + 1| + |x - 3|, \quad -\infty < x < \infty$

COMPUTER EXPLORATIONS

In Exercises 75–80, you will use a CAS to help find the absolute extrema of the given function over the specified closed interval. Perform the following steps.

- Plot the function over the interval to see its general behavior there.
- Find the interior points where $f' = 0$. (In some exercises, you may have to use the numerical equation solver to approximate a solution.) You may want to plot f' as well.
- Find the interior points where f' does not exist.
- Evaluate the function at all points found in parts (b) and (c) and at the endpoints of the interval.
- Find the function's absolute extreme values on the interval and identify where they occur.

75. $f(x) = x^4 - 8x^2 + 4x + 2, \quad [-20/25, 64/25]$

76. $f(x) = -x^4 + 4x^3 - 4x + 1, \quad [-3/4, 3]$

77. $f(x) = x^{2/3}(3 - x), \quad [-2, 2]$

78. $f(x) = 2 + 2x - 3x^{2/3}, \quad [-1, 10/3]$

79. $f(x) = \sqrt{x} + \cos x, \quad [0, 2\pi]$

80. $f(x) = x^{3/4} - \sin x + \frac{1}{2}, \quad [0, 2\pi]$